



Volume 1
All previous year
TIFR CSE Questions

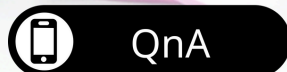
GATE OVERFLOW FOR TIFR CSE

Edition 1 2025

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led by

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QnA



Chat



FB Group

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1.0.1 TIFR CSE 2014 | Part B | Question: 6



Consider the problem of computing the minimum of a set of n distinct numbers. We choose a permutation uniformly at random (i.e., each of the $n!$ permutations of $\langle 1, \dots, n \rangle$ is chosen with probability $(1/n!)$ and we inspect the numbers in the order given by this permutation. We maintain a variable MIN that holds the minimum value seen so far. MIN is initialized to ∞ and if we see a value smaller than MIN during our inspection, then MIN is updated. For example, in the inspection given by the following sequence, MIN is updated four times.

5 9 4 2 6 8 0 3 1 7

What is the expected number of times MIN is updated?

- A. $O(1)$ B. $H_n = \sum_{i=1}^n 1/i$ C. $\sqrt{n}$ D. $n/2$
E. n

tifr2014 algorithms maximum-minimum

Answer key

1.0.2 TIFR CSE 2014 | Part B | Question: 9



Given a set of n distinct numbers, we would like to determine the smallest three numbers in this set using comparisons. Which of the following statements is TRUE?

- A. These three elements can be determined using $O(\log^2 n)$ comparisons.
B. $O(\log^2 n)$ comparisons do not suffice, however these three elements can be determined using $n + O(1)$ comparisons.
C. $n + O(1)$ comparisons do not suffice, however these three elements can be determined using $n + O(\log n)$ comparisons.
D. $n + O(\log n)$ comparisons do not suffice, however these three elements can be determined using $O(n)$ comparisons.
E. None of the above.

tifr2014 algorithms maximum-minimum

Answer key

1.0.3 TIFR CSE 2014 | Part B | Question: 10



Given a set of n distinct numbers, we would like to determine both the smallest and the largest number. Which of the following statements is TRUE?

- A. These two elements can be determined using $O(\log^{100} n)$ comparisons.
B. $O(\log^{100} n)$ comparisons do not suffice, however these two elements can be determined using $n + O(\log n)$ comparisons.
C. $n + O(\log n)$ comparisons do not suffice, however these two elements can be determined using $3\lceil n/2 \rceil$ comparisons.
D. $3\lceil n/2 \rceil$ comparisons do not suffice, however these two elements can be determined using $2(n-1)$ comparisons.
E. None of the above.

Answer key



1.0.4 TIFR CSE 2025 | Part B | Question: 9

Consider the following pseudocode. The program should take as input n , and when it ends, the element $A[i]$ of the vector A should contain the binomial coefficient $\binom{n}{i}$ (with the index of A starting from 0). In order to do so, what should replace the _____ in the pseudocode?

```

procedure BINOMIALCOEFFICIENT
  input n
  for i ← 0 to n do
    A[i] ← 0
  end for
  A[0] ← 1
  for i ← 1 to n do
    B ← A
    for j ← 1 to n do
      A[j] ← _____
    end for
  end for
end procedure

```

- A. $B(j-1) + B(j)$
 B. $A(j-1) + B(j-1)$
 C. $\frac{B(j) + B(n-j)}{2}$
 D. $\frac{B(j-1) \times B(j)}{2}$
 E. $\frac{A(j-1)^2 + B(j-1)}{2}$

Answer key

1.1

Algorithm Design (1)

1.1.1 Algorithm Design: TIFR CSE 2024 | Part B | Question: 7



Consider the following algorithm that takes as input a positive integer n .

```

if (n == 1) {
  return "Neither prime nor composite."
}
m=2
while (m < n) {
  if (m divides n){
    return "Composite."
  }
  m=m+1
}
return "Prime."

```

If n is a number of the form $n = p^2 q^3 r^4$ where p, q, r are natural numbers greater than 1, how many times does the while loop in the algorithm run?

In the options below, for any real number m , $\lceil m \rceil$ denotes the least integer greater than or equal to m .

- A. The while loop runs at most $\lceil n^{1/9} \rceil$ times for all natural numbers p, q, r greater than one.
 B. The while loop runs at most $\lceil n^{1/9} \rceil$ times only if p, q, r are all distinct.
 C. The while loop runs at most $\lceil n^{1/9} \rceil$ times only if at least two of p, q, r are distinct.
 D. The while loop runs at most $\lceil n^{1/9} \rceil$ times only if p, q, r are distinct primes.
 E. The while loop runs at most $\lceil n^{1/9} \rceil$ times only if p, q, r are distinct primes or distinct prime powers.

Answer key 

1.2

Asymptotic Notations (11)

1.2.1 Asymptotic Notations: TIFR CSE 2011 | Part B | Question: 27



Let n be a large integer. Which of the following statements is **TRUE**?

- A. $n^{\frac{1}{\sqrt{\log_2 n}}} < \sqrt{\log_2 n} < n^{\frac{1}{100}}$
- B. $n^{\frac{1}{100}} < n^{\frac{1}{\sqrt{\log_2 n}}} < \sqrt{\log_2 n}$
- C. $n^{\frac{1}{\sqrt{\log_2 n}}} < n^{\frac{1}{100}} < \sqrt{\log_2 n}$
- D. $\sqrt{\log_2 n} < n^{\frac{1}{\sqrt{\log_2 n}}} < n^{\frac{1}{100}}$
- E. $\sqrt{\log_2 n} < n^{\frac{1}{100}} < n^{\frac{1}{\sqrt{\log_2 n}}}$

tifr2011 asymptotic-notations

Answer key 

1.2.2 Asymptotic Notations: TIFR CSE 2012 | Part B | Question: 6



Let n be a large integer. Which of the following statements is **TRUE**?

- A. $2^{\sqrt{2 \log n}} < \frac{n}{\log n} < n^{1/3}$
- B. $\frac{n}{\log n} < n^{1/3} < 2^{\sqrt{2 \log n}}$
- C. $2^{\sqrt{2 \log n}} < n^{1/3} < \frac{n}{\log n}$
- D. $n^{1/3} < 2^{\sqrt{2 \log n}} < \frac{n}{\log n}$
- E. $\frac{n}{\log n} < 2^{\sqrt{2 \log n}} < n^{1/3}$

tifr2012 algorithms asymptotic-notations

Answer key 

1.2.3 Asymptotic Notations: TIFR CSE 2014 | Part B | Question: 8



Which of these functions grows fastest with n ?

- A. e^n/n .
- B. $e^{n-0.9 \log n}$.
- C. 2^n .
- D. $(\log n)^{n-1}$.
- E. None of the above.

tifr2014 algorithms asymptotic-notations

Answer key 

1.2.4 Asymptotic Notations: TIFR CSE 2016 | Part B | Question: 7



Let $n = m!$. Which of the following is **TRUE**?

- A. $m = \Theta(\log n / \log \log n)$
- B. $m = \Omega(\log n / \log \log n)$ but not $m = O(\log n / \log \log n)$
- C. $m = \Theta(\log^2 n)$
- D. $m = \Omega(\log^2 n)$ but not $m = O(\log^2 n)$
- E. $m = \Theta(\log^{1.5} n)$

tifr2016 algorithms asymptotic-notations

Answer key

1.2.5 Asymptotic Notations: TIFR CSE 2017 | Part A | Question: 4



Which of the following functions asymptotically grows the fastest as n goes to infinity?

- A. $(\log \log n)!$
- B. $(\log \log n)^{\log n}$
- C. $(\log \log n)^{\log \log \log n}$
- D. $(\log n)^{\log \log n}$
- E. $2^{\sqrt{\log \log n}}$

tifr2017 algorithms asymptotic-notations

Answer key

1.2.6 Asymptotic Notations: TIFR CSE 2018 | Part A | Question: 3



Which of the following statements is TRUE for all sufficiently large integers n ?

- A. $2^{2^{\sqrt{\log \log n}}} < 2^{\sqrt{\log n}} < n$
- B. $2^{\sqrt{\log n}} < n < 2^{2^{\sqrt{\log \log n}}}$
- C. $n < 2^{\sqrt{\log n}} < 2^{2^{\sqrt{\log \log n}}}$
- D. $n < 2^{2^{\sqrt{\log \log n}}} < 2^{\sqrt{\log n}}$
- E. $2^{\sqrt{\log n}} < 2^{2^{\sqrt{\log \log n}}} < n$

tifr2018 algorithms asymptotic-notations

Answer key

1.2.7 Asymptotic Notations: TIFR CSE 2018 | Part B | Question: 5



Which of the following functions, given by their recurrence, grows the fastest asymptotically?

- A. $T(n) = 4T\left(\frac{n}{2}\right) + 10n$
- B. $T(n) = 8T\left(\frac{n}{3}\right) + 24n^2$
- C. $T(n) = 16T\left(\frac{n}{4}\right) + 10n^2$
- D. $T(n) = 25T\left(\frac{n}{5}\right) + 20(n \log n)^{1.99}$
- E. They all are asymptotically the same

tifr2018 asymptotic-notations recurrence-relation

Answer key

1.2.8 Asymptotic Notations: TIFR CSE 2019 | Part B | Question: 5



Stirling's approximation for $n!$ states for some constants c_1, c_2

$$c_1 n^{n+\frac{1}{2}} e^{-n} \leq n! \leq c_2 n^{n+\frac{1}{2}} e^{-n}.$$

What are the tightest asymptotic bounds that can be placed on $n!$?

- A. $n! = \Omega(n^n)$ and $n! = \mathcal{O}(n^{n+\frac{1}{2}})$
- B. $n! = \Theta(n^{n+\frac{1}{2}})$
- C. $n! = \Theta\left(\left(\frac{n}{e}\right)^n\right)$
- D. $n! = \Theta\left(\left(\frac{n}{e}\right)^{n+\frac{1}{2}}\right)$
- E. $n! = \Theta(n^{n+\frac{1}{2}} 2^{-n})$

tifr2019 algorithms asymptotic-notations

Answer key

1.2.9 Asymptotic Notations: TIFR CSE 2020 | Part B | Question: 10



Among the following asymptotic expressions, which of these functions grows the slowest (as a function of n) asymptotically?

- A. $2^{\log n}$
 B. n^{10}
 C. $(\sqrt{\log n})^{\log^2 n}$
 D. $(\log n)^{\sqrt{\log n}}$
 E. $2^{2^{\sqrt{\log \log n}}}$

tifr2020 algorithms asymptotic-notations time-complexity

Answer key 

1.2.10 Asymptotic Notations: TIFR CSE 2024 | Part B | Question: 1



Consider the following three functions defined for all positive integers $n > 0$.

$$f(n) = |\sin(n) + n|,$$

$$g(n) = n,$$

$$h(n) = |\sin(n)|.$$

Which of the following statements about these functions is / are true?

- i. $f(n) = O(g(n))$
- ii. $g(n) = O(f(n))$
- iii. $h(n) = O(g(n))$

- A. Only (i) is true.
B. Only (ii) is true.
C. Only (i) and (ii) are true.
D. Only (ii) and (iii) are true.
E. All of (i), (ii), and (iii) are true.

tifr2024 algorithms asymptotic-notations

1.2.11 Asymptotic Notations: TIFR CSE 2024 | Part B | Question: 4



Consider functions f and g from the set of positive real numbers to itself, recursively defined as follows.

$$\begin{aligned} \forall n \leq 1 \quad f(n), g(n) &= 1, \\ \forall n > 1 \quad f(n) &= 3f(n/3) + n^2, \\ \forall n > 1 \quad g(n) &= n^{2/3}g(n^{1/3}) + n. \end{aligned}$$

Which of the following options is correct about their asymptotic behaviour?

- A. $f(n) = \Theta(n^2)$ and $g(n) = \Theta(n \log \log n)$ ✓
 B. $f(n) = \Theta(n^2)$ and $g(n) = \Theta(n \log n)$
 C. $f(n) = \Theta(n^2 \log n)$ and $g(n) = \Theta(n \log \log \log n)$
 D. $f(n) = \Theta(n^2 \log n)$ and $g(n) = \Theta(n \log \log n)$
 E. $f(n) = \Theta(n^2 \log n)$ and $g(n) = \Theta(n \log n)$

Answer key

1.3

Binary Search (3)

1.3.1 Binary Search: TIFR CSE 2010 | Part B | Question: 29



Suppose you are given an array A with $2n$ numbers.

The numbers in odd positions are sorted in ascending order, that is, $A[1] \leq A[3] \leq \dots \leq A[2n-1]$.

The numbers in even positions are sorted in descending order, that is, $A[2] \geq A[4] \geq \dots \geq A[2n]$.

What is the method you would recommend for determining if a given number is in the array?

- A. Sort the array using quick-sort and then use binary search.
- B. Merge the sorted lists and perform binary search.
- C. Perform a single binary search on the entire array.
- D. Perform separate binary searches on the odd positions and the even positions.
- E. Search sequentially from the end of the array.

Answer key

1.3.2 Binary Search: TIFR CSE 2012 | Part B | Question: 11



Consider the following three version of the binary search program. Assume that the elements of type T can be compared with each other; also assume that the array is sorted.

```

i, j, k : integer;
a : array [1...N] of T;
x : T;

Program 1 : i := 1; j := N;
repeat
    k := (i + j) div 2;
    if a[k] < x then i := k else j := k
until (a[k] = x) or (i > j)
Program 2 : i := 1; j := N;
repeat
    k := (i + j) div 2;
    if x < a[k] then j := k - 1;
    if a[k] < x then i := k + 1;
until i > j
Program 3 : i := 1; j := N;
repeat
    k := (i + j) div 2;
    if x < a[k] then j := k else i := k + 1
until i > j

```

A binary search program is called correct provided it terminates with $a[k] = x$ whenever such an element exists, or it terminates with $a[k] \neq x$ if there exists no array element with value x . Which of the following statements is correct?

- A. Only Program 1 is correct
- B. Only Program 2 is correct
- C. Only Program 1 and 2 are correct.
- D. Both Program 2 and 3 are correct
- E. All the three programs are wrong

Answer key

1.3.3 Binary Search: TIFR CSE 2019 | Part A | Question: 5



Asha and Lata play a game in which Lata first thinks of a natural number between 1 and 1000. Asha must find out that number by asking Lata questions, but Lata can only reply by saying “Yes” or “no”. Assume that Lata always tells the truth. What is the least number of questions that Asha needs to ask within which she can always find out the number Lata has thought of?

- A. 10
- B. 32
- C. 100
- D. 999
- E. None of the above

tifr2019 algorithm-design binary-search

Answer key

1.4

Combinatory (1)

1.4.1 Combinatory: TIFR CSE 2024 | Part A | Question: 3



There is a 100 cm long ruler that has 11 ants on positions 0 cm, 10 cm, 20 cm, 30 cm, ..., 100 cm. The ant at the 0 cm mark is facing towards the 100 cm mark, and the ant at the 100 cm is facing towards the 0 cm mark; all other ants is either facing towards 0 cm or towards the 100 cm mark.

All ants start moving in the direction they are facing at a speed of 1 cm per second. Whenever an ant collides with another ant, the two instantly reverse their direction and continue moving in the other direction. All ants continue moving until they fall off the ruler when they move past one of the two ends (at 0 cm or 100 cm marks).

How long will it take for all ants to fall off the ruler?

- A. At most 50 seconds, but cannot be determined without knowing the directions of all ants.
- B. 50 seconds.
- C. At least 50 seconds and at most 100 seconds, but cannot be determined without knowing the directions of all ants.
- D. 100 seconds.
- E. More than 100 seconds, but cannot be determined without knowing the directions of all ants.

tifr2024 analytical-aptitude combinatory

Answer key

1.5

Complexity Theory (1)

1.5.1 Complexity Theory: TIFR CSE 2024 | Part B | Question: 13



Suppose we are given a graph $G=(V, E)$ with non-negative edge weights $\{w_e\}_{e \in E}$.

Consider the following problems:

- P1: Finding a minimum spanning tree of G .
- P2: Finding a maximum spanning tree of G .
- P3: Finding a cycle of smallest weight in G .
- P4: Finding a cycle of largest weight in G .

Note: a cycle consists of distinct vertices.

Assuming $P \neq NP$, which of the above problems can be solved in polynomial time?

- A. P1, P2 but not P3, P4.
- B. P1, P3 but not P2, P4.
- C. P1 but not P2, P3, P4.
- D. P1, P2, P3 but not P4.
- E. P1, P4 but not P2, P3.

$L = \{(G, H) | G \text{ and } H \text{ are undirected graphs such that a subgraph of } G \text{ is isomorphic to } H\}$

Then which of the following are true?

- i. $L \in NP$.
- ii. L is NP -hard.
- iii. L is undecidable.

- A. Only (i)
- B. Only (ii)
- C. Only (iii)
- D. (i) and (ii)
- E. (ii) and (iii)

tifr2015 graph-theory graph-isomorphism p-np-npc-nph non-gatecse

Answer key 

1.8

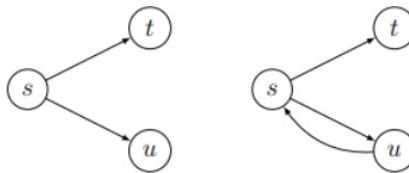
Graph Traversal (1)

1.8.1 Graph Traversal: TIFR CSE 2025 | Part B | Question: 5



Given a directed graph G and an initial vertex s , we would like to *explore* the graph from s , that is, starting from s see all vertices along a path.

For example, the graph on the left side below cannot be explored from s since there does not exist a path along which we can visit all vertices starting from s while the graph on the right side can be explored from s by following the path $s \rightarrow u \rightarrow s \rightarrow t$.



Which of the following is TRUE?

- A. There exists a polynomial time algorithm to explore the graph from s .
- B. There exist s an exponential time algorithm to explore the graph from s but there does not exist any polynomial time algorithm.
- C. There exists an algorithm to explore the graph from s but it is not known if it runs in exponential time.
- D. The decidability of the problem is open.
- E. The halting problem of Turing machine can be reduced to checking if a graph can be explored from s .

tifr2025 algorithms graph-theory graph-traversal

Answer key 

1.9

Greedy Algorithms (1)

1.9.1 Greedy Algorithms: TIFR CSE 2023 | Part B | Question: 3



Given a set $\mathcal{F}$ of intervals $(s_i, t_i)_{i=1}^n$ on the integer line (assume all s_i, t_i are distinct), a subset S of $\mathcal{F}$ is said to be *independent* if no two intervals in S have a non-empty intersection.

Consider greedy algorithms of the following type:


```

1 Order the intervals in  $\mathcal{F}$  according to an ordering  $\sigma$  (soon to be specified).
2  $S \leftarrow \emptyset$ ; //  $\emptyset$  denotes the empty set.
3 for  $I$  in  $\mathcal{F}$  (in the order specified by  $\sigma$ ) do
4     if  $I$  does not intersect any of the intervals already present in  $S$  then
5          $S \leftarrow S \cup \{I\}$ 
6 return  $S$ 

```

Here are three possible choices for the ordering σ :

- (C1) σ arranges the intervals in $\mathcal{F}$ in increasing order of s_i .
- (C2) σ arranges the intervals in $\mathcal{F}$ in increasing order of t_i .
- (C3) σ arranges the intervals in $\mathcal{F}$ in increasing order of $|s_i - t_i|$.

For which of these choices of the ordering σ does the algorithm always produce an independent set of $\mathcal{F}$ of maximum size?

- A. Choice (C1) only
- B. Choice (C2) only
- C. Choice (C3) only
- D. Choices (C2) and (C3), but not choice (C1)
- E. Choices (C1) and (C2), but not choice (C3)

tifr2023 algorithms greedy-algorithms

1.10

Identify Function (8)

1.10.1 Identify Function: TIFR CSE 2010 | Part B | Question: 24



Consider the following program operating on four variables u, v, x, y , and two constants X and Y .

```

x, y, u, v := X, Y, Y, X;
While (x ≠ y)
do
    if (x > y) then x, v := x - y, v + u;
    else if (y > x) then y, u := y - x, u + v;
od;
print ((x + y) / 2); print ((u + v) / 2);

```

Given $X > 0 \wedge Y > 0$, pick the true statement out of the following:

- A. The program prints $\gcd(X, Y)$ and the first prime larger than both X and Y .
- B. The program prints $\gcd(X, Y)$ followed by $\text{lcm}(X, Y)$.
- C. The program prints $\gcd(X, Y)$ followed by $\frac{1}{2} \times \text{lcm}(X, Y)$.
- D. The program prints $\frac{1}{2} \times \gcd(X, Y)$ followed by $\frac{1}{2} \times \text{lcm}(X, Y)$.
- E. The program does none of the above.

tifr2010 algorithms identify-function

Answer key

1.10.2 Identify Function: TIFR CSE 2014 | Part B | Question: 2



Consider the following code.

```

def brian(n):
    count = 0

    while (n != 0)
        n = n & (n - 1)
        count = count + 1

    return count

```

Here n is meant to be an unsigned integer. The operator $\&$ considers its arguments in binary and

computes their bit wise AND. For example, 22 & 15 gives 6, because the binary (say 8-bit) representation of 22 is 00010110 and the binary representation of 15 is 00001111, and the bit-wise AND of these binary strings is 00000110, which is the binary representation of 6. What does the function **brian** return?

- A. The highest power of 2 dividing n , but zero if n is zero.
- B. The number obtained by complementing the binary representation of n .
- C. The number of ones in the binary representation of n .
- D. The code might go into an infinite loop for some n .
- E. The result depends on the number of bits used to store unsigned integers.

tifr2014 algorithms identify-function

Answer key

1.10.3 Identify Function: TIFR CSE 2014 | Part B | Question: 20



Consider the following game. There is a list of distinct numbers. At any round, a player arbitrarily chooses two numbers a, b from the list and generates a new number c by subtracting the smaller number from the larger one. The numbers a and b are put back in the list. If the number c is non-zero and is not yet in the list, c is added to the list. The player is allowed to play as many rounds as the player wants. The score of a player at the end is the size of the final list.

Suppose at the beginning of the game the list contains the following numbers: 48, 99, 120, 165 and 273. What is the score of the best player for this game?

- A. 40
- B. 16
- C. 33
- D. 91
- E. 123

tifr2014 algorithms identify-function

Answer key

1.10.4 Identify Function: TIFR CSE 2015 | Part B | Question: 3



Consider the following code fragment in the C programming language when run on a non-negative integer n .

```
int f (int n)
{
    if (n==0 || n==1)
        return 1;
    else
        return f (n - 1) + f(n - 2);
}
```

Assuming a typical implementation of the language, what is the running time of this algorithm and how does it compare to the optimal running time for this problem?

- A. This algorithm runs in polynomial time in n but the optimal running time is exponential in n .
- B. This algorithm runs in exponential time in n and the optimal running time is exponential in n .
- C. This algorithm runs in exponential time in n but the optimal running time is polynomial in n .
- D. This algorithm runs in polynomial time in n and the optimal running time is polynomial in n .
- E. The algorithm does not terminate.

tifr2015 algorithms identify-function time-complexity

Answer key

1.10.5 Identify Function: TIFR CSE 2016 | Part B | Question: 12



A computer program computes a function $f: \{0,1\}^* \times \{0,1\}^* \rightarrow \{0,1\}^*$. Suppose $f(a,b)$ has length $|b|^2$, where $|a|$ and $|b|$ are the lengths of a and b . Suppose, using this program, the following computation is performed.

```
x="01"
for i=1, ..., n do
  x=f("01", x)
```

Suppose at the end, the length of the string x is t . Which of the following is TRUE (assume $n \geq 10$)?

- A. $t \leq 2n$
- B. $n < t \leq n^2$
- C. $n^2 < t \leq n^{\log_2 n}$
- D. $n^{\log_2 n} < t \leq 2^{(2^n)}$
- E. $2^{(2^n)} < t$

tifr2016 algorithms identify-function

Answer key

1.10.6 Identify Function: TIFR CSE 2017 | Part A | Question: 12



Consider the following program modifying an $n \times n$ square matrix A :

```
for i=1 to n:
  for j=1 to n:
    temp=A[i][j]+10
    A[i][j]=A[j][i]
    A[j][i]=temp-10
  end for
end for
```

Which of the following statements about the contents of matrix A at the end of this program must be TRUE?

- A. the new A is the transpose of the old A
- B. all elements above the diagonal have their values increased by 10 and all the values below have their values decreased by 10
- C. all elements above the diagonal have their values decreased by 10 and all the values below have their values increased by 10
- D. the new matrix A is symmetric, that is, $A[i][j] = A[j][i]$ for all $1 \leq i, j \leq n$
- E. A remains unchanged

tifr2017 algorithms identify-function

Answer key

1.10.7 Identify Function: TIFR CSE 2020 | Part B | Question: 7



Consider the following algorithm (Note: For positive integers, $p, q, p/q$ denotes the floor of the rational number $\frac{p}{q}$, assume that given $p, q, p/q$ can be computed in one step):

Input: Two positive integers $a, b, a \geq b$.

Output: A positive integers g .

```
while(b>0) {
  x = a - (a/b)*b;
  a = b;
  b = x;
}
g = a;
```

Suppose K is an upper bound on a . How many iterations does the above algorithm take in the worst case?

- A. $\Theta(\log K)$ B. $\Theta(K)$ C. $\Theta(K \log K)$ D. $\Theta(K^2)$
 E. $\Theta(2^K)$

tifr2020 algorithms identify-function time-complexity

Answer key

1.10.8 Identify Function: TIFR CSE 2022 | Part B | Question: 4



Consider the following algorithm for computing the factorial of a positive integer n , specified in binary:

```
prod ← 1
for i from 1 to n
    prod ← prod × i
output prod
```

Assume that the number of bit operations required to multiply a k -bit positive integer with an ℓ -bit positive integer is at least $\Omega(k + \ell)$ and at most $O(k\ell)$. Then, the number of bit operations required by this algorithm is

- A. $O(n)$ B. $O(n \log n)$ but $\omega(n)$
 C. $O(n^2)$ but $\omega(n \log n)$ D. $O(n^3)$ but $\omega(n^2)$
 E. None of the above

tifr2022 algorithms identify-function time-complexity

Answer key

1.11 Insertion Sort (1)

1.11.1 Insertion Sort: TIFR CSE 2025 | Part B | Question: 11



Recall the insertion sort algorithm:

```
procedure INSERTIONSORT(A)
  for i ← 1 to A.length - 1 do
    key ← A[i]
    j ← i - 1
    while j ≥ 0 and A[j] > key do
      A[j + 1] ← A[j]
      j ← j - 1
    end while
    A[j + 1] ← key
  end for
end procedure
```

Suppose for some parameter k (possibly a function of the length of the array n), we are guaranteed that the array has k inversions, i.e., there are exactly k pairs (i, j) such that $A[i] > A[j]$ and $i < j$. Then, the running time of insertion sort for A is:

- A. $\Theta(n + k)$ B. $\Theta(n \log n + k)$
 C. $\Theta(n^2 + k)$ D. $\Theta(k \log n)$
 E. $\Theta(n \log k)$

tifr2025 algorithms sorting insertion-sort time-complexity

Answer key

1.12 Linear Probing (1)

1.12.1 Linear Probing: TIFR CSE 2025 | Part B | Question: 10



Suppose 3 elements are hashed independently and uniformly at random one by one to slots in a hash table of size 6 (assume that in case of a collision, the element is hashed to the first free slot). What is the expected number of elements that **DO NOT** collide with an existing element in the table?

- A. $25/18$ B. $3/2$ C. $59/36$ D. 2
E. $5/2$

tifr2025 algorithms hashing expectation linear-probing

Answer key

1.13 Minimum Spanning Tree (9)

1.13.1 Minimum Spanning Tree: TIFR CSE 2011 | Part B | Question: 35



Let G be a connected simple graph (no self-loops or parallel edges) on $n \geq 3$ vertices, with distinct edge weights. Let $e_1, e_2, \dots, e_m$ be an ordering of the edges in decreasing order of weight. Which of the following statements is FALSE?

- A. The edge e_1 has to be present in every maximum weight spanning tree.
B. Both e_1 and e_2 have to be present in every maximum weight spanning tree.
C. The edge e_m has to be present in every minimum weight spanning tree.
D. The edge e_m is never present in any maximum weight spanning tree.
E. G has a unique maximum weight spanning tree.

tifr2011 algorithms graph-algorithms minimum-spanning-tree

Answer key

1.13.2 Minimum Spanning Tree: TIFR CSE 2013 | Part B | Question: 17



In a connected weighted graph with n vertices, all the edges have distinct positive integer weights. Then, the maximum number of minimum weight spanning trees in the graph is

- A. 1 B. n
C. equal to number of edges in the graph. D. equal to maximum weight of an edge of the graph.
E. n^{n-2}

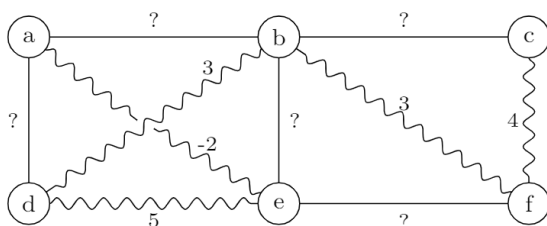
tifr2013 algorithms minimum-spanning-tree

Answer key

1.13.3 Minimum Spanning Tree: TIFR CSE 2014 | Part B | Question: 4



Consider the following undirected graph with some edge costs missing.



Suppose the wavy edges form a Minimum Cost Spanning Tree for G . Then, which of the following inequalities **NEED NOT** hold?

- A. $\text{cost}(a,b) \geq 6$.
 B. $\text{cost}(b,e) \geq 5$.
 C. $\text{cost}(e,f) \geq 5$.
 D. $\text{cost}(a,d) \geq 4$.
 E. $\text{cost}(b,c) \geq 4$.

tifr2014 algorithms graph-algorithms minimum-spanning-tree

Answer key

1.13.4 Minimum Spanning Tree: TIFR CSE 2014 | Part B | Question: 5



Let $G = (V, E)$ be an undirected connected simple (i.e., no parallel edges or self-loops) graph with the weight function $w : E \rightarrow \mathbb{R}$ on its edge set. Let $w(e_1) < w(e_2) < \dots < w(e_m)$, where $E = \{e_1, e_2, \dots, e_m\}$. Suppose T is a minimum spanning tree of G . Which of the following statements is FALSE?

- A. The tree T has to contain the edge e_1 .
- B. The tree T has to contain the edge e_2 .
- C. The minimum weight edge incident on each vertex has to be present in T .
- D. T is the unique minimum spanning tree in G .
- E. If we replace each edge weight $w_i = w(e_i)$ by its square w_i^2 , then T must still be a minimum spanning tree of this new instance.

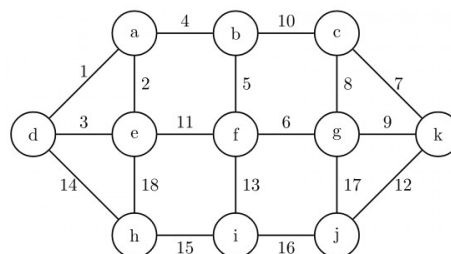
tifr2014 algorithms minimum-spanning-tree

Answer key

1.13.5 Minimum Spanning Tree: TIFR CSE 2015 | Part B | Question: 2



Consider the following undirected connected graph G with weights on its edges as given in the figure below. A minimum spanning tree is a spanning tree of least weight and a maximum spanning tree is one with largest weight. A second best minimum spanning tree whose weight is the smallest among all spanning trees that are not minimum spanning trees in G .



Which of the following statements is TRUE in the above graph? (Note that all the edge weights are distinct in the above graph)

- A. There is more than one minimum spanning tree and similarly, there is more than one maximum spanning tree here.
- B. There is a unique minimum spanning tree, however there is more than one maximum spanning tree here.
- C. There is more than one minimum spanning tree, however there is a unique maximum spanning tree here.
- D. There is more than one minimum spanning tree and similarly, there is more than one second-best minimum spanning tree here.
- E. There is unique minimum spanning tree, however there is more than one second-best minimum spanning tree here.

tifr2015 algorithms graph-algorithms minimum-spanning-tree

Answer key

1.13.6 Minimum Spanning Tree: TIFR CSE 2018 | Part B | Question: 13



Let $n \geq 3$, and let G be a simple, connected, undirected graph with the same number n of vertices and edges. Each edge of G has a distinct real weight associated with it. Let T be the minimum weight spanning tree of G . Which of the following statements is NOT ALWAYS TRUE ?

- A. The minimum weight edge of G is in T .
- B. The maximum weight edge of G is not in T .
- C. G has a unique cycle C and the minimum weight edge of C is also in T .
- D. G has a unique cycle C and the maximum weight edge of C is not in T .
- E. T can be found in $O(n)$ time from the adjacency list representation of G .

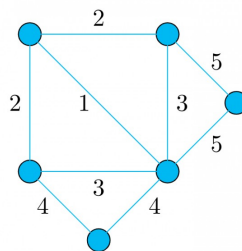
tifr2018 graph-algorithms minimum-spanning-tree

Answer key

1.13.7 Minimum Spanning Tree: TIFR CSE 2018 | Part B | Question: 3



How many distinct minimum weight spanning trees does the following undirected, weighted graph have ?



- A. 1
- B. 2
- C. 4
- D. 6
- E. 8

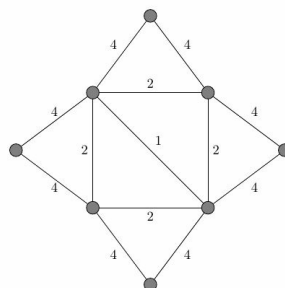
tifr2018 algorithms minimum-spanning-tree

Answer key

1.13.8 Minimum Spanning Tree: TIFR CSE 2019 | Part B | Question: 2



How many distinct minimum weight spanning trees does the following undirected, weighted graph have ?



- A. 8
- B. 16
- C. 32
- D. 64
- E. None of the above

tifr2019 algorithms minimum-spanning-tree

Answer key

1.13.9 Minimum Spanning Tree: TIFR CSE 2021 | Part B | Question: 10



Let G be a connected bipartite simple graph (i.e., no parallel edges) with distinct edge weights. Which of the following statements on **MST** (minimum spanning tree) need **NOT** be true?

- A. G has a unique **MST**.
- B. Every **MST** in G contains the lightest edge.
- C. Every **MST** in G contains the second lightest edge.
- D. Every **MST** in G contains the third lightest edge.
- E. No **MST** in G contains the heaviest edge.

tifr2021 algorithms minimum-spanning-tree

Answer key

1.14 P NP NPC NPH (6)

1.14.1 P NP NPC NPH: TIFR CSE 2010 | Part B | Question: 39



Suppose a language L is **NP** complete. Then which of the following is FALSE?

- A. $L \in \mathbf{NP}$
- B. Every problem in **P** is polynomial time reducible to L .
- C. Every problem in **NP** is polynomial time reducible to L .
- D. The Hamilton cycle problem is polynomial time reducible to L .
- E. $\mathbf{P} \neq \mathbf{NP}$ and $L \in \mathbf{P}$.

tifr2010 algorithms p-np-npc-nph

Answer key

1.14.2 P NP NPC NPH: TIFR CSE 2011 | Part B | Question: 37



Given an integer $n \geq 3$, consider the problem of determining if there exist integers $a, b \geq 2$ such that $n = a^b$. Call this the forward problem. The reverse problem is: given a and b , compute $a^b \pmod{b}$. Note that the input length for the forward problem is $\lfloor \log n \rfloor + 1$, while the input length for the reverse problem is $\lfloor \log a \rfloor + \lfloor \log b \rfloor + 2$. Which of the following statements is TRUE?

- A. Both the forward and reverse problems can be solved in time polynomial in the lengths of their respective inputs.
- B. The forward problem can be solved in polynomial time, however the reverse problem is **NP**-hard.
- C. The reverse problem can be solved in polynomial time, however the forward problem is **NP**-hard.
- D. Both the forward and reverse problem are **NP**-hard.
- E. None of the above.

tifr2011 algorithms p-np-npc-nph

Answer key

1.14.3 P NP NPC NPH: TIFR CSE 2013 | Part B | Question: 7



Which of the following is not implied by $P = \mathbf{NP}$?

- A. **3SAT** can be solved in polynomial time.
- B. Halting problem can be solved in polynomial time.

- C. Factoring can be solved in polynomial time.
- D. Graph isomorphism can be solved in polynomial time.
- E. Travelling salesman problem can be solved in polynomial time.

tifr2013 algorithms p-np-npc-nph

Answer key

1.14.4 P NP NPC NPH: TIFR CSE 2017 | Part B | Question: 15



A multivariate polynomial in n variables with integer coefficients has a binary root if it is possible to assign each variable either 0 or 1, so that the polynomial evaluates to 0. For example, the multivariate polynomial $-2x_1^3 - x_1x_2 + 2$ has the binary root $(x_1 = 1, x_2 = 0)$. Then determining whether a multivariate polynomial, given as the sum of monomials, has a binary root:

- A. is trivial: every polynomial has a binary root
- B. can be done in polynomial time
- C. is NP-hard, but not in NP
- D. is in NP, but not in P and not NP-hard
- E. is both in NP and NP-hard

tifr2017 algorithms p-np-npc-nph

Answer key

1.14.5 P NP NPC NPH: TIFR CSE 2017 | Part B | Question: 2



Consider the following statements:

- i. Checking if a given *undirected* graph has a cycle is in P
- ii. Checking if a given *undirected* graph has a cycle is in NP
- iii. Checking if a given *directed* graph has a cycle is in P
- iv. Checking if a given *directed* graph has a cycle is in NP

Which of the above statements is/are TRUE? Choose from the following options.

- A. Only i and ii
- B. Only ii and iv
- C. Only ii, iii, and iv
- D. Only i, ii and iv
- E. All of them

tifr2017 algorithms graph-algorithms p-np-npc-nph

Answer key

1.14.6 P NP NPC NPH: TIFR CSE 2022 | Part B | Question: 10



Consider the assertions

(A1) Given a directed graph G with positive weights on the edges, two special vertices s and t , and an integer k - it is NP-complete to determine if G has an $s - t$ path of length at most k .

(A2) $P = NP$

Then, which of the following is true?

- A. A1 implies A2 and A2 implies A1
- B. A1 implies A2 and A2 does not imply A1
- C. A1 does not imply A2 and A2 implies A1
- D. A1 does not imply A2 and A2 does not imply A1
- E. None of the above.

tifr2022 algorithms graph-connectivity p-np-npc-nph

1.15.1 Probability: TIFR CSE 2024 | Part B | Question: 8



In the following pseudocode, assume that for any pair of integers $x \leq y$, the function `random (x, y)` produces an integer uniformly chosen from the set $\{x, x + 1, \dots, y\}$.

```
n=9
for (i=1 to n){
  A[i]=i
}
for (i=1 to n){
  r= random (i, n)
  temp =A[i]
  A [i]= A [r]
  A[r]=temp
  print A[i]
}
```

Which of the following statements is **TRUE** of the output of the code?

- A. It outputs all permutations of 123456789 with equal probability.
- B. It never outputs 123456789.
- C. It outputs all cyclic permutations of 123456789 with equal probability, and does not print any other output.
- D. The output is always 987654321.
- E. The output may not be a permutation of 123456789.

tifr2024 algorithms data-structures probability

Answer key

1.16

Quick Sort (3)

1.16.1 Quick Sort: TIFR CSE 2012 | Part B | Question: 14



Consider the quick sort algorithm on a set of n numbers, where in every recursive subroutine of the algorithm, the algorithm chooses the median of that set as the pivot. Then which of the following statements is TRUE?

- A. The running time of the algorithm is $\Theta(n)$.
- B. The running time of the algorithm is $\Theta(n \log n)$.
- C. The running time of the algorithm is $\Theta(n^{1.5})$.
- D. The running time of the algorithm is $\Theta(n^2)$.
- E. None of the above.

tifr2012 algorithms sorting quick-sort time-complexity

Answer key

1.16.2 Quick Sort: TIFR CSE 2013 | Part B | Question: 18



Let S be a set of numbers. For $x \in S$, the rank of x is the number of elements in S that are less than or equal to x . The procedure **Select**(S, r) takes a set S of numbers and a rank r ($1 \leq r \leq |S|$) and returns the element in S of rank r . The procedure **MultiSelect**(S, R) takes a set of numbers S and a list of ranks $R = \{r_1 < r_2 < \dots < r_k\}$, and returns the list $\{x_1 < x_2 < \dots < x_k\}$ of elements of S , such that the rank of x_i is r_i . Suppose there is an implementation for **Select**(S, r) that uses at most (constant $\cdot |S|$) binary comparisons between elements of S . The minimum number of comparisons needed to implement **MultiSelect**(S, R) is

- A. $\text{constant} \cdot |S| \log |S|$
 C. $\text{constant} \cdot |S| |R|$
 E. $\text{constant} \cdot |S| (1 + \log |R|)$

- B. $\text{constant} \cdot |S|$
 D. $\text{constant} \cdot |R| \log |S|$

tifr2013 algorithms quick-sort time-complexity

Answer key

1.16.3 Quick Sort: TIFR CSE 2018 | Part B | Question: 7



Consider the recursive quicksort algorithm with "random pivoting". That is, in each recursive call, a pivot is chosen uniformly at random from the sub-array being sorted. When this randomized algorithm is applied to an array of size n all whose elements are distinct, what is the probability that the smallest and the largest elements in the array are compared during a run of the algorithm ?

- A. $\left(\frac{1}{n}\right)$
 C. $\Theta\left(\frac{1}{n \log n}\right)$
 E. $\Theta\left(\frac{1}{n \log^2 n}\right)$
 B. $\left(\frac{2}{n}\right)$
 D. $O\left(\frac{1}{n^2}\right)$

tifr2018 algorithms sorting quick-sort

Answer key

1.17 Randomized Algorithms (1)

1.17.1 Randomized Algorithms: TIFR CSE 2025 | Part B | Question: 13



There are n people in a house and n pairs of shoes such that the i -th shoe fits the i -th person's feet (and no one else's). A burglar comes and throws the shoes around in some indescribable manner. The people want to determine whose feet fit which shoe and they can only use the following oracles:

- **Fit** : Takes as input a person and a shoe and outputs $<$ if the person's feet are too small for the shoe, $=$ if they fit, and $>$ if they are too large.
- **Compare** : Takes as input two people or two shoes and outputs $<$ if the first is smaller than the second, $=$ if they are the same size, and $>$ if the first is larger than the second.

Which of the following statements are **TRUE** (assuming access to randomness)?

1. If they only have the **Fit** oracle, the determination can be made in $O(n \log n)$ calls to the oracle in expectation.
2. If they only have the **Compare** oracle, the determination can be made in $O(n \log n)$ calls to the oracle in expectation.
3. If they have both oracles, the determination can be made in $O(\sqrt{n})$ calls to the oracles in expectation.

- A. Only 1.
 C. Only 3.
 E. All three statements are true.
 B. Only 2.
 D. Only 1 and 2.

tifr2025 algorithms randomized-algorithms sorting time-complexity

1.18 Recurrence Relation (4)

1.18.1 Recurrence Relation: TIFR CSE 2014 | Part B | Question: 11



Consider the following recurrence relation:

$$T(n) = \begin{cases} T\left(\frac{n}{k}\right) + T\left(\frac{3n}{4}\right) + n & \text{if } n \geq 2 \\ 1 & \text{if } n = 1 \end{cases}$$

Which of the following statements is FALSE?

- A. $T(n)$ is $O(n^{3/2})$ when $k = 3$.
- B. $T(n)$ is $O(n \log n)$ when $k = 3$.
- C. $T(n)$ is $O(n \log n)$ when $k = 4$.
- D. $T(n)$ is $O(n \log n)$ when $k = 5$.
- E. $T(n)$ is $O(n)$ when $k = 5$.

tifr2014 algorithms recurrence-relation

Answer key

1.18.2 Recurrence Relation: TIFR CSE 2015 | Part B | Question: 1



Consider the following recurrence relation:

$$T(n) = \begin{cases} 2T(\lfloor \sqrt{n} \rfloor) + \log n & \text{if } n \geq 2 \\ 1 & \text{if } n = 1 \end{cases}$$

Which of the following statements is TRUE?

- A. $T(n)$ is $O(\log n)$.
- B. $T(n)$ is $O(\log n \cdot \log \log n)$ but not $O(\log n)$.
- C. $T(n)$ is $O(\log^{3/2} n)$ but not $O(\log n \cdot \log \log n)$.
- D. $T(n)$ is $O(\log^2 n)$ but not $O(\log^{3/2} n)$.
- E. $T(n)$ is $O(\log^2 n \cdot \log \log n)$ but not $O(\log^2 n)$.

tifr2015 algorithms recurrence-relation time-complexity

Answer key

1.18.3 Recurrence Relation: TIFR CSE 2017 | Part A | Question: 15



Let $T(a, b)$ be the function with two arguments (both nonnegative integral powers of 2) defined by the following recurrence:

- $T(a, b) = T\left(\frac{a}{2}, b\right) + T\left(a, \frac{b}{2}\right)$ if $a, b \geq 2$;
- $T(a, 1) = T\left(\frac{a}{2}, 1\right)$ if $a \geq 2$;
- $T(1, b) = T\left(1, \frac{b}{2}\right)$ if $b \geq 2$;
- $T(1, 1) = 1$.

What is $T(2^r, 2^s)$?

- A. rs
- B. $r + s$
- C. $\binom{2^r + 2^s}{2^r}$
- D. $\binom{r + s}{r}$
- E. 2^{r-s} if $r \geq s$, otherwise 2^{s-r}

tifr2017 algorithms recurrence-relation

Answer key

1.18.4 Recurrence Relation: TIFR CSE 2023 | Part B | Question: 6



What is the solution to the following recurrence?

$$T(n) = \begin{cases} 1 & \text{if } n \leq 10, \\ \sqrt{n} \cdot T(\sqrt{n}) + n & \text{if } n > 10. \end{cases}$$

- A. $T(n) = \Theta(n^2)$
 C. $T(n) = \Theta(n\sqrt{\log n})$
 E. None of the above

- B. $T(n) = \Theta(n \log n)$
 D. $T(n) = \Theta(n \log \log n)$

tifr2023 algorithms recurrence-relation time-complexity

Answer key

1.19

Recursion (1)

1.19.1 Recursion: TIFR CSE 2025 | Part B | Question: 1



What is the solution to the following recursion?

$$T(n) = T\left(\frac{n}{2}\right) + T\left(\frac{n}{3}\right) + T\left(\frac{n}{6}\right) + O(n),$$

$$T(n) = 5 \quad \forall n < 100.$$

- A. $T(n) = \Theta(n \log n)$
 C. $T(n) = \Theta(n^2)$
 E. $T(n) = 2^{\Theta(n)}$

- B. $T(n) = \Theta(n \log^2 n)$
 D. $T(n) = \Theta(n^2 \log n)$

tifr2025 algorithms time-complexity recursion

Answer key

1.20

Searching (1)

1.20.1 Searching: TIFR CSE 2024 | Part B | Question: 10



Arun has a non-empty subset S of the numbers $\{1, 2, 3, \dots, 1000\}$. Bela wants to find any number x in Arun's set S .

To do this, Arun and Bela decide to play a game which proceeds in rounds. In each round, Bela can give Arun a subset $T \subseteq \{1, 2, 3, \dots, 1000\}$, and Arun answers with a "YES" if $S \cap T$ is non-empty, and with "NO" otherwise.

In the worst case, how many rounds will Bela need to find out some x in Arun's set S ?

- A. 9 B. 10 C. 11 D. 1023
 E. 1024

tifr2024 combinatory searching algorithm-design probability

Answer key

1.21

Shortest Path (3)

1.21.1 Shortest Path: TIFR CSE 2018 | Part B | Question: 9



Let $G = (V, E)$ be a DIRECTED graph, where each edge e has a positive weight $\omega(e)$, and all vertices can be reached from vertex s . For each vertex v , let $\phi(v)$ be the length of the shortest path from s to v . Let $G' = (V, E)$ be a new weighted graph with the same vertices and edges, but with the edge weight of every edge $e = (u \rightarrow v)$ changed to

$\omega'(e) = \omega(e) + \phi(v) - \phi(u)$. Let P be a path from s to a vertex v , and let $\omega(P) = \sum_{e \in P} \omega_e$, and $\omega'(P) = \sum_{e \in P} \omega'_e$.

Which of the following options is NOT NECESSARILY TRUE ?

- A. If P is a shortest path in G , then P is a shortest path in G' .
- B. If P is a shortest path in G' , then P is a shortest path in G .
- C. If P is a shortest path in G , then $\omega'(P) = 2 \times \omega(P)$.
- D. If P is NOT a shortest path in G , then $\omega'(P) < 2 \times \omega(P)$.
- E. All of the above options are necessarily TRUE.

tifr2018 graph-algorithms shortest-path

Answer key

1.21.2 Shortest Path: TIFR CSE 2022 | Part B | Question: 13



Consider a directed graph $G = (V, E)$, where each edge $e \in E$ has a positive edge weight c_e . Determine the appropriate choices for the blanks below so that the value of the following linear program is the length of the shortest directed path in G from s to t . (Assume that the graph has at least one path from s to t .)

$$\begin{aligned} & \text{(blank 1)imize} && X_t \\ & \text{s.t.} && X_s = 0 \\ & && X_w - X_v \text{ (blank 2)} && c_e \quad (\text{for each edge } e = (v, w) \in E). \end{aligned}$$

- A. blank 1 : max, blank 2 : $\leq$
- B. blank 1 : max, blank 2 : $\geq$
- C. blank 1 : min, blank 2 : $\leq$
- D. blank 1 : min, blank 2 : $\geq$
- E. blank 1 : min, blank 2 : $=$

tifr2022 algorithms shortest-path

1.21.3 Shortest Path: TIFR CSE 2025 | Part B | Question: 8



Let $G = (V, E)$ be a weighted, undirected and connected graph, with weight $1 \leq \text{wt}_G(e) \leq 99$ for edge $e \in E$. Suppose G' is the graph with the same set of vertices and edges, but with edge weights $\text{wt}_{G'}(e) = 100 - \text{wt}_G(e)$. Assume that $|V|$ is an even number.

Consider the following statements.

- i. Any minimum-weight spanning-tree of G is also a maximum-weight spanning-tree of G' .
- ii. For any pair of vertices $s, t \in V$, any shortest path from s to t in G is also a longest path from s to t in G' .
- iii. Any minimum-weight perfect matching of G is also a maximum-weight perfect matching of G' .

Which of the above statements ALWAYS TRUE for any such graphs G, G' ?

- A. Only (i) and (ii)
- B. Only (i) and (iii)
- C. Only (ii) and (iii)
- D. (i), (ii) and (iii)
- E. None of (i), (ii) or (iii) is always true.

tifr2025 algorithms minimum-spanning-tree shortest-path graph-theory

1.22.1 Sorting: TIFR CSE 2010 | Part B | Question: 23



Suppose you are given n numbers and you sort them in descending order as follows:

First find the maximum. Remove this element from the list and find the maximum of the remaining elements, remove this element, and so on, until all elements are exhausted. How many comparisons does this method require in the worst case?

- A. Linear in n .
- B. $O(n^2)$ but not better.
- C. $O(n \log n)$
- D. Same as heap sort.
- E. $O(n^{1.5})$ but not better.

tifr2010 algorithms time-complexity sorting

Answer key

1.22.2 Sorting: TIFR CSE 2010 | Part B | Question: 27



Consider the Insertion Sort procedure given below, which sorts an array L of size $n (\geq 2)$ in ascending order:

```
begin
  for xindex:= 2 to n do
    x := L[xindex];
    j:= xindex - 1;
    while j > 0 and L[j] > x do
      L[j + 1] := L[j];
      j:= j - 1;
    end {while}
    L[j + 1] := x;
  end {for}
end
```

It is known that insertion sort makes at most $n(n-1)/2$ comparisons. Which of the following is true?

- A. There is no input on which insertion Sort makes $n(n-1)/2$ comparisons.
- B. Insertion Sort makes $n(n-1)/2$ comparisons when the input is already sorted in ascending order.
- C. Insertion Sort makes $n(n-1)/2$ comparisons only when the input is sorted in descending order.
- D. There are more than one input orderings where insertion sort makes $n(n-1)/2$ comparisons.
- E. Insertion Sort makes $n(n-1)/2$ comparisons whenever all the elements of L are not distinct.

tifr2010 algorithms sorting

Answer key

1.22.3 Sorting: TIFR CSE 2011 | Part B | Question: 21



Let $S = \{x_1, \dots, x_n\}$ be a set of n numbers. Consider the problem of storing the elements of S in an array $A[1..n]$ such that the following min-heap property is maintained for all $2 \leq i \leq n : A[\lfloor i/2 \rfloor] \leq A[i]$. (Note that $\lfloor x \rfloor$ is the largest integer that is at most x). Which of the following statements is TRUE?

- A. This problem can be solved in $O(\log n)$ time.
- B. This problem can be solved in $O(n)$ time but not in $O(\log n)$ time.
- C. This problem can be solved in $O(n \log n)$ time but not in $O(n)$ time.
- D. This problem can be solved in $O(n^2)$ time but not in $O(n \log n)$ time.
- E. None of the above.

tifr2011 algorithms sorting

Answer key

1.22.4 Sorting: TIFR CSE 2011 | Part B | Question: 31



Given a set of $n = 2^k$ distinct numbers, we would like to determine the smallest and the second smallest using comparisons. Which of the following statements is TRUE?

- A. Both these elements can be determined using $2k$ comparisons.
- B. Both these elements can be determined using $n - 2$ comparisons.
- C. Both these elements can be determined using $n + k - 2$ comparisons.
- D. $2n - 3$ comparisons are necessary to determine these two elements.
- E. nk comparisons are necessary to determine these two elements.

tifr2011 algorithms sorting

Answer key

1.22.5 Sorting: TIFR CSE 2011 | Part B | Question: 39



The first n cells of an array L contain positive integers sorted in decreasing order, and the remaining $m - n$ cells all contain 0. Then, given an integer x , in how many comparisons can one find the position of x in L ?

- A. At least n comparisons are necessary in the worst case.
- B. At least $\log m$ comparisons are necessary in the worst case.
- C. $O(\log(m - n))$ comparisons suffice.
- D. $O(\log n)$ comparisons suffice.
- E. $O(\log(m/n))$ comparisons suffice.

tifr2011 algorithms sorting

Answer key

1.22.6 Sorting: TIFR CSE 2012 | Part B | Question: 13



An array A contains n integers. We wish to sort A in ascending order. We are told that initially no element of A is more than a distance k away from its final position in the sorted list. Assume that n and k are large and k is much smaller than n . Which of the following is true for the worst case complexity of sorting A ?

- A. A can be sorted with constant $\cdot kn$ comparison but not with fewer comparisons.
- B. A cannot be sorted with less than constant $\cdot n \log n$ comparisons.
- C. A can be sorted with constant $\cdot n$ comparisons.
- D. A can be sorted with constant $\cdot n \log k$ comparisons but not with fewer comparisons.
- E. A can be sorted with constant $\cdot k^2 n$ comparisons but not fewer.

tifr2012 algorithms sorting

Answer key

1.22.7 Sorting: TIFR CSE 2013 | Part B | Question: 12



It takes $O(n)$ time to find the median in a list of n elements, which are not necessarily in sorted order while it takes only $O(1)$ time to find the median in a list of n sorted elements. How much time does it take to find the median of $2n$ elements which are given as two lists of n sorted elements each?

- A. $O(1)$
- C. $O(\sqrt{n})$ but not $O(\log n)$
- E. $O(n \log n)$ but not $O(n)$

- B. $O(\log n)$ but not $O(1)$
- D. $O(n)$ but not $O(\sqrt{n})$

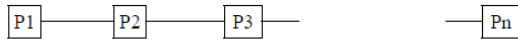
tifr2013 algorithms sorting time-complexity

Answer key 

1.22.8 Sorting: TIFR CSE 2013 | Part B | Question: 20



Suppose n processors are connected in a linear array as shown below. Each processor has a number. The processors need to exchange numbers so that the numbers eventually appear in ascending order (the processor P1 should have the minimum value and the the processor P n should have the maximum value).



The algorithm to be employed is the following. Odd numbered processors and even numbered processors are activated alternate steps; assume that in the first step all the even numbered processors are activated. When a processor is activated, the number it holds is compared with the number held by its right-hand neighbour (if one exists) and the smaller of the two numbers is retained by the activated processor and the bigger stored in its right hand neighbour.

How long does it take for the processors to sort the values?

- A. $n \log n$ steps
- C. n steps
- E. The algorithm is not guaranteed to sort
- B. n^2 steps
- D. $n^{1.5}$ steps

tifr2013 algorithms sorting

Answer key 

1.22.9 Sorting: TIFR CSE 2017 | Part B | Question: 7



An array of n distinct elements is said to be un-sorted if for every index i such that $2 \leq i \leq n-1$, either $A[i] > \max\{A[i-1], A[i+1]\}$, or $A[i] < \min\{A[i-1], A[i+1]\}$.

What is the time-complexity of the fastest algorithm that takes as input a sorted array A with n distinct elements, and un-sorts A ?

- A. $O(n \log n)$ but not $O(n)$
- C. $O(\sqrt{n})$ but not $O(\log n)$
- E. $O(1)$
- B. $O(n)$ but not $O(\sqrt{n})$
- D. $O(\log n)$ but not $O(1)$

tifr2017 algorithms sorting time-complexity

Answer key 

1.22.10 Sorting: TIFR CSE 2021 | Part B | Question: 15



Let $A[i] : i = 0, 1, 2, \dots, n-1$ be an array of n distinct integers. We wish to sort A in ascending order. We are given that each element in the array is at a position that is at most k away from its position in the sorted array, that is, we are given that $A[i]$ will move to a position in $\{i-k, i-k+1, \dots, i, \dots, i+k-1, i+k\}$ after the array is sorted in ascending order. Suppose insertion sort is used to sort this array: that is, in the i th iteration, $A[i]$ is compared with the elements in positions $A[i-1], A[i-2], \dots$ until one that is smaller is found and $A[i]$ is inserted after that element. Note that elements can be moved back when later insertions are made before them. Let $t(n)$ be the worst-case number of comparisons made by insertion sort for such inputs. Then,

- A. $t(n) = \Theta(n^2)$
- B. $t(n) = \Theta(n \log_2 n)$
- C. $t(n) = \Theta(nk \log k)$

- D. $t(n) = \Theta(n \log_2 k)$
 E. $t(n) = \Theta(nk)$

tifr2021 algorithms sorting time-complexity

Answer key

1.22.11 Sorting: TIFR CSE 2022 | Part B | Question: 11



Consider the following function **count**, that takes as input a , an array of integers, and N , the size of the array.

```
int count(int a[], int N) {
    int i, j, count_FN;
    count_FN = 0;
    for (i=1 ; i<N ; i++) {
        j=i-1 ;
        while (a[j]>a[i]) {
            count_FN++;
            j--;
        }
    }
    return count_FN;
}
```

Further, let **count_IS** be the number of comparisons made by the insertion sort algorithm on the array a .

Which of the following statements is TRUE for some constant c ?

- A. For all $N \geq c$, there exists an array of size N for which $\text{count_IS} \geq N^2/c$, while $\text{count_FN} \leq cN$
 B. For all $N \geq c$, there exists an array of size N for which $\text{count_FN} \geq N^2/c$, while $\text{count_IS} \leq cN$
 C. For all $N \geq c$, for all arrays of size N , $\text{count_FN} \leq \text{count_IS} \leq c \times \text{count_FN}$
 D. For all $N \geq c$, for all arrays of size N , $\text{count_FN} \geq N^2/c$
 E. None of the above

tifr2022 algorithms sorting

1.22.12 Sorting: TIFR CSE 2022 | Part B | Question: 3



Consider the problem of sorting n single digit integers (base 10). This problem can be solved in time

- A. $O(n \log n)$ but not $O(n \log \log n)$
 B. $O(n \log \log n)$ but not $O(n)$
 C. $O(n)$ but not $O(n/\log \log n)$
 D. $O(n/\log \log n)$
 E. None of the above.

tifr2022 algorithms sorting time-complexity

Answer key

1.23

Time Complexity (1)

1.23.1 Time Complexity: TIFR CSE 2014 | Part B | Question: 7



Which of the following statements is TRUE for all sufficiently large n ?

- A. $(\log n)^{\log \log n} < 2^{\sqrt{\log n}} < n^{1/4}$
 B. $2^{\sqrt{\log n}} < n^{1/4} < (\log n)^{\log \log n}$

C. $n^{1/4} < (\log n)^{\log \log n} < 2^{\sqrt{\log n}}$

D. $(\log n)^{\log \log n} < n^{1/4} < 2^{\sqrt{\log n}}$

E. $2^{\sqrt{\log n}} < (\log n)^{\log \log n} < n^{1/4}$

tifr2014 algorithms time-complexity

Answer key 

1.24

Tower of Hanoi (1)

1.24.1 Tower of Hanoi: TIFR CSE 2011 | Part B | Question: 29



You are given ten rings numbered from 1 to 10, and three pegs labeled A , B , and C . Initially all the rings are on peg A , arranged from top to bottom in ascending order of their numbers. The goal is to move all the rings to peg B in the minimum number of moves obeying the following constraints:

- In one move, only one ring can be moved.
- A ring can only be moved from the top of its peg to the top of a new peg.
- At no point can a ring be placed on top of another ring with a lower number.

How many moves are required?

- A. 501 B. 1023 C. 2011 D. 10079
- E. None of the above.

tifr2011 data-structures stack tower-of-hanoi

Answer key 

Answer Keys

1.0.1	B	1.0.2	C	1.0.3	C	1.0.4	A	1.1.1	A
1.2.1	D	1.2.2	C	1.2.3	D	1.2.4	A	1.2.5	B
1.2.6	A	1.2.7	C	1.2.8	D	1.2.9	E	1.2.10	E
1.2.11	A	1.3.1	D	1.3.2	E	1.3.3	A	1.4.1	D
1.5.1	D	1.6.1	C	1.6.2	D	1.6.3	D	1.7.1	D
1.8.1	A	1.9.1	B	1.10.1	B	1.10.2	C	1.10.3	D
1.10.4	C	1.10.5	D	1.10.6	E	1.10.7	A	1.10.8	D
1.11.1	A	1.12.1	E	1.13.1	D	1.13.2	A	1.13.3	A
1.13.4	E	1.13.5	E	1.13.6	B	1.13.7	C	1.13.8	D
1.13.9	E	1.14.1	E	1.14.2	A	1.14.3	B	1.14.4	E
1.14.5	E	1.14.6	A	1.15.1	A	1.16.1	B	1.16.2	E
1.16.3	B	1.17.1	D	1.18.1	B	1.18.2	B	1.18.3	D
1.18.4	D	1.19.1	A	1.20.1	B	1.21.1	E	1.21.2	A
1.21.3	B	1.22.1	B	1.22.2	D	1.22.3	B	1.22.4	C

1.22.5	D
1.22.10	E

1.22.6	D
1.22.11	A

1.22.7	B
1.22.12	C

1.22.8	C
1.23.1	A

1.22.9	B
1.24.1	B



2.1 Expression Evaluation (1)

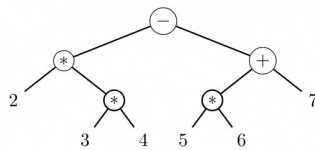
2.1.1 Expression Evaluation: TIFR CSE 2015 | Part B | Question: 15



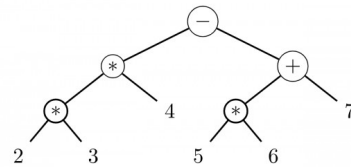
Consider the following grammar (the start symbol is E) for generating expressions.

- $E \rightarrow T - E \mid T + E \mid T$
- $T \rightarrow T * F \mid F$
- $F \rightarrow 0 \mid 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9$

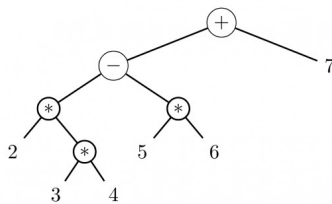
With respect to this grammar, which of the following trees is the valid evaluation tree for the expression $2 * 3 * 4 - 5 * 6 + 7$?



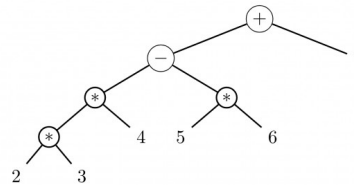
A.



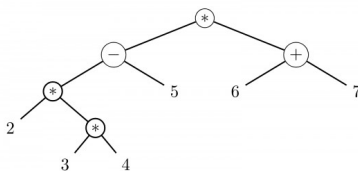
B.



C.



D.



E.

tifr2015 parsing expression-evaluation

Answer key

2.2

Grammar (1)

2.2.1 Grammar: TIFR CSE 2022 | Part B | Question: 7



Consider the following grammar: P, Q, R are non-terminals; c, d are terminals; P is the start symbol; and the production rules follow.

$P ::= QR$

$Q ::= c$

$Q ::= RcR$

$R ::= ddQ$

Which of the following is **False**:

- A. The length of every string produced by the grammar is even
- B. No string produced by the grammar has an odd number of consecutive d 's
- C. No string produced by the grammar has four consecutive d 's

- D. No string produced by the grammar has three consecutive c 's
 E. Every string produced by the grammar has at least as many d 's as c 's

tifr2022 compiler-design grammar

Answer key

2.3 LR Parser (1)

2.3.1 LR Parser: TIFR CSE 2012 | Part B | Question: 17



Which of the following correctly describes $LR(k)$ parsing?

- A. The input string is alternately scanned left to right and right to left with k reversals.
- B. Input string is scanned once left to right with rightmost derivation and k symbol look-ahead.
- C. $LR(k)$ grammars are expressively as powerful as context-free grammars.
- D. Parser makes k left-to-right passes over input string.
- E. Input string is scanned from left to right once with k symbol to the right as look-ahead to give left-most derivation.

tifr2012 compiler-design parsing lr-parser

Answer key

2.4 Lexical Analysis (1)

2.4.1 Lexical Analysis: TIFR CSE 2020 | Part B | Question: 9



A particular Panini-Backus-Naur Form definition for a $\langle \text{word} \rangle$ is given by the following rules:

- $\langle \text{word} \rangle ::= \langle \text{letter} \rangle | \langle \text{letter} \rangle \langle \text{pairlet} \rangle | \langle \text{letter} \rangle \langle \text{pairedig} \rangle$
- $\langle \text{pairlet} \rangle ::= \langle \text{letter} \rangle \langle \text{letter} \rangle | \langle \text{pairlet} \rangle \langle \text{letter} \rangle \langle \text{letter} \rangle$
- $\langle \text{pairedig} \rangle ::= \langle \text{digit} \rangle \langle \text{digit} \rangle | \langle \text{pairedig} \rangle \langle \text{digit} \rangle \langle \text{digit} \rangle$
- $\langle \text{letter} \rangle ::= a | b | c | \dots | y | z$
- $\langle \text{digit} \rangle ::= 0 | 1 | 2 | \dots | 9$

Which of the following lexical entities can be derived from $\langle \text{word} \rangle$?

- I. word
- II. words
- III. c22

- A. None of I, II or III
- B. I and II only
- C. I and III only
- D. II and III only
- E. I, II and III

tifr2020 compiler-design lexical-analysis

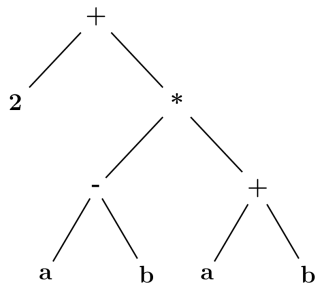
Answer key

2.5 Operator Precedence (1)

2.5.1 Operator Precedence: TIFR CSE 2012 | Part B | Question: 8



Consider the parse tree



Assume that $*$ has higher precedence than $+$, $-$ and operators associate right to left (i.e. $(a + b + c = (a + (b + c)))$). Consider

- i. $2 + a - b$
- ii. $2 + a - b * a + b$
- iii. $(2 + ((a - b) * (a + b)))$
- iv. $2 + (a - b) * (a + b)$

The parse tree corresponds to

- A. Expression (i)
- B. Expression (ii)
- C. Expression (iv) only
- D. Expression (ii), (iii), and (iv)
- E. Expression (iii) and (iv) only

tifr2012 compiler-design parsing operator-precedence

Answer key 

Answer Keys

2.1.1	B	2.2.1	C	2.3.1	B	2.4.1	D	2.5.1	E
-------	---	-------	---	-------	---	-------	---	-------	---



3.1.1 Network Security: TIFR CSE 2011 | Part B | Question: 36



Consider malware programs. Which of the following is true?

- A. A worm is a parasite.
- B. A virus cannot affect a linux operating system.
- C. A trojan can be in the payload of only a worm.
- D. A worm and virus are self replicating programs.
- E. There is no difference between a virus and a worm.

tifr2011 computer-networks network-security

Answer key 

Answer Keys

3.1.1	D
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4.1.1 Relational Algebra: TIFR CSE 2010 | Part B | Question: 33



In a relational database there are three relations:

- $Customers = C(CName)$
- $Shops = S(SName)$
- $Buys = B(CName, SName)$

Then the Relational Algebra expression (Π is the projection operator).

$$C - \Pi_{CName}((C \times S) - B)$$

returns the names of

- A. Customers who buy from at least one shop.

C. Customers who buy from all shops.

E. None of the above.

B. Customers who buy from at least two shops.

D. Customers who do not buy anything at all.

tifr2010 databases relational-algebra

Answer key

4.1.2 Relational Algebra: TIFR CSE 2013 | Part B | Question: 19



In a relational database there are three relations:

- $Customers = C(CName)$,
- $Shops = S(SName)$,
- $Buys = B(CName, SName)$.

Which of the following relational algebra expressions returns the names of shops that have no customers at all? [Here Π is the projection operator.]

- A. $\Pi_{SName} B$

C. $S - \Pi_{SName} B$

E. None of the above

B. $S - B$

D. $S - \Pi_{SName}((C \times S) - B)$

tifr2013 databases relational-algebra

Answer key

Answer Keys

4.1.1

C

4.1.2

C



5.1

Binomial Theorem (1)

5.1.1 Binomial Theorem: TIFR CSE 2021 | Part A | Question: 12



How many numbers in the range $0, 1, \dots, 1365$ have exactly four 1's in their binary representation? (Hint: 1365_{10} is 1010101010_2 , that is,

$$1365 = 2^{10} + 2^8 + 2^6 + 2^4 + 2^2 + 2^0.)$$

In the following, the binomial coefficient $\binom{n}{k}$ counts the number of k -element subsets of an n -element set.

- A. $\binom{6}{4}$
- B. $\binom{10}{4}$
- C. $\binom{10}{4} + \binom{8}{3} + \binom{6}{2} + \binom{5}{1}$
- D. $\binom{11}{4} + \binom{9}{3} + \binom{7}{2} + \binom{5}{1}$
- E. 1024

tifr2021 digital-logic number-representation binomial-theorem

Answer key

5.2

Boolean Algebra (5)

5.2.1 Boolean Algebra: TIFR CSE 2010 | Part B | Question: 21



For $x \in \{0, 1\}$, let $\neg x$ denote the negation of x , that is

$$\neg x = \begin{cases} 1 & \text{iff } x = 0 \\ 0 & \text{iff } x = 1 \end{cases}$$

.

If $x \in \{0, 1\}^n$, then $\neg x$ denotes the component wise negation of x ; that is:

$$(\neg x)_i = (\neg x_i \mid i \in [1..n])$$

Consider a circuit C , computing a function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ using **AND** ($\wedge$), **OR** ($\vee$), and **NOT** ($\neg$) gates. Let D be the circuit obtained from C by replacing each **AND** gate by an **OR** gate and replacing each **OR** gate by an **AND**. Suppose D computes the function g . Which of the following is true for all inputs x ?

- A. $g(x) = \neg f(x)$
- B. $g(x) = f(x) \wedge f(\neg x)$
- C. $g(x) = f(x) \vee f(\neg x)$
- D. $g(x) = \neg f(\neg x)$
- E. None of the above.

tifr2010 digital-logic boolean-algebra

Answer key

5.2.2 Boolean Algebra: TIFR CSE 2014 | Part B | Question: 17



Let $f : \{0, 1\}^n \rightarrow \{0, 1\}$ be a boolean function computed by a logical circuit comprising just binary AND and binary OR gates (assume that the circuit does not have any feedback). Let $\text{PARITY} : \{0, 1\}^n \rightarrow \{0, 1\}$ be the boolean function that outputs 1 if the total number of input bits set

to 1 is odd. Similarly, let MAJORITY be the boolean function that outputs 1 if the number of input bits that are set to 1 is at least as large as the number of input bits that are set to 0. Then, which of the following is NOT possible?

- A. $f(0, 0, \dots, 0) = f(1, 1, \dots, 1) = 0$.
- B. $f(0, 0, \dots, 0) = f(1, 1, \dots, 1) = 1$
- C. f is the MAJORITY function.
- D. f is the PARITY function.
- E. f outputs 1 at exactly one assignment of the input bits.

tifr2014 boolean-algebra

Answer key

5.2.3 Boolean Algebra: TIFR CSE 2017 | Part B | Question: 8



For any natural number n , an ordering of all binary strings of length n is a Gray code if it starts with 0^n , and any successive strings in the ordering differ in exactly one bit (the first and last string must also differ by one bit). Thus, for $n = 3$, the ordering $(000, 100, 101, 111, 110, 010, 011, 001)$ is a Gray code. Which of the following must be TRUE for all Gray codes over strings of length n ?

- A. the number of possible Gray codes is even
- B. the number of possible Gray codes is odd
- C. In any Gray code, if two strings are separated by k other strings in the ordering, then they must differ in exactly $k + 1$ bits
- D. In any Gray code, if two strings are separated by k other strings in the ordering, then they must differ in exactly k bits
- E. none of the above

tifr2017 digital-logic boolean-algebra

Answer key

5.2.4 Boolean Algebra: TIFR CSE 2020 | Part B | Question: 1



Consider the following Boolean valued function on n Boolean variables: $f(x_1, \dots, x_n) = x_1 + \dots + x_n \pmod{2}$, where addition is over integers, mapping 'FALSE' to 0 and 'TRUE' to 1. Consider Boolean circuits (with no feedback) that use only logical **AND** and **OR** gates, and where each gate has two input bits, each of which is either an input bit of f or the output bit of some other gate of the circuit. The circuit has a distinguished gate whose value is the output of the circuit. The minimum size of such a circuit computing f (asymptotically in n) is :

- A. $2^{o(\log n)}$
- B. n^c , for some fixed constant c
- C. $n^{\omega(1)}$, but $n^{O(\log n)}$
- D. $2^{\Theta(n)}$
- E. None of the others

tifr2020 digital-logic boolean-algebra time-complexity

5.2.5 Boolean Algebra: TIFR CSE 2020 | Part B | Question: 15



Suppose $X_{1a}, X_{1b}, X_{2a}, X_{2b}, \dots, X_{5a}, X_{5b}$ are ten Boolean variables each of which can take the value TRUE or FLASE. Recall the Boolean XOR $X \oplus Y := (X \wedge \neg Y) \vee (\neg X \wedge Y)$. Define the Boolean logic formulas

$$F := (X_{1a} \vee X_{1b}) \wedge (X_{2a} \vee X_{2b}) \wedge (X_{3a} \vee X_{3b}) \wedge (X_{4a} \vee X_{4b}) \wedge (X_{5a} \vee X_{5b}),$$

$$G_i := (X_{i,a} \oplus X_{i+1,a}) \vee (X_{i,b} \oplus X_{i+1,b}), \quad 1 \leq i \leq 4$$

$$G_5 := (X_{5a} \oplus X_{1a}) \vee (X_{5b} \oplus X_{1b}),$$

$$H := F \wedge G_1 \wedge G_2 \wedge G_3 \wedge G_4 \wedge G_5.$$

A truth assignment to the ten Boolean variables X_{ia}, X_{ib} , $1 \leq i \leq 5$ is said to be a satisfying assignment if H takes the value TRUE for example,

$$(X_{1a}, X_{1b}, X_{2a}, X_{2b}, \dots, X_{5a}, X_{5b}) = (F, T, T, F, F, T, T, T, T, F)$$

is a satisfying assignment,

$$(X_{1a}, X_{1b}, X_{2a}, X_{2b}, \dots, X_{5a}, X_{5b}) = (F, T, T, T, F, T, T, T, T, F)$$

is another satisfying assignment, while

$$(X_{1a}, X_{1b}, X_{2a}, X_{2b}, \dots, X_{5a}, X_{5b}) = (F, T, T, F, F, T, T, F, T, F)$$

is not a satisfying assignment.

How many satisfying assignments does H have?

- A. 20 B. 30 C. 32 D. 160
E. 1024

tifr2020 digital-logic boolean-algebra

Answer key 

5.3

Canonical Normal Form (1)

5.3.1 Canonical Normal Form: TIFR CSE 2015 | Part B | Question: 9



A Boolean expression is an expression made out of propositional letters (such as p, q, r) and operators $\wedge, \vee$ and $\neg$; e.g. $p \wedge \neg(q \vee \neg r)$. An expression is said to be in sum of product form (also called disjunctive normal form) if all $\neg$ occur just before letters and no $\vee$ occurs in scope of $\wedge$; e.g. $(p \wedge \neg q) \vee (\neg p \wedge q)$. The expression is said to be in product of sum form (also called conjunctive normal form) if all negations occur just before letters and no $\wedge$ occurs in the scope of $\vee$; e.g. $(p \vee \neg q) \wedge (\neg p \vee q)$. Which of the following is not correct?

- A. Every Boolean expression is equivalent to an expression in sum of product form.
B. Every Boolean expression is equivalent to an expression in product of sum form.
C. Every Boolean expression is equivalent to an expression without $\vee$ operator.
D. Every Boolean expression is equivalent to an expression without $\wedge$ operator.
E. Every Boolean expression is equivalent to an expression without $\neg$ operator.

tifr2015 canonical-normal-form

Answer key 

5.4

Combinational Circuit (1)

5.4.1 Combinational Circuit: TIFR CSE 2015 | Part A | Question: 4



The Boolean function obtained by adding an inverter to each and every input of an AND gate is:

- A. OR B. XOR C. NAND D. NOR
E. None of the above

Answer key

5.5 Number Representation (5)

5.5.1 Number Representation: TIFR CSE 2010 | Part A | Question: 9



A table contains 287 entries. When any one of the entries is requested, it is encoded into a binary string and transmitted. The number of bits required is.

- A. 8
- B. 9
- C. 10
- D. Cannot be determined from the given information.
- E. None of the above.

tifr2010 digital-logic number-representation

Answer key

5.5.2 Number Representation: TIFR CSE 2011 | Part A | Question: 16



A variable that takes thirteen possible values can be communicated using?

- A. Thirteen bits.
- B. Three bits.
- C. $\log_2 13$ bits.
- D. Four bits.
- E. None of the above.

tifr2011 number-representation

Answer key

5.5.3 Number Representation: TIFR CSE 2012 | Part A | Question: 11



Let N be the sum of all numbers from 1 to 1023 except the five primes numbers: 2, 3, 11, 17, 31. Suppose all numbers are represented using two bytes (sixteen bits). What is the value of the least significant byte (the least significant eight bits) of N ?

- A. 00000000
- B. 10101110
- C. 01000000
- D. 10000000
- E. 11000000

tifr2012 digital-logic number-representation

Answer key

5.5.4 Number Representation: TIFR CSE 2019 | Part B | Question: 1



Which of the following decimal numbers can be exactly represented in binary notation with a finite number of bits ?

- A. 0.1
- B. 0.2
- C. 0.4
- D. 0.5
- E. All the above

tifr2019 digital-logic number-representation

Answer key

5.5.5 Number Representation: TIFR CSE 2020 | Part B | Question: 3



Consider the (decimal) number 182, whose binary representation is 10110110. How many positive integers are there in the following set?

$$\{n \in \mathbb{N} : n \leq 182 \text{ and } n \text{ has exactly four ones in its binary representation}\}$$

A. 91

B. 70

C. 54

D. 35

E. 27

tifr2020 digital-logic number-representation

Answer key 

Answer Keys

5.1.1	C
5.2.5	B
5.5.3	E

5.2.1	D
5.3.1	E
5.5.4	D

5.2.2	D
5.4.1	D
5.5.5	C

5.2.3	A
5.5.1	D

5.2.4	E
5.5.2	D



6.1

Balls In Bins (4)

6.1.1 Balls In Bins: TIFR CSE 2012 | Part A | Question: 7



It is required to divide the $2n$ members of a club into n disjoint teams of 2 members each. The teams are not labelled. The number of ways in which this can be done is:

- A. $\frac{(2n)!}{2^n}$ B. $\frac{(2n)!}{n!}$ C. $\frac{(2n)!}{2^n \cdot n!}$ D. $\frac{n!}{2}$
E. None of the above

tifr2012 combinatory balls-in-bins

Answer key

6.1.2 Balls In Bins: TIFR CSE 2013 | Part A | Question: 9



There are n kingdoms and $2n$ champions. Each kingdom gets 2 champions. The number of ways in which this can be done is:

- A. $\frac{(2n)!}{2^n}$ B. $\frac{(2n)!}{n!}$ C. $\frac{(2n)!}{2^n \cdot n!}$ D. $\frac{n!}{2}$
E. None of the above

tifr2013 combinatory discrete-mathematics normal balls-in-bins

Answer key

6.1.3 Balls In Bins: TIFR CSE 2015 | Part A | Question: 8



There is a set of $2n$ people: n male and n female. A good party is one with equal number of males and females (including the one where none are invited). The total number of good parties is.

- A. 2^n B. n^2 C. $\binom{n}{\lfloor n/2 \rfloor}^2$ D. $\binom{2n}{n}$
E. None of the above

tifr2015 combinatory discrete-mathematics normal balls-in-bins

Answer key

6.1.4 Balls In Bins: TIFR CSE 2017 | Part A | Question: 5



How many distinct ways are there to split 50 identical coins among three people so that each person gets at least 5 coins?

- A. 3^{35} B. $3^{50} - 2^{50}$
C. $\binom{35}{2}$ D. $\binom{50}{15} \cdot 3^{35}$
E. $\binom{37}{2}$

tifr2017 combinatory discrete-mathematics normal balls-in-bins

Answer key

6.2

Binomial Theorem (2)

6.2.1 Binomial Theorem: TIFR CSE 2016 | Part A | Question: 13



Let $n \geq 2$ be any integer. Which of the following statements is not necessarily true?

- A. $\binom{n}{i} = \binom{n-1}{i} + \binom{n-1}{i-1}$, where $1 \leq i \leq n-1$
- B. $n!$ divides the product of any n consecutive integers
- C. $\sum_{i=0}^n \binom{n}{i} = 2^n$
- D. n divides $\binom{n}{i}$, for all $i \in \{1, 2, \dots, n-1\}$
- E. If n is an odd prime, then n divides $2^{n-1} - 1$

tifr2016 combinatory binomial-theorem

Answer key

6.2.2 Binomial Theorem: TIFR CSE 2022 | Part B | Question: 9



Let $n \geq 2$ be any integer. Which of the following statements is FALSE?

- A. $n!$ divides the product of any n consecutive integers
- B. $\sum_{i=0}^n \binom{n}{i} = 2^n$
- C. $\binom{n}{i} = \binom{n-1}{i} + \binom{n-1}{i-1}$, where $1 \leq i \leq n-1$
- D. If n is an odd prime, then n divides $2^{n-1} - 1$
- E. n divides $\binom{2n}{n}$

tifr2022 combinatory binomial-theorem

Answer key

6.3 Counting (13)

6.3.1 Counting: TIFR CSE 2011 | Part A | Question: 2



In how many ways can the letters of the word ABACUS be rearranged such that the vowels always appear together?

- A. $\frac{(6+3)!}{2!}$
- B. $\frac{6!}{2!}$
- C. $\frac{3!3!}{2!}$
- D. $\frac{4!3!}{2!}$
- E. None of the above

tifr2011 combinatory counting

Answer key

6.3.2 Counting: TIFR CSE 2012 | Part A | Question: 10



In how many different ways can r elements be picked from a set of n elements if

- Repetition is not allowed and the order of picking matters?
- Repetition is allowed and the order of picking does not matter?

- A. $\frac{n!}{(n-r)!}$ and $\frac{(n+r-1)!}{r!(n-1)!}$, respectively.
- B. $\frac{n!}{(n-r)!}$ and $\frac{n!}{r!(n-1)!}$, respectively.

- C. $\frac{n!}{r!(n-r)!}$ and $\frac{(n-r+1)!}{r!(n-1)!}$, respectively.
 E. $\frac{n!}{r!}$ and $\frac{r!}{n!}$, respectively.

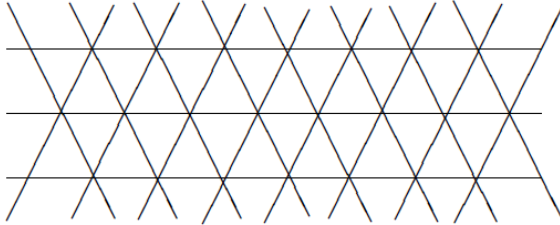
tifr2012 combinatory counting normal

Answer key 

6.3.3 Counting: TIFR CSE 2013 | Part A | Question: 1



An infinite two-dimensional pattern is indicated below.



The smallest closed figure made by the lines is called a unit triangle. Within every unit triangle, there is a mouse.

At every vertex there is a laddoo. What is the average number of laddoos per mouse?

- A. 3 B. 2 C. 1 D. $\left(\frac{1}{2}\right)$
 E. $\left(\frac{1}{3}\right)$

tifr2013 combinatory counting

Answer key 

6.3.4 Counting: TIFR CSE 2015 | Part A | Question: 7



A 1×1 chessboard has one square, a 2×2 chessboard has five squares. Continuing along this fashion, what is the number of squares on the regular 8×8 chessboard?

- A. 64 B. 65 C. 204 D. 144
 E. 256

tifr2015 combinatory counting

Answer key 

6.3.5 Counting: TIFR CSE 2016 | Part A | Question: 7



Let S be the 4×4 square grid $\{(x, y) : x, y \in \{0, 1, 2, 3\}\}$. A *monotone path* in this grid starts at $(0, 0)$ and at each step either moves one unit up or one unit right. For example, from the point (x, y) one can in one step either move to $(x + 1, y) \in S$ or $(x, y + 1) \in S$, but never leave S . Let the number of distinct monotone paths to reach point $(2, 2)$ starting from $(0, 0)$ be z . How many distinct monotone paths are there to reach point $(3, 3)$ starting from $(0, 0)$?

- A. $2z + 6$ B. $3z + 6$ C. $2z + 8$ D. $3z + 8$
 E. $3z + 4$

tifr2016 combinatory counting

Answer key 

6.3.6 Counting: TIFR CSE 2017 | Part A | Question: 6



How many distinct words can be formed by permuting the letters of the word ABRACADABRA?

- A. $\frac{11!}{5! 2! 2!}$ B. $\frac{11!}{5! 4!}$ C. $11! 5! 2! 2!$ D. $11! 5! 4!$
E. $11!$

tifr2017 combinatory counting easy

Answer key

6.3.7 Counting: TIFR CSE 2019 | Part B | Question: 13



A row of 10 houses has to be painted using the colours red, blue, and green so that each house is a single colour, and any house that is immediately to the right of a red or a blue house must be green. How many ways are there to paint the houses?

- A. 199 B. 683 C. 1365 D. $3^{10} - 2^{10}$
E. 3^{10}

tifr2019 combinatory counting

Answer key

6.3.8 Counting: TIFR CSE 2021 | Part A | Question: 14



Five married couples attended a party. In the party, each person shook hands with those they did not know. Everyone knows his or her spouse. At the end of the party, Shyamal, one of the attendees, listed the number of hands that other attendees including his spouse shook. He got every number from 0 to 8 once in the list. How many persons shook hands with Shyamal at the party?

- A. 2 B. 4
C. 6 D. 8
E. Insufficient information

tifr2021 combinatory counting

Answer key

6.3.9 Counting: TIFR CSE 2022 | Part A | Question: 1



A snail crawls up a vertical pole 75 feet high, starting from the ground. Each day it crawls up 5 feet, and each night it slides down 4 feet. When will it first reach the top of the pole?

- A. 75th day B. 74th day C. 73rd day D. 72nd day
E. 71st day

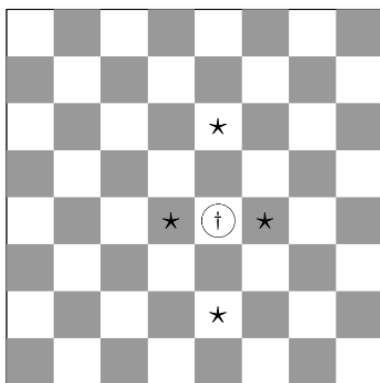
tifr2022 combinatory counting

Answer key

6.3.10 Counting: TIFR CSE 2023 | Part A | Question: 12



Suppose $\oplus$ is a piece on a chess board that attacks squares that are exactly two steps in the vertical direction, and squares that are adjacent horizontally (as marked with a "*" figure in the image below)



What is the maximum number of these pieces that can be placed on the squares of a standard 8×8 chess board such that no two of these pieces attack each other? (Note that each square is allowed to contain at most one piece.)

- A. 4 B. 8 C. 16 D. 24
- E. 32

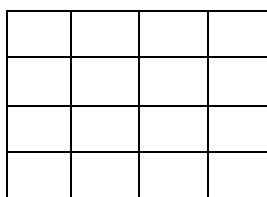
tifr2023 combinatory counting

Answer key

6.3.11 Counting: TIFR CSE 2023 | Part A | Question: 2



How many distinct rectangles can be formed using the vertices in the grid shown below? Squares are also counted as rectangles, and two rectangles are distinct if either their top-left vertices are different or their bottom-right vertices are different.



- A. 16 B. 25 C. 36 D. 64
- E. 100

tifr2023 combinatory counting

Answer key

6.3.12 Counting: TIFR CSE 2024 | Part A | Question: 8



A palindrome is a string that reads the same in reverse (e.g. ABBA or KAYAK or MALAYALAM). How many strings of length 5 using the letters from $\{A, B, C, D, E\}$ have no palindromic substring of length at least 2?

- A. 243 B. 405 C. 540 D. 675
- E. 1280

tifr2024 combinatory counting

Answer key

6.3.13 Counting: TIFR CSE 2025 | Part A | Question: 11



How many distinct ways are there to paint the 6 faces of a cube using six colors (1, 2, 3, 4, 5, 6), with each face painted a different color, such that no two painted cubes are identical (i.e., cannot be transformed into each other by any rotation)?

- A. 30 B. 60 C. 120 D. 360
E. 720

tifr2025 combinatory counting

Answer key

6.4

Generating Functions (2)

6.4.1 Generating Functions: TIFR CSE 2010 | Part A | Question: 12



The coefficient of x^3 in the expansion of $(1+x)^3(2+x^2)^{10}$ is.

- A. 2^{14} B. 31
C. $\left(\frac{3}{3}\right) + \left(\frac{10}{1}\right)$ D. $\left(\frac{3}{3}\right) + 2\left(\frac{10}{1}\right)$
E. $\left(\frac{3}{3}\right)\left(\frac{10}{1}\right)2^9$

tifr2010 generating-functions

Answer key

6.4.2 Generating Functions: TIFR CSE 2021 | Part A | Question: 11



Find the following sum.

$$\frac{1}{2^2-1} + \frac{1}{4^2-1} + \frac{1}{6^2-1} + \dots + \frac{1}{40^2-1}$$

- A. $\frac{20}{41}$
B. $\frac{10}{41}$
C. $\frac{10}{21}$
D. $\frac{20}{21}$
E. 1

tifr2021 combinatory generating-functions

Answer key

6.5

Modular Arithmetic (1)

6.5.1 Modular Arithmetic: TIFR CSE 2014 | Part A | Question: 14



Let m and n be any two positive integers. Then, which of the following is FALSE?

- A. $m+1$ divides $m^{2n}-1$.
B. For any prime p , $m^p \equiv m \pmod{p}$.
C. If one of m, n is prime, then there are integers x, y such that $mx + ny = 1$.
D. If $m < n$, then $m!$ divides $n(n-1)(n-2)\dots(n-m+1)$.
E. If $2^n - 1$ is prime, then n is prime.

Answer key

6.6

Parity (1)

6.6.1 Parity: TIFR CSE 2024 | Part A | Question: 4



Let $z_1, z_2, z_3, \dots, z_{2023}$ be a permutation of the numbers $1, 2, 3, \dots, 2023$. Which of the following is true about the product $\prod_{i=1}^{2023} (z_i - i)$?

Note: The parity of an integer n just denotes whether n is even or odd. Formally, the parity of n is said to be odd if n is odd, and even if n is even.

- A. The above product is always even.
- B. The above product is always odd.
- C. The parity of the above product always changes if we swap the values of any two variables among $z_1, z_2, \dots, z_{2023}$.
- D. There always exist two variables among $z_1, z_2, \dots, z_{2023}$ such that the parity of the above product changes if we swap their values, but there may also exist two variables among the $z_1, z_2, \dots, z_{2023}$ such that swapping their values does not change the parity of the above product.
- E. None of the above statements is true.

Answer key

6.7

Pigeonhole Principle (3)

6.7.1 Pigeonhole Principle: TIFR CSE 2014 | Part A | Question: 5



The rules for the University of Bombay five-a-side cricket competition specify that the members of each team must have birthdays in the same month. What is the minimum number of mathematics students needed to be enrolled in the department to guarantee that they can raise a team of students?

- A. 23
- B. 91
- C. 60
- D. 49
- E. None of the above

Answer key

6.7.2 Pigeonhole Principle: TIFR CSE 2016 | Part A | Question: 15



In a tournament with 7 teams, each team plays one match with every other team. For each match, the team earns two points if it wins, one point if it ties, and no points if it loses. At the end of all matches, the teams are ordered in the descending order of their total points (the order among the teams with the same total are determined by a whimsical tournament referee). The first three teams in this ordering are then chosen to play in the next round. What is the minimum total number of points a team must earn in order to be guaranteed a place in the next round?

- A. 13
- B. 12
- C. 11
- D. 10
- E. 9

Answer key

6.7.3 Pigeonhole Principle: TIFR CSE 2018 | Part A | Question: 6



What is the minimum number of students needed in a class to guarantee that there are at least 6 students whose birthdays fall in the same month ?

- A. 6 B. 23 C. 61 D. 72
E. 91

tifr2018 pigeonhole-principle combinatory

Answer key

6.8 Qod Cse 18 (1)

6.8.1 Qod Cse 18: TIFR CSE 2025 | Part A | Question: 5



There are three boxes. One box contains ten red socks, another contains ten blue socks, and the third contains a mix five red and five blue socks. The three boxes are labelled R (for red), B (for blue), or M (for mixed), but the labels are permuted so that for EACH drawer the label attached to it does NOT match its contents.

How many total socks do you have to look at from the drawers, in order to correctly identify each drawer? (You are not allowed to look inside the drawers. You can only pick socks one at a time and observe its colour.)

- A. One sock. B. Two socks. C. Three socks. D. Four socks.
E. Six socks.

tifr2025 combinatory qod-cse qod-cse-18

Answer key

6.9 Qod Cse 2 (1)

6.9.1 Qod Cse 2: TIFR CSE 2021 | Part A | Question: 1



A box contains 5 red marbles, 8 green marbles, 11 blue marbles, and 15 yellow marbles. We draw marbles uniformly at random without replacement from the box. What is the minimum number of marbles to be drawn to ensure that out of the marbles drawn, at least 7 are of the same colour?

- A. 7 B. 8 C. 23 D. 24
E. 39

tifr2021 combinatory pigeonhole-principle qod-cse-2

Answer key

6.10 Recurrence Relation (2)

6.10.1 Recurrence Relation: TIFR CSE 2014 | Part A | Question: 3



The Fibonacci sequence is defined as follows: $F_0 = 0, F_1 = 1$, and for all integers $n \geq 2, F_n = F_{n-1} + F_{n-2}$. Then which of the following statements is FALSE?

- A. $F_{n+2} = 1 + \sum_{i=0}^n F_i$ for any integer $n \geq 0$

- B. $F_{n+2} \geq \phi^n$ for any integer $n \geq 0$, where $\phi = (\sqrt{5} + 1)/2$ is the positive root of $x^2 - x - 1 = 0$.
 C. F_{3n} is even, for every integer $n \geq 0$.
 D. F_{4n} is a multiple of 3, for every integer $n \geq 0$.
 E. F_{5n} is a multiple of 4, for every integer $n \geq 0$.

tifr2014 recurrence-relation easy

Answer key 

6.10.2 Recurrence Relation: TIFR CSE 2017 | Part A | Question: 7



Consider the sequence $S_0, S_1, S_2, \dots$ defined as follows: $S_0 = 0, S_1 = 1$ and $S_n = 2S_{n-1} + S_{n-2}$ for $n \geq 2$. Which of the following statements is FALSE?

- A. for every $n \geq 1$, S_{2n} is even
 B. for every $n \geq 1$, S_{2n+1} is odd
 C. for every $n \geq 1$, S_{3n} is multiple of 3
 D. for every $n \geq 1$, S_{4n} is multiple of 6
 E. none of the above

tifr2017 recurrence-relation

Answer key 

Answer Keys

6.1.1	C	6.1.2	A	6.1.3	D	6.1.4	E	6.2.1	D
6.2.2	E	6.3.1	D	6.3.2	A	6.3.3	D	6.3.4	C
6.3.5	C	6.3.6	A	6.3.7	C	6.3.8	B	6.3.9	E
6.3.10	E	6.3.11	E	6.3.12	C	6.3.13	A	6.4.1	A
6.4.2	A	6.5.1	C	6.6.1	A	6.7.1	D	6.7.2	D
6.7.3	C	6.8.1	A	6.9.1	D	6.10.1	E	6.10.2	C



7.0.1 TIFR CSE 2024 | Part B | Question: 9



Given m vectors $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_m$ in $\mathbb{R}^d$, we construct an undirected graph $G = (V, E)$ as follows. Each vector $\vec{x}_i$ is represented by a vertex v_i . We add an edge between vertices v_i and v_j if the corresponding pair of vectors $\vec{x}_i$ and $\vec{x}_j$ are linearly independent.

Which of the following statements regarding the graph G must be TRUE, for any such vectors $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_m$? Recall that a clique in a graph is a set of vertices $S \subseteq V$ that have an edge between every pair of vertices in S .

- A. If $m > d$, the graph must be disconnected.
- B. Any clique has size at most d
- C. Any clique has size at most $m/2$
- D. The maximum degree of any vertex in G is at most d
- E. None of the above.

tifr2024 graph-theory linear-algebra

Answer key

7.1

Bipartite Graph (2)

7.1.1 Bipartite Graph: TIFR CSE 2017 | Part B | Question: 13



For an undirected graph $G = (V, E)$, the line graph $G' = (V', E')$ is obtained by replacing each edge in E by a vertex, and adding an edge between two vertices in V' if the corresponding edges in G are incident on the same vertex. Which of the following is TRUE of line graphs?

- A. the line graph for a complete graph is complete
- B. the line graph for a connected graph is connected
- C. the line graph for a bipartite graph is bipartite
- D. the maximum degree of any vertex in the line graph is at most the maximum degree in the original graph
- E. each vertex in the line graph has degree one or two

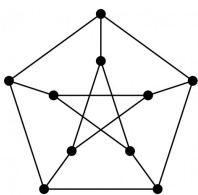
tifr2017 graph-theory bipartite-graph

Answer key

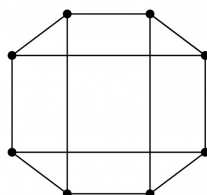
7.1.2 Bipartite Graph: TIFR CSE 2020 | Part B | Question: 11



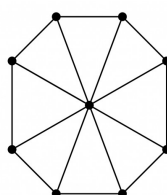
Which of the following graphs are bipartite?



(1)



(2)



(3)

- A. Only (1)
- B. Only (2)
- C. Only (2) and (3)
- D. None of (1), (2), (3)
- E. All of (1), (2), (3)

Answer key

7.2

Combinatory (1)

7.2.1 Combinatory: TIFR CSE 2024 | Part B | Question: 14



For an undirected graph G , let $\bar{G}$ refer to the complement (a graph on the same vertex set as G , with (i, j) as an edge in $\bar{G}$ if and only if it is not an edge in G). Consider the following statements.

- G has a vertex-cover of size at most k .
- $\bar{G}$ has an independent set of size at least k .
- G has an independent set of size at least $n - k$.
- G has a clique of size at least k .
- $\bar{G}$ has a clique of size at least $n - k$.

Which of the following is true for any graph G and its complement graph $\bar{G}$?

- (i) is equivalent to (iii) and (iv).
- (i) is equivalent to (iii) and (v).
- (i) is equivalent to (ii) and (iv).
- (i) is equivalent to (ii) and (v).
- None of the five statements are equivalent to each other.

Answer key

7.3

Degree of Graph (5)

7.3.1 Degree of Graph: TIFR CSE 2010 | Part B | Question: 36



In a directed graph, every vertex has exactly seven edges coming in. What can one always say about the number of edges going out of its vertices?

- Exactly seven edges leave every vertex.
- Exactly seven edges leave some vertex.
- Some vertex has at least seven edges leaving it.
- The number of edges coming out of vertex is odd.
- None of the above.

Answer key

7.3.2 Degree of Graph: TIFR CSE 2012 | Part B | Question: 2



In a graph, the degree of a vertex is the number of edges incident (connected) on it. Which of the following is true for every graph G ?

- There are even number of vertices of even degree.
- There are odd number of vertices of even degree.
- There are even number of vertices of odd degree.
- There are odd number of vertices of odd degree.
- All the vertices are of even degree.

Answer key 

7.3.3 Degree of Graph: TIFR CSE 2018 | Part B | Question: 8



In an undirected graph G with n vertices, vertex 1 has degree 1, while each vertex $2, \dots, n-1$ has degree 10 and the degree of vertex n is unknown, Which of the following statement must be TRUE on the graph G ?

- A. There is a path from vertex 1 to vertex n .
- B. There is a path from vertex 1 to each vertex $2, \dots, n-1$.
- C. Vertex n has degree 1.
- D. The diameter of the graph is at most $\frac{n}{10}$
- E. All of the above choices must be TRUE

Answer key 

7.3.4 Degree of Graph: TIFR CSE 2019 | Part B | Question: 3



A graph is d -regular if every vertex has degree d . For a d -regular graph on n vertices, which of the following must be TRUE?

- A. d divides n
- B. Both d and n are even
- C. Both d and n are odd
- D. At least one of d and n is odd
- E. At least one of d and n is even

Answer key 

7.3.5 Degree of Graph: TIFR CSE 2023 | Part B | Question: 10



A d -regular graph is one in which every vertex has degree d . Also, a *minimum cut* in a graph is a smallest set of edges which, upon removal, disconnects the graph, so that there are vertices in the resulting graph with no path between them. We are given two graphs. $G_1 = (V_1, E_1)$ is a *connected*, 3-regular graph. $G_2 = (V_2, E_2)$ is a *connected*, 4-regular graph. The vertex sets V_1 and V_2 are disjoint. We are further told that the size of any minimum cut in G_1 is the *same* as the size of any minimum cut in G_2 . Which of the following must be the size of this minimum cut?

- A. 0
- B. 1
- C. 2
- D. 3
- E. 4

Answer key 

7.4 Euler Graph (3)

7.4.1 Euler Graph: TIFR CSE 2016 | Part B | Question: 9



Which of the following graphs DOES NOT have an Eulerian circuit? (Recall that an Eulerian circuit in an undirected graph is a walk in the graph that starts at a vertex and returns to the vertex after traveling on each edge exactly once.)

- A. $K_{9,9}$
- B. $K_{8,8}$

- C. $K_{12,12}$
D. K_9
E. The graph G on vertex set $\{1, 2, \dots, 9\}$ with edge set

$$E(G) = \{\{i, j\} : 1 \leq i < j \leq 5 \text{ or } 5 \leq i < j \leq 9\}.$$

tifr2016 discrete-mathematics graph-theory euler-graph normal

Answer key

7.4.2 Euler Graph: TIFR CSE 2020 | Part B | Question: 13



Let G be an undirected graph. An Eulerian cycle of G is a cycle that traverses each edge of G exactly once. A Hamiltonian cycle of G is a cycle that traverses each vertex of G exactly once. Which of the following must be true?

- A. Checking if G has a Eulerian cycle can be done in polynomial time
B. Deciding if G has a Hamiltonian cycle is not NP-complete
C. If G has an Eulerian cycle, then it has a Hamiltonian cycle
D. A complete graph always has both an Eulerian cycle and a Hamiltonian cycle
E. All of the other statements are true

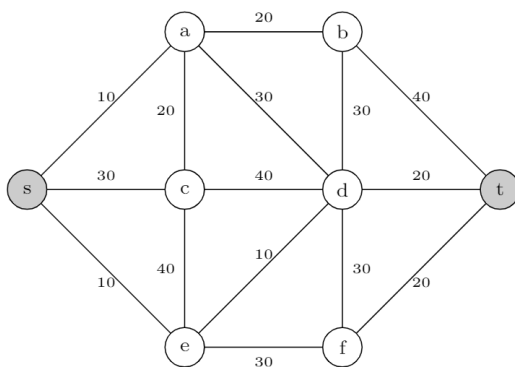
tifr2020 graph-theory euler-graph

Answer key

7.4.3 Euler Graph: TIFR CSE 2020 | Part B | Question: 14



The figure below describes the network of streets in a city where Motabhai sells pakoras from his cart. The number next to an edge is the time (in minutes) taken to traverse the corresponding street.



At present, the cart is required to start at point s and, after visiting each street at least once, reach point t . For example, Motabhai can visit the streets in the following order

$$s - a - c - s - e - c - d - a - b - d - f - e - d - b - t - f - d - t$$

in order to go from s to t . Note that the streets $\{b, d\}$ and $\{d, f\}$ are both visited twice in this strategy. The total time taken for this trip is 440 minutes [which is, 380 (the sum of traversal times of all streets in the network) + 60 (the sum of the traversal times of streets $\{b, d\}$ and $\{d, f\}$)].

Motabhai now wants the cart to return to s at the end of the trip. So the previous strategy is not valid, and he must find a new strategy. How many minutes will Motabhai now take if he uses an optimal strategy?

Hint: s, t, b and f are the only odd degree nodes in the figure above.

- A. 430 B. 440 C. 460 D. 470
E. 480

tifr2020 graph-theory euler-graph

Answer key 

7.5

Graph Coloring (9)

7.5.1 Graph Coloring: TIFR CSE 2013 | Part B | Question: 1



Let $G = (V, E)$ be a simple undirected graph on n vertices. A colouring of G is an assignment of colours to each vertex such that endpoints of every edge are given different colours. Let $\chi(G)$ denote the chromatic number of G , i.e. the minimum numbers of colours needed for a valid colouring of G . A set $B \subseteq V$ is an independent set if no pair of vertices in B is connected by an edge. Let $\alpha(G)$ be the number of vertices in a largest possible independent set in G . In the absence of any further information about G we can conclude.

- A. $\chi(G) \geq \alpha(G)$ B. $\chi(G) \leq \alpha(G)$
C. $\alpha(G) \geq \frac{n}{\chi(G)}$ D. $\alpha(G) \leq \frac{n}{\chi(G)}$
E. None of the above

tifr2013 graph-theory graph-coloring

Answer key 

7.5.2 Graph Coloring: TIFR CSE 2016 | Part B | Question: 13



An undirected graph $G = (V, E)$ is said to be k -colourable if there exists a mapping $c : V \rightarrow \{1, 2, \dots, k\}$ such that for every edge $\{u, v\} \in E$ we have $c(u) \neq c(v)$. Which of the following statements is FALSE?

- A. G is $|V|$ -colourable
B. G is 2-colourable if there are no odd cycles in G
C. G is $(\Delta + 1)$ -colourable where Δ is the maximum degree in G
D. There is a polynomial time algorithm to check if G is 2-colourable
E. If G has no triangle then it is 3-colourable

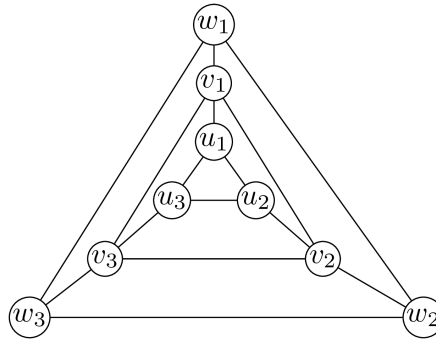
tifr2016 graph-theory graph-coloring

Answer key 

7.5.3 Graph Coloring: TIFR CSE 2017 | Part B | Question: 1



A vertex colouring with three colours of a graph $G = (V, E)$ is a mapping $c : V \rightarrow \{R, G, B\}$ so that adjacent vertices receive distinct colours. Consider the following undirected graph.



How many vertex colouring with three colours does this graph have?

- A. 3^9 B. 6^3 C. 3×2^8 D. 27
E. 24

tifr2017 graph-theory graph-coloring

Answer key

7.5.4 Graph Coloring: TIFR CSE 2017 | Part B | Question: 10



A vertex colouring of a graph $G = (V, E)$ with k colours is a mapping $c : V \rightarrow \{1, \dots, k\}$ such that $c(u) \neq c(v)$ for every $(u, v) \in E$. Consider the following statements:

- If every vertex in G has degree at most d then G admits a vertex colouring using $d + 1$ colours.
- Every cycle admits a vertex colouring using 2 colours
- Every tree admits a vertex colouring using 2 colours

Which of the above statements is/are TRUE? Choose from the following options:

- A. only i B. only i and ii C. only i and iii D. only ii and iii
E. i, ii, and iii

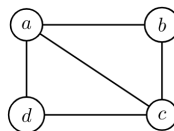
tifr2017 graph-theory graph-coloring

Answer key

7.5.5 Graph Coloring: TIFR CSE 2018 | Part A | Question: 9



How many ways are there to assign colours from range $\{1, 2, \dots, r\}$ to vertices of the following graph so that adjacent vertices receive distinct colours?



- A. r^4 B. $r^4 - 4r^3$
C. $r^4 - 5r^3 + 8r^2 - 4r$ D. $r^4 - 4r^3 + 9r^2 - 3r$
E. $r^4 - 5r^3 + 10r^2 - 15r$

tifr2018 graph-theory graph-coloring

Answer key

7.5.6 Graph Coloring: TIFR CSE 2021 | Part B | Question: 13



Let A be a 3×6 matrix with real-valued entries. Matrix A has rank 3. We construct a graph with

6 vertices where each vertex represents distinct column in A , and there is an edge between two vertices if the two columns represented by the vertices are linearly independent. Which of the following statements **MUST** be true of the graph constructed?

- A. Each vertex has degree at most 2.
- B. The graph is connected.
- C. There is a clique of size 3.
- D. The graph has a cycle of length 4.
- E. The graph is 3-colourable.

tifr2021 graph-theory graph-coloring matrix

Answer key 

7.5.7 Graph Coloring: TIFR CSE 2021 | Part B | Question: 14



Consider the following greedy algorithm for colouring an n -vertex undirected graph G with colours $c_1, c_2, \dots$: consider the vertices of G in any sequence and assign the chosen vertex the first colour that has not already been assigned to any of its neighbours. Let $m(n, r)$ be the minimum number of edges in a graph that causes this greedy algorithm to use r colours. Which of the following is correct?

- A. $m(n, r) = \Theta(r)$
- B. $m(n, r) = \Theta(r \lceil \log_2 r \rceil)$
- C. $m(n, r) = \binom{r}{2}$
- D. $m(n, r) = nr$
- E. $m(n, r) = n \binom{r}{2}$

tifr2021 graph-theory graph-coloring

Answer key 

7.5.8 Graph Coloring: TIFR CSE 2023 | Part B | Question: 13



You have a regular tetrahedron and 4 distinct colours. You wish to paint the faces of the tetrahedron such that each face gets a different colour. How many ways can you colour the tetrahedron? Recall that a regular tetrahedron is a three-dimensional solid with exactly four faces, all of which are equilateral triangles: see the sketch given below (not drawn to scale) for a rough view of such a solid. Two colourings of the tetrahedron are considered the same if they are identical after possibly rotating the tetrahedron.



- A. 24
- B. 12
- C. 8
- D. 6
- E. 2

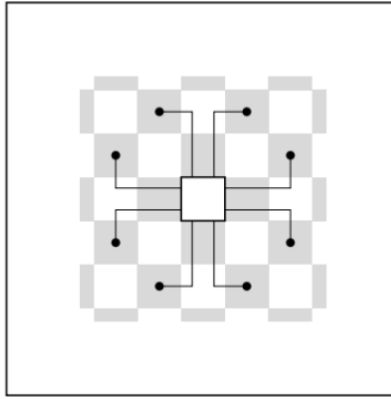
tifr2023 graph-theory graph-coloring

Answer key 

7.5.9 Graph Coloring: TIFR CSE 2025 | Part A | Question: 9



What is the largest number of knights that can be placed on a 60×60 chessboard such that no two knights attack each other (two knights attack each other if they are two steps away vertically and one step away horizontally, or vice versa).



The figure above shows a part of a chessboard showing all those squares that a knight placed in the central white square attacks.

- A. 1200 B. 1500 C. 1800 D. 2100
E. 2400

tifr2025 graph-theory graph-coloring combinatory

Answer key 

7.6 Graph Connectivity (11)

7.6.1 Graph Connectivity: TIFR CSE 2011 | Part B | Question: 33



Which of the following is NOT a sufficient and necessary condition for an undirected graph G to be a tree?

- A. G is connected and has $n - 1$ edges.
B. G is acyclic and connected.
C. G is acyclic and has $n - 1$ edges.
D. G is acyclic, connected and has $n - 1$ edges.
E. G has $n - 1$ edges.

tifr2011 graph-theory graph-connectivity

Answer key 

7.6.2 Graph Connectivity: TIFR CSE 2015 | Part B | Question: 11



Let K_n be the complete graph on n vertices labeled $\{1, 2, \dots, n\}$ with $m = \frac{n(n-1)}{2}$ edges. What is the number of spanning trees of K_n ?

- A. $\frac{m}{n-1}$ B. m^{n-1} C. n^{n-2} D. n^{n-1}
E. None of the above

tifr2015 graph-theory graph-connectivity

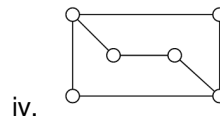
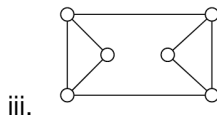
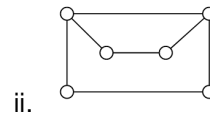
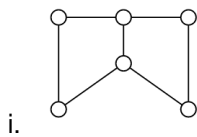
Answer key 

7.6.3 Graph Connectivity: TIFR CSE 2015 | Part B | Question: 5



Suppose
$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

is the adjacency matrix of an undirected graph with six vertices: that is, the rows and columns are indexed by vertices of the graph, and an entry is 1 if the corresponding vertices are connected by an edge and is 0 otherwise; the same order of vertices is used for the rows and columns. Which of the graphs below has the above adjacency matrix?



- A. Only (i)
B. Only (ii)
C. Only (iii)
D. Only (iv)
E. (i) and (ii)

tifr2015 graph-connectivity graph-theory

Answer key

7.6.4 Graph Connectivity: TIFR CSE 2016 | Part A | Question: 14



A *diagonal* in a polygon is a straight line segment that connects two non-adjacent vertices, and is contained in the interior of the polygon (except for its points). Two such diagonals are said to cross if they have a point in common in the interior of the polygon. In one such polygon with n vertices, a certain number (say k) of non-crossing diagonals were drawn to cut up the inside of the polygon into regions, each of which was a quadrilateral. how many diagonals were drawn, that is, what is k ?

- A. cannot be determined from the information given
B. $\frac{n}{2} - 2$
C. $\frac{n}{4} - 1$
D. $n - 4$
E. $n^2 - 9.5n + 22$

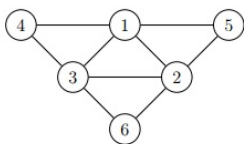
tifr2016 graph-theory graph-connectivity

Answer key

7.6.5 Graph Connectivity: TIFR CSE 2016 | Part A | Question: 2



Consider the graph shown below:



The following experiment is performed using this graph. First, an edge $e = \{i, j\}$ of the graph is chosen uniformly at random from the set of 9 possibilities. Next, a common neighbour k of i and j is chosen, again uniformly from the set of possibilities. (Note that the set of possibilities is always non-empty.) Thus, $\{i, j, k\}$ is a triangle in the graph. What is the probability that the triangle finally picked is $\{1, 2, 3\}$?

- A. $\frac{1}{6}$ B. $\frac{1}{4}$ C. $\frac{1}{3}$ D. $\frac{2}{3}$
E. $\frac{5}{6}$

tifr2016 graph-theory graph-connectivity probability

Answer key

7.6.6 Graph Connectivity: TIFR CSE 2016 | Part B | Question: 11



Let $n \geq 4$ be an integer. Regard the set $\mathbb{R}^n$ as a vector space over $\mathbb{R}$. Consider the following undirected graph H .

$$V(H) = \{S \subseteq \mathbb{R}^n : S \text{ is a basis for } \mathbb{R}^n\};$$

$$E(H) = \{\{S, T\} : |S \setminus T| = 1 \text{ and } |T \setminus S| = 1\},$$

where $S \setminus T = \{x \in S : x \notin T\}$. Which of the following statements is FALSE?

- A. H has an infinite number of vertices B. The diameter of H is infinite
C. H is connected D. H contains an infinite clique
E. H contains an infinite independent set

tifr2016 graph-theory graph-connectivity

Answer key

7.6.7 Graph Connectivity: TIFR CSE 2016 | Part B | Question: 15



Let G be an undirected graph. For a pair (x, y) of distinct vertices of G , let $\text{mincut}(x, y)$ be the least number of edges that should be deleted from G so that the resulting graph has no $x - y$ path. Let a, b, c be three vertices in G such that $\text{mincut}(a, b) \leq \text{mincut}(b, c) \leq \text{mincut}(c, a)$. Consider the following possibilities:

- $\text{mincut}(a, b) < \text{mincut}(b, c) < \text{mincut}(c, a)$
- $\text{mincut}(a, b) = \text{mincut}(b, c) < \text{mincut}(c, a)$
- $\text{mincut}(a, b) < \text{mincut}(b, c) = \text{mincut}(c, a)$
- $\text{mincut}(a, b) = \text{mincut}(b, c) = \text{mincut}(c, a)$

Which of the following is TRUE?

- A. All of i, ii, iii, iv are possible
B. i, ii, iii are possible but not iv
C. i and iv are possible but neither ii nor iii
D. ii and iv are possible but neither i nor iii
E. iii and iv are possible but neither i nor ii

Answer key

7.6.8 Graph Connectivity: TIFR CSE 2017 | Part B | Question: 12



An undirected graph is complete if there is an edge between every pair of vertices. Given a complete undirected graph on n vertices, in how many ways can you choose a direction for the edges so that there are no directed cycles?

- A. n
 B. $\frac{n(n-1)}{2}$
 C. $n!$
 D. 2^{n^2}
 E. 2^m , where $m = \frac{n(n-1)}{2}$

tifr2017 graph-theory graph-connectivity

Answer key

7.6.9 Graph Connectivity: TIFR CSE 2019 | Part B | Question: 12



Let $G = (V, E)$ be a directed graph with $n(\geq 2)$ vertices, including a special vertex r . Each edge $e \in E$ has a strictly positive edge weight $w(e)$. An arborescence in G rooted at r is a subgraph H of G in which every vertex $u \in V \setminus \{r\}$ has a directed path to the special vertex r . The weight of an arborescence H is the sum of the weights of the edges in H .

Let H^* be a minimum arborescence rooted at r , and w^* the weight of H^* . Which of the following is *NOT* always true?

- A. $w^* \geq \sum_{u \in V \setminus \{r\}} \min_{(u,v) \in E} w((u,v))$
 B. $w^* \geq \sum_{u \in V \setminus \{r\}} \min_{(v,u) \in E} w((v,u))$
 C. H^* has exactly $n - 1$ edges
 D. H^* is acyclic
 E. w^* is less than the weight of the minimum weight directed Hamiltonian cycle in G , when G has a directed Hamiltonian cycle

tifr2019 graph-connectivity graph-theory difficult

Answer key

7.6.10 Graph Connectivity: TIFR CSE 2019 | Part B | Question: 15



Consider directed graphs on n labelled vertices $\{1, 2, \dots, n\}$, where each vertex has exactly one edge coming in and exactly one edge going out. We allow self-loops. How many graphs have exactly two cycles?

- A. $\sum_{k=1}^{n-1} k!(n-k)!$
 B. $\frac{n!}{2} \left[\sum_{k=1}^{n-1} \frac{1}{k(n-k)} \right]$
 C. $n! \left[\sum_{k=1}^{n-1} \frac{1}{k} \right]$
 D. $\frac{n!(n-1)}{2}$
 E. None of the above

tifr2019 graph-connectivity graph-theory

Answer key

7.6.11 Graph Connectivity: TIFR CSE 2021 | Part B | Question: 12



Let G be an undirected graph. For any two vertices u, v in G , let $\text{cut}(u, v)$ be the minimum number of edges that should be deleted from G so that there is no path between u and v in the resulting graph. Let a, b, c, d be 4 vertices in G . Which of the following statements is impossible?

- A. $\text{cut}(a, b) = 3, \text{cut}(a, c) = 2$ and $\text{cut}(a, d) = 1$
- B. $\text{cut}(a, b) = 3, \text{cut}(b, c) = 1$ and $\text{cut}(b, d) = 1$
- C. $\text{cut}(a, b) = 3, \text{cut}(a, c) = 2$ and $\text{cut}(b, c) = 2$
- D. $\text{cut}(a, c) = 2, \text{cut}(b, c) = 2$ and $\text{cut}(c, d) = 2$
- E. $\text{cut}(b, d) = 2, \text{cut}(b, c) = 2$ and $\text{cut}(c, d) = 1$

tifr2021 graph-theory graph-connectivity

Answer key

7.7

Graph Matching (4)

7.7.1 Graph Matching: TIFR CSE 2012 | Part B | Question: 20



This question concerns the classes P and NP . If you are familiar with them, you may skip the definitions and go directly to the question.

Let L be a set. We say that L is in P if there is some algorithm which given input x decides if x is in L or not in time bounded by a polynomial in the length of x . For example, the set of all connected graphs is in P , because there is an algorithm which, given a graph, can decide if it is connected or not in time roughly proportional to the number of edges of the graph.

The class NP is a superset of class P . It contains those sets that have membership witnesses that can be verified in polynomial time. For example, the set of composite numbers is in NP . To see this take the witness for a composite number to be one of its divisors. Then the verification process consists of performing just one division using two reasonable size numbers. Similarly, the set of those graphs that have a Hamilton cycle, i.e. a cycle containing all the vertices of the graph, is in NP . To verify that the graph has a Hamilton cycle we just check if the witnessing sequence of vertices indeed a cycle of the graph that passes through all the vertices of the graph. This can be done in time that is polynomial in the size of the graph.

More precisely, if L is a set in P consisting of elements of the form (x, w) , then the set

$$M = \{x : \exists w, |w| \leq |x|^k \text{ and } (x, w) \in L\},$$

is in NP .

Let $G = (V, E)$ be a graph. G is said to have perfect matching if there is a subset M of the edges of G so that

- i. No two edges in M intersect (have a vertex in common); and
- ii. Every vertex of G has an edge in M .

Let MATCH be the set of all graphs that have a perfect matching. Let $\overline{\text{MATCH}}$ be the set of graphs that do not have a perfect matching. Let $o(G)$ be the number of components of G that have an odd number of vertices.

Tutte's Theorem: $G \in \text{MATCH}$ if and only if for all subsets S of V , the number of components in $G - S$ (the graph formed by deleting the vertices in S) with an odd number of vertices is at most $|S|$. That is,

$$G \in \text{MATCH} \leftrightarrow \forall S \subseteq V, o(G - S) \leq |S|.$$

Which of the following is true?

- A. $\text{MATCH} \in NP$ and $\overline{\text{MATCH}} \notin NP$ B. $\overline{\text{MATCH}} \in NP$ and $\text{MATCH} \notin NP$
C. $\text{MATCH} \in NP$ and $\overline{\text{MATCH}} \in NP$ D. $\text{MATCH} \notin P$ and $\overline{\text{MATCH}} \notin P$
E. none of the above

tifr2012 graph-theory graph-matching p-np-npc-nph

Answer key

7.7.2 Graph Matching: TIFR CSE 2021 | Part A | Question: 6



A matching in a graph is a set of edges such that no two edges in the set share a common vertex. Let G be a graph on n vertices in which there is a subset M of m edges which is a matching. Consider a random process where each vertex in the graph is independently selected with probability $0 < p < 1$ and let B be the set of vertices so obtained. What is the probability that there exists at least one edge from the matching M with both end points in the set B ?

- A. p^2 B. $1 - (1 - p^2)^m$
C. p^{2m} D. $(1 - p^2)^m$
E. $1 - (1 - p(1 - p))^m$

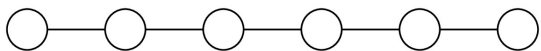
tifr2021 graph-theory graph-matching probability

Answer key

7.7.3 Graph Matching: TIFR CSE 2022 | Part B | Question: 2



Let $G = (V, E)$ be an undirected simple graph. A subset $M \subseteq E$ is a *matching* in G if distinct edges in M do not share a vertex. A matching is *maximal* if no strict superset of M is a matching. How many maximal matchings does the following graph have?



- A. 1 B. 2 C. 3 D. 4
E. 5

tifr2022 graph-theory graph-matching

Answer key

7.7.4 Graph Matching: TIFR CSE 2022 | Part B | Question: 6



We are given a graph G along with a matching M and a vertex cover C in it such that $|M| = |C|$. Consider the following statements:

1. M is a maximum matching in G .
2. C is a minimum vertex cover in G .
3. G is a bipartite graph.

Which of the following is TRUE?

- A. Only statement (1) is correct B. Only statement (2) is correct
C. Only statement (3) is correct D. Only statements (1) and (2) are correct
E. All the three statements (1), (2), and (3) are correct

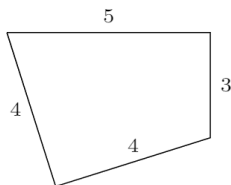
tifr2022 graph-theory graph-matching

Answer key

7.8.1 Graph Planarity: TIFR CSE 2021 | Part A | Question: 15



Let P be a convex polygon with sides 5, 4, 4, 3. For example, the following:



Consider the shape in the plane that consists of all points within distance 1 from some point in P . If ℓ is the perimeter of the shape, which of the following is always correct?

- A. ℓ cannot be determined from the given information. B. $20 \leq \ell < 21$
 C. $21 \leq \ell < 22$ D. $22 \leq \ell < 23$
 E. $23 \leq \ell < 24$

tifr2021 graph-theory graph-planarity

Answer key

7.9.1 Minimum Cut: TIFR CSE 2025 | Part B | Question: 14



Let G be an undirected graph. For any pair of vertices s, t in G , let $\text{MinCut}(s, t)$ be the least number of edges that have to be deleted from G so that there is no $s - t$ path in the resulting graph. Let a, b, c be three vertices in G such that $\text{MinCut}(a, b) \geq \text{MinCut}(b, c) \geq \text{MinCut}(a, c)$. Consider the following statements.

- i. $\text{MinCut}(a, b) > \text{MinCut}(b, c) > \text{MinCut}(a, c)$.
 ii. $\text{MinCut}(a, b) = \text{MinCut}(b, c) > \text{MinCut}(a, c)$.
 iii. $\text{MinCut}(a, b) > \text{MinCut}(b, c) = \text{MinCut}(a, c)$.
 iv. $\text{MinCut}(a, b) = \text{MinCut}(b, c) = \text{MinCut}(a, c)$.

Which of the above statements are possible?

- A. Only (i) and (ii) B. Only (ii) and (iii)
 C. Only (iii) and (iv) D. Only (ii), (iii), (iv)
 E. All of (i), (ii), (iii), (iv)

tifr2025 graph-theory minimum-cut network-flow

7.10.1 P NP NPC NPH: TIFR CSE 2022 | Part B | Question: 12



Given an undirected graph G , an ordering σ of its vertices is called a *perfect ordering* if for every vertex v , the neighbours of v which precede v in σ form a clique in G .

Recall that given an undirected graph G , a *clique* in G is a subset of vertices every two of which are connected by an edge, while a *perfect colouring* of G with k colours is an assignment of labels from the set $\{1, 2, \dots, k\}$ to the vertices of G such that no two vertices which are adjacent in G receive the same label.

Consider the following problems.

Problem SPECIAL-CLIQUE

INPUT: An undirected graph G , a positive integer k , and a perfect ordering σ of the vertices of G .

OUTPUT: Yes, if G has a clique of size at least k , No otherwise.

Problem SPECIAL-COLOURING

INPUT: An undirected graph G , a positive integer k , and a perfect ordering σ of the vertices of G .

OUTPUT: Yes, if G has a proper colouring with at most k colours, No otherwise.

Assume that $P \neq NP$. Which of the following statements is true?

- A. Both SPECIAL-CLIQUE and SPECIAL-COLOURING are undecidable
- B. Only SPECIAL-CLIQUE is in P
- C. Only SPECIAL-COLOURING is in P
- D. Both SPECIAL-CLIQUE and SPECIAL-COLOURING are in P
- E. Neither of SPECIAL-CLIQUE and SPECIAL-COLOURING is in P , but both are decidable

tifr2022 graph-theory graph-coloring p-np-npc-nph

Answer key

7.10.2 P NP NPC NPH: TIFR CSE 2023 | Part B | Question: 12



A graph $G = (V, E)$ is said to be k -colourable if the set V of vertices can be coloured with k colours such that no edge has both its endpoints of the same colour. It is known that the following language 3COL is NP-complete.

$$3COL = \{G \mid G \text{ is 3-colourable} \}.$$

Consider the following algorithmic problems (assume that the input graph G is provided as its adjacency matrix):

- (P1) Given a graph G as input, check if G is 2-colourable.
- (P2) Given a graph G as input, find the minimum value of k such that G is k -colourable.
- (P3) Given a graph G , check if G is 3-colourable.
- (P4) Given a graph G , along with a guarantee that the minimum value of k such that G is k -colourable is either 2 or 3, decide which of the two is the case.

Assume $NP \neq P$. Then, which of the above problems can be solved in polynomial time?

- A. Only problem (P1)
- B. Only problems (P1) and (P4)
- C. Only problems (P2) and (P3)
- D. Only problems (P1), (P3) and (P4)
- E. Only problems (P1), (P2) and (P4)

tifr2023 graph-theory graph-coloring p-np-npc-nph

Answer key

7.11

Rank of Matrix (1)

7.11.1 Rank of Matrix: TIFR CSE 2022 | Part B | Question: 14



Let G be a directed graph (with no self-loops or parallel edges) with $n \geq 2$ vertices and m edges. Consider the $n \times m$ incidence matrix M of G , whose rows are indexed by the vertices of G and the columns by the edges of G . The entry $m_{v,e}$ is defined as follows.

$$m_{v,e} = \begin{cases} -1 & \text{if } e = (v, w) \text{ for some vertex } w, \\ +1 & \text{if } e = (u, v) \text{ for some vertex } u, \\ 0 & \text{otherwise.} \end{cases}$$

Suppose every vertex of G is reachable from a special source vertex of G . Then, what is the rank of M ?

- A. $m - 1$ B. $m - n + 1$ C. $\lceil m/2 \rceil$ D. $n - 1$
E. $\lceil n/2 \rceil$

tifr2022 graph-theory graph-connectivity rank-of-matrix

Answer key 

7.12

Vertex Cover (1)

7.12.1 Vertex Cover: TIFR CSE 2016 | Part B | Question: 10



A *vertex cover* in an undirected graph G is a subset $C \subseteq V(G)$ such that every edge of G has an endpoint in C . An independent set in G is a subset $I \subseteq V(G)$ such that no edge has both its endpoints in I . Which of the following is TRUE of every graph G and every vertex cover C of G ?

- A. There exists an independent set of size $|C|$
B. $V(G) - C$ is an independent set
C. $|C| \geq |E(G)| / 2$
D. $|C| \geq |V(G)| / 2$
E. C intersects every independent set

tifr2016 graph-theory vertex-cover

Answer key 

Answer Keys

7.0.1	E	7.1.1	B	7.1.2	B	7.2.1	B	7.3.1	C
7.3.2	C	7.3.3	A	7.3.4	E	7.3.5	C	7.4.1	A
7.4.2	A	7.4.3	A	7.5.1	C	7.5.2	E	7.5.3	E
7.5.4	C	7.5.5	C	7.5.6	C	7.5.7	C	7.5.8	E
7.5.9	C	7.6.1	E	7.6.2	C	7.6.3	E	7.6.4	B
7.6.5	A	7.6.6	B	7.6.7	D	7.6.8	C	7.6.9	B
7.6.10	B	7.6.11	E	7.7.1	C	7.7.2	B	7.7.3	D
7.7.4	D	7.8.1	D	7.9.1	C	7.10.1	D	7.10.2	B
7.11.1	D	7.12.1	B						



8.1.1 First Order Logic: TIFR CSE 2010 | Part A | Question: 8



Which of the following is NOT necessarily true? { Notation: The symbol " $\neg$ " notes negation; $P(x, y)$ means that for given x and y , the property $P(x, y)$ is true }.

- A. $(\forall x \forall y P(x, y)) \Rightarrow (\forall y \forall x P(x, y))$
- B. $(\forall x \exists y \neg P(x, y)) \Rightarrow \neg(\exists x \forall y P(x, y))$
- C. $(\exists x \exists y P(x, y)) \Rightarrow (\exists y \exists x P(x, y))$
- D. $(\exists x \forall y P(x, y)) \Rightarrow (\forall y \exists x P(x, y))$
- E. $(\forall x \exists y P(x, y)) \Rightarrow (\exists y \forall x P(x, y))$

tifr2010 mathematical-logic first-order-logic

Answer key

8.1.2 First Order Logic: TIFR CSE 2012 | Part A | Question: 2



If $Mr. M$ is guilty, then no witness is lying unless he is afraid. There is a witness who is afraid. Which of the following statements is true?

(Hint: Formulate the problem using the following predicates

- G — $Mr. M$ is guilty
- $W(x)$ — x is a witness
- $L(x)$ — x is lying
- $A(x)$ — x is afraid)

- | | |
|---|-------------------------------------|
| A. $Mr. M$ is guilty. | B. $Mr. M$ is not guilty. |
| C. From these facts one cannot conclude that $Mr. M$ is guilty. | D. There is a witness who is lying. |
| E. No witness is lying. | |

tifr2012 mathematical-logic first-order-logic

Answer key

8.1.3 First Order Logic: TIFR CSE 2012 | Part B | Question: 3



For a person p , let $w(p)$, $A(p, y)$, $L(p)$ and $J(p)$ denote that p is a woman, p admires y , p is a lawyer and p is a judge respectively. Which of the following is the correct translation in first order logic of the sentence: "All woman who are lawyers admire some judge"?

- A. $\forall x : [(w(x) \wedge L(x)) \Rightarrow (\exists y : (J(y) \wedge w(y) \wedge A(x, y)))]$
- B. $\forall x : [(w(x) \Rightarrow L(x)) \Rightarrow (\exists y : (J(y) \wedge A(x, y)))]$
- C. $\forall x \forall y : [(w(x) \wedge L(x)) \Rightarrow (J(y) \wedge A(x, y))]$
- D. $\exists y \forall x : [(w(x) \wedge L(x)) \Rightarrow (J(y) \wedge A(x, y))]$
- E. $\forall x : [(w(x) \wedge L(x)) \Rightarrow (\exists y : (J(y) \wedge A(x, y)))]$

tifr2012 mathematical-logic first-order-logic

Answer key

8.1.4 First Order Logic: TIFR CSE 2014 | Part A | Question: 8



All that glitters is gold. No gold is silver.

Claims:

1. No silver glitters.
2. Some gold glitters.

Then, which of the following is TRUE?

- | | |
|--|---|
| A. Only claim 1 follows. | B. Only claim 2 follows. |
| C. Either claim 1 or claim 2 follows but not both. | D. Neither claim 1 nor claim 2 follows. |
| E. Both claim 1 and claim 2 follow. | |

tifr2014 mathematical-logic first-order-logic

Answer key

8.1.5 First Order Logic: TIFR CSE 2016 | Part B | Question: 4



In the following, A stands for a set of apples, and $S(x, y)$ stands for " x is sweeter than y ". Let

$$\Psi \equiv \exists x : x \in A$$

$$\Phi \equiv \forall x \in A : \exists y \in A : S(x, y).$$

Which of the following statements implies that there are infinitely many apples (i.e., A is an infinite set)?

- A. $\Psi \wedge \Phi \wedge [\forall x \in A : \neg S(x, x)]$
- B. $\Psi \wedge \Phi \wedge [\forall x \in A : S(x, x)]$
- C. $\Psi \wedge \Phi \wedge [\forall x, y \in A : S(x, x) \wedge S(x, y) \rightarrow S(y, y)]$
- D. $\Psi \wedge \Phi \wedge [\forall x \in A : \neg S(x, x)] \wedge [\forall x, y, z \in A : S(x, y) \wedge S(y, z) \rightarrow S(y, x)]$
- E. $\Psi \wedge \Phi \wedge [\forall x \in A : \neg S(x, x)] \wedge [\forall x, y, z \in A : S(x, y) \wedge S(y, z) \rightarrow S(x, z)]$

tifr2016 mathematical-logic first-order-logic

Answer key

8.1.6 First Order Logic: TIFR CSE 2017 | Part B | Question: 11



Given that

- $B(x)$ means " x is a bat",
- $F(x)$ means " x is a fly", and
- $E(x, y)$ means " x eats y ",

what is the best English translation of

$$\forall x(F(x) \rightarrow \forall y(E(y, x) \rightarrow B(y)))?$$

- | | |
|------------------------|-----------------------------------|
| A. all flies eat bats | B. every fly is eaten by some bat |
| C. bats eat only flies | D. every bat eats flies |
| E. only bats eat flies | |

tifr2017 first-order-logic

Answer key

8.1.7 First Order Logic: TIFR CSE 2017 | Part B | Question: 6



Consider the First Order Logic (FOL) with equality and suitable function and relation symbols. Which of the following is FALSE?

- A. Partial orders cannot be axiomatized in FOL
- B. FOL has a complete proof system
- C. Natural numbers cannot be axiomatized in FOL
- D. Real numbers cannot be axiomatized in FOL
- E. Relational numbers cannot be axiomatized in FOL

tifr2017 first-order-logic normal

Answer key

8.2

Propositional Logic (11)

8.2.1 Propositional Logic: TIFR CSE 2010 | Part A | Question: 4



- If the bank receipt is forged, then Mr. M is liable.
- If Mr. M is liable, he will go bankrupt.
- If the bank will loan him money, he will not go bankrupt.
- The bank will loan him money.

Which of the following can be concluded from the above statements?

- | | |
|---------------------------|------------------------------|
| A. Mr. M is liable | B. The receipt is not forged |
| C. Mr. M will go bankrupt | D. The bank will go bankrupt |
| E. None of the above | |

tifr2010 mathematical-logic propositional-logic

Answer key

8.2.2 Propositional Logic: TIFR CSE 2011 | Part A | Question: 1



- If either wages or prices are raised, there will be inflation.
- If there is inflation, then either the government must regulate it or the people will suffer.
- If the people suffer, the government will be unpopular.
- Government will not be unpopular.

Which of the following can be validly concluded from the above statements.

- A. People will not suffer
- B. If the inflation is not regulated, then wages are not raised
- C. Prices are not raised
- D. If the inflation is not regulated, then the prices are not raised
- E. Wages are not raised

tifr2011 mathematical-logic propositional-logic normal

Answer key

8.2.3 Propositional Logic: TIFR CSE 2011 | Part A | Question: 12



The action for this problem takes place in an island of Knights and Knaves, where Knights always make true statements and Knaves always make false statements and everybody is either a Knight or a Knave. Two friends A and B lives in a house. The census taker (an outsider) knocks on the door and it is opened by A. The census taker says "I need information about you and your friend.

Which if either is a Knight and which if either is a Knave?". "We are both Knaves" says A angrily and slams the door. What, if any thing can the census taker conclude?

- A. A is a Knight and B is a Knave.
- B. A is a Knave and B is a Knight.
- C. Both are Knaves.
- D. Both are Knights.
- E. No conclusion can be drawn.

tifr2011 mathematical-logic propositional-logic

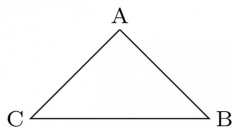
Answer key

8.2.4 Propositional Logic: TIFR CSE 2012 | Part A | Question: 3



Long ago, in a planet far far away, there lived three races of intelligent inhabitants: the blues (who always tell the truth), the whites (who always lie), and the pinks (who, when asked a series of questions, start with a lie and then tell the truth and lie alternately). To three creatures, chosen from the planet and seated facing each other at A , B , and C (see figure), the following three questions are put:

- i. What race is your left-hand neighbour?
- ii. What race is your right-hand neighbour?
- iii. What race are you?



Here are their answers:

- A. (i) White (ii) Pink (iii) Blue
- B. (i) Pink (ii) Pink (iii) Blue
- C. (i) White (ii) Blue (iii) Blue
- What is the actual race of each of the three creatures?
- A. A is Pink, B is White, C is Blue.
- B. A is Blue, B is Pink, C is White.
- C. A is Pink, B is Blue, C is Pink.
- D. A is White, B is Pink, C is Blue.
- E. Cannot be determined from the above data.

tifr2012 mathematical-logic propositional-logic

Answer key

8.2.5 Propositional Logic: TIFR CSE 2015 | Part A | Question: 5



What is logically equivalent to "If Kareena and Parineeti go to the shopping mall then it is raining":

- A. If Kareena and Parineeti do not go to the shopping mall then it is not raining.
- B. If Kareena and Parineeti do not go to the shopping mall then it is raining.
- C. If it is raining then Kareena and Parineeti go to the shopping mall.
- D. If it is not raining then Kareena and Parineeti do not go to the shopping mall.
- E. None of the above.

tifr2015 mathematical-logic propositional-logic

Answer key

8.2.6 Propositional Logic: TIFR CSE 2016 | Part B | Question: 1



A Boolean formula is said to be a *tautology* if it evaluates to TRUE for all assignments to its variables. Which one of the following is NOT a tautology?

- A. $((p \vee q) \wedge (r \vee s)) \Rightarrow ((p \wedge r) \vee q \vee s)$
 B. $((p \vee q) \wedge (r \vee s)) \Rightarrow (q \vee s)$
 C. $((p \vee q) \wedge (r \vee s)) \Rightarrow (r \vee q \vee s)$
 D. $((p \vee q) \wedge (r \vee s)) \Rightarrow (p \vee q \vee s)$
 E. $((p \vee q) \wedge (r \vee s)) \Rightarrow (p \vee q)$

tifr2016 mathematical-logic propositional-logic

Answer key 

8.2.7 Propositional Logic: TIFR CSE 2018 | Part B | Question: 4



The notation " $\Rightarrow$ " denotes "implies" and " $\wedge$ " denotes "and" in the following formulae.

- Let X denote the formula: $(b \Rightarrow a) \Rightarrow (a \Rightarrow b)$
- Let Y denote the formula: $(a \Rightarrow b) \wedge b$

Which of the following is TRUE?

- A. X is satisfiable and Y is not satisfiable.
 B. X is satisfiable and Y is tautology.
 C. X is not tautology and Y is not satisfiable.
 D. X is not tautology and Y is satisfiable.
 E. X is a tautology and Y is satisfiable,

tifr2018 mathematical-logic propositional-logic

Answer key 

8.2.8 Propositional Logic: TIFR CSE 2019 | Part B | Question: 4



Let φ be a propositional formula on a set of variables A and ψ be a propositional formula on a set of variables B , such that $\varphi \Rightarrow \psi$. A *Craig interpolant* of φ and ψ is a propositional formula μ on variables $A \cap B$ such that $\varphi \Rightarrow \mu$ and $\mu \Rightarrow \psi$. Given propositional formula $\varphi = q \vee (r \wedge s)$ on the set of variables $A = \{q, r, s\}$ and propositional formula $\psi = \neg q \rightarrow (s \vee t)$ on the set of variables $B = \{q, s, t\}$, which of the following is a Craig interpolant for φ and ψ ?

- A. q
 B. φ itself
 C. $q \vee s$
 D. $q \vee r$
 E. $\neg q \wedge s$

tifr2019 mathematical-logic propositional-logic

Answer key 

8.2.9 Propositional Logic: TIFR CSE 2021 | Part B | Question: 1



Consider the following statements about propositional formulas.

- $(p \wedge q) \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$ are *not* logically equivalent.
- $(\neg a \rightarrow b) \wedge (\neg b \vee (\neg a \vee \neg b))$ and $\neg(a \leftrightarrow b)$ are *not* logically equivalent.

- A. Both (i) and (ii) are true.
 B. (i) is true and (ii) is false.
 C. (i) is false and (ii) is true.
 D. Both (i) and (ii) are false.
 E. Depending on the values p and q , (i) can be either true or false, while (ii) is always false.

Answer key

8.2.10 Propositional Logic: TIFR CSE 2023 | Part B | Question: 1



Consider the following two statements:

- (P) The current population of Bhutan is greater than the current population of India.
- (Q) The Moon is smaller than the Earth.

Clearly, (P) is false, while (Q) is true. Which of the following logical statements evaluates to true?

- | | |
|----------------------------|--------------------------|
| A. $\neg P \Rightarrow Q$ | B. $Q \Rightarrow P$ |
| C. $\neg(P \Rightarrow Q)$ | D. $P \Leftrightarrow Q$ |
| E. None of the above | |

Answer key

8.2.11 Propositional Logic: TIFR CSE 2025 | Part B | Question: 2



Let φ be a propositional formula on n variables, with $n \geq 1$. Consider the following statements.

- φ is satisfiable
- $\neg\varphi$ is unsatisfiable
- φ is a tautology
- $\neg\varphi$ is a contradiction

Which of the following is ALWAYS TRUE for all such formulas φ ?

- (i) $\Leftrightarrow$ (ii), and (ii) $\Leftrightarrow$ (iii) $\Leftrightarrow$ (iv)
- (i) $\Leftrightarrow$ (ii), and (ii) $\Leftrightarrow$ (iii), and (iii) $\Leftrightarrow$ (iv)
- (i) $\Leftrightarrow$ (ii), and (iii) $\Leftrightarrow$ (iv), and (ii) $\Leftrightarrow$ (iii)
- (i) $\Leftrightarrow$ (ii), and (ii) $\Leftrightarrow$ (iii), and (iii) $\Leftrightarrow$ (iv)
- (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iii) $\Leftrightarrow$ (iv)

Answer key

8.3

Set Theory (1)

8.3.1 Set Theory: TIFR CSE 2013 | Part A | Question: 3



Three candidates, Amar, Birendra and Chanchal stand for the local election. Opinion polls are conducted and show that fraction a of the voters prefer Amar to Birendra, fraction b prefer Birendra to Chanchal and fraction c prefer Chanchal to Amar. Which of the following is impossible?

- | | |
|--------------------------------------|--------------------------------------|
| A. $(a, b, c) = (0.51, 0.51, 0.51);$ | B. $(a, b, c) = (0.61, 0.71, 0.67);$ |
| C. $(a, b, c) = (0.68, 0.68, 0.68);$ | D. $(a, b, c) = (0.49, 0.49, 0.49);$ |
| E. None of the above. | |

Answer key

Answer Keys

8.1.1	E
8.1.6	E
8.2.4	C
8.2.9	B

8.1.2	C
8.1.7	A
8.2.5	D
8.2.10	A

8.1.3	E
8.2.1	B
8.2.6	B
8.2.11	A

8.1.4	E
8.2.2	X
8.2.7	D
8.3.1	C

8.1.5	E
8.2.3	B
8.2.8	C



9.1

Convex Sets Functions (1)

9.1.1 Convex Sets Functions: TIFR CSE 2019 | Part A | Question: 6



A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is said to be *convex* if for all $x, y \in \mathbb{R}$ and λ such that $0 \leq \lambda \leq 1$,

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a convex function, and define the following functions:

$$p(x) = f(-x), \quad q(x) = -f(-x), \quad \text{and} \quad r(x) = f(1 - x).$$

Which of the functions p, q and r must be convex?

- A. Only p B. Only q C. Only r D. Only p and r
E. Only q and r

tifr2019 set-theory&algebra functions convex-sets-functions non-gatecse

Answer key

9.2

Countable Uncountable Set (1)

9.2.1 Countable Uncountable Set: TIFR CSE 2025 | Part B | Question: 3



Consider the following sets.

- i. The set of all natural numbers $\mathbb{N}$.
- ii. The set of all rational numbers $\mathbb{Q}$.
- iii. The set of all functions from $\mathbb{N}$ to $\{1, 2, 3, 4, 5\}$.
- iv. The set of all functions from $\{1, 2, 3, 4, 5\}$ to $\mathbb{N}$.

Which of the above sets are countable?

- A. Only (i) B. Only (i) and (ii)
C. Only (i) and (iv) D. Only (i), (ii) and (iii)
E. Only (i), (ii) and (iv)

tifr2025 set-theory&algebra countable-uncountable-set

Answer key

9.3

Functions (9)

9.3.1 Functions: TIFR CSE 2012 | Part B | Question: 1



For $x, y \in \{0, 1\}^n$, let $x \oplus y$ be the element of $\{0, 1\}^n$ obtained by the component-wise exclusive-or of x and y . A Boolean function $F : \{0, 1\}^n \rightarrow \{0, 1\}$ is said to be linear if $F(x \oplus y) = F(x) \oplus F(y)$, for all x and y . The number of linear functions from $\{0, 1\}^n$ to $\{0, 1\}$ is.

- A. 2^{2^n} B. 2^{n+1} C. $2^{n-1} + 1$ D. $n!$
E. 2^n

tifr2012 set-theory&algebra functions

Answer key

9.3.2 Functions: TIFR CSE 2013 | Part B | Question: 16



Consider a function $T_{k,n} : \{0, 1\}^n \rightarrow \{0, 1\}$ which returns 1 if at least k of its n inputs are

1. Formally, $T_{k,n}(x) = 1$ if $\sum_1^n x_i \geq k$. Let $y \in \{0,1\}^n$ be such that y has exactly k ones. Then, the function $T_{k,n-1}(y_1, y_2, \dots, y_{i-1}, y_{i+1}, \dots, y_n)$ (where y_i is omitted) is equivalent to

- A. $T_{k-1,n}(y)$ B. $T_{k,n}(y)$ C. y_i D. $\neg y_i$
E. None of the above

tifr2013 set-theory&algebra functions

Answer key 

9.3.3 Functions: TIFR CSE 2014 | Part B | Question: 18



Let k be an integer at least 4 and let $[k] = \{1, 2, \dots, k\}$. Let $f : [k]^4 \rightarrow \{0, 1\}$ be defined as follows: $f(y_1, y_2, y_3, y_4) = 1$ if and only if the y_i 's are all distinct. For each choice $z = (z_1, z_2, z_3) \in [k]^3$, let $g_z : [k] \rightarrow \{0, 1\}$ be defined by $g_z(Y) = f(Y, z_1, z_2, z_3)$. Let N be the number of distinct functions g_z that are obtained as z varies in $\{1, 2, \dots, k\}^3$, that is, $N = |\{g_z : z \in \{1, 2, \dots, k\}^3\}|$. What is N ?

- A. $k^3 + 1$ B. $2^{\binom{k}{3}}$ C. $\binom{k}{3}$ D. $\binom{k}{3} + 1$
E. $4 \binom{k}{3}$

tifr2014 set-theory&algebra functions

Answer key 

9.3.4 Functions: TIFR CSE 2017 | Part A | Question: 10



For a set A define $P(A)$ to be the set of all subsets of A . For example, if $A = \{1, 2\}$ then $P(A) = \{\emptyset, \{1, 2\}, \{1\}, \{2\}\}$. Let $A \rightarrow P(A)$ be a function and A is not empty. Which of the following must be TRUE?

- A. f cannot be one-to-one (injective) B. f cannot be onto (surjective)
C. f is both one-to-one and onto (bijective) D. there is no such f possible
E. if such a function f exists, then A is infinite

tifr2017 set-theory&algebra functions easy

Answer key 

9.3.5 Functions: TIFR CSE 2017 | Part A | Question: 11



Let $f \circ g$ denote function composition such that $(f \circ g)(x) = f(g(x))$. Let $f : A \rightarrow B$ such that for all $g : B \rightarrow A$ and $h : B \rightarrow A$ we have $f \circ g = f \circ h \Rightarrow g = h$. Which of the following must be true?

- A. f is onto (surjective) B. f is one-to-one (injective)
C. f is both one-to-one and onto (bijective) D. the range of f is finite
E. the domain of f is finite

tifr2017 set-theory&algebra functions

Answer key 

9.3.6 Functions: TIFR CSE 2018 | Part B | Question: 10



For two n bit strings $x, y \in \{0, 1\}^n$, define $z = x \oplus y$ to be the bitwise XOR of the two strings (that is, if x_i, y_i, z_i denote the i^{th} bits of x, y, z respectively, then $z_i = x_i + y_i \pmod{2}$). A

function $h : \{0, 1\}^n \rightarrow \{0, 1\}^n$ is called linear if $h(x \oplus y) = h(x) \oplus h(y)$, for every $x, y \in \{0, 1\}^n$. The number of such linear functions for $n \geq 2$ is:

- A. 2^n B. 2^{n^2} C. $2^{\frac{n}{2}}$ D. 2^{4n}
E. 2^{n^2+n}

tifr2018 set-theory&algebra functions

Answer key 

9.3.7 Functions: TIFR CSE 2019 | Part A | Question: 12



Let f be a function with both input and output in the set $\{0, 1, 2, \dots, 9\}$, and let the function g be defined as $g(x) = f(9 - x)$. The function f is non-decreasing, so that $f(x) \geq f(y)$ for $x \geq y$. Consider the following statements:

- There exists $x \in \{0, \dots, 9\}$ so that $x = f(x)$
- There exists $x \in \{0, \dots, 9\}$ so that $x = g(x)$
- There exists $x \in \{0, \dots, 9\}$ so that $x = (f(x) + g(x)) \bmod 10$

Which of the above statements must be TRUE for ALL such functions f and g ?

- A. Only (i) B. Only (i) and (ii)
C. Only (iii) D. None of them
E. All of them

tifr2019 engineering-mathematics discrete-mathematics set-theory&algebra functions

Answer key 

9.3.8 Functions: TIFR CSE 2021 | Part A | Question: 5



Let n, m and k be three positive integers such that $n \geq m \geq k$. Let S be a subset of $\{1, 2, \dots, n\}$ of size k . Consider sampling a function f uniformly at random from the set of all functions mapping $\{1, \dots, n\}$ to $\{1, \dots, m\}$. What is the probability that f is not injective on the set S , i.e., there exist $i, j \in S$ such that $f(i) = f(j)$?

In the following, the binomial coefficient $\binom{n}{k}$ counts the number of k -element subsets of an n -element set.

- A. $1 - \frac{k!}{k^k}$
B. $1 - \frac{m!}{m^k}$
C. $1 - \frac{k! \binom{m}{k}}{m^k}$
D. $1 - \frac{k! \binom{n}{k}}{n^k}$
E. $1 - \frac{k! \binom{n}{k}}{m^k}$

tifr2021 set-theory&algebra functions probability

Answer key 

9.3.9 Functions: TIFR CSE 2023 | Part B | Question: 4



Amar, Balu, and Chhaya are three friends and they play the following game. Chhaya first chooses a number $k \in U$ where $U = \{1, 2, \dots, 127\}$. She either gives k to both Amar and Balu, or else she gives k to Amar and $k + 1$ to Balu. Amar computes a function $f : U \rightarrow \{0, 1\}^n$ on

the number he receives from Chhaya and sends the output of this function to Balu. Using this output and his own input, Balu has to decide whether Chhaya gave the same number to Amar and him, or if she gave numbers that differed by 1. What is the minimal n that allows Balu to distinguish between these two cases?

- A. 1 B. 2 C. 7 D. 127
E. 128

tifr2023 set-theory&algebra functions

Answer key 

9.4

Lattice (3)

9.4.1 Lattice: TIFR CSE 2012 | Part B | Question: 4



Let $\wedge, \vee$ denote the meet and join operations of lattice. A lattice is called distributive if for all x, y, z ,

$$x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$$

It is called complete if meet and join exist for every subset. It is called modular if for all x, y, z

$$z \leq x \Rightarrow x \wedge (y \vee z) = (x \wedge y) \vee z$$

The positive integers under divisibility ordering i.e. $p \leq q$ if p divides q forms a.

- A. Complete lattice.
B. Modular, but not distributive lattice.
C. Distributive lattice.
D. Lattice but not a complete lattice.
E. Under the give ordering positive integers do not form a lattice.

tifr2012 set-theory&algebra lattice

Answer key 

9.4.2 Lattice: TIFR CSE 2014 | Part B | Question: 15



Consider the set N^* of finite sequences of natural numbers with $x \leq_p y$ denoting that sequence x is a prefix of sequence y . Then, which of the following is true?

- A. N^* is uncountable.
B. $\leq_p$ is a total order.
C. Every non-empty subset of N^* has a least upper bound.
D. Every non-empty subset of N^* has a greatest lower bound.
E. Every non-empty finite subset of N^* has a least upper bound.

tifr2014 set-theory&algebra partial-order lattice

Answer key 

9.4.3 Lattice: TIFR CSE 2014 | Part B | Question: 16



Consider the ordering relation $x | y \subseteq N \times N$ over natural numbers N such that $x | y$ if there exists $z \in N$ such that $x \bullet z = y$. A set is called lattice if every finite subset has a least upper bound and greatest lower bound. It is called a complete lattice if every subset has a least upper bound and greatest lower bound. Then,

- A. $|$ is an equivalence relation. B. Every subset of N has an upper bound under $|$.

- C. $|$ is a total order. D. $(N, |)$ is a complete lattice.
 E. $(N, |)$ is a lattice but not a complete lattice.

tifr2014 set-theory&algebra partial-order lattice

Answer key 

9.5

Modular Arithmetic (1)

9.5.1 Modular Arithmetic: TIFR CSE 2025 | Part B | Question: 15



Let m be a positive integer and a be an integer such that $a \in \{0, 1, \dots, m-1\}$. Let $S_a = \{b : b \in \{0, 1, \dots, m-1\}, b^2 \equiv a \pmod{m}\}$ be the set of square roots of a modulo m . Consider the following statements.

- For all natural numbers m and for all $a \in \{0, 1, \dots, m-1\}$, $|S_a| \leq 2$.
- For all natural numbers m , there exists an $a \in \{0, 1, \dots, m-1\}$, $|S_a| > 2$.
- If m is a prime number, then for all $a \in \{0, 1, \dots, m-1\}$, $|S_a| \leq 2$.
- For composite m , $|S_a|$ is not necessarily bounded above by 2.

Which of the above statements are TRUE ?

- A. Only (i)
B. Both (ii), (iv)
C. Only (iii)
D. Only (iv)
E. Both (iii), (iv)

tifr2025 number-theory modular-arithmetic set-theory&algebra set-theory

9.6

Partial Order (2)

9.6.1 Partial Order: TIFR CSE 2012 | Part B | Question: 5



Let R be a binary relation over a set S . The binary relation R is called an equivalence relation if it is reflexive, transitive and symmetric. The relation is called partial order if it is reflexive, transitive and anti symmetric. (Notation: Let aRb denote that order pair $(a, b) \in R$.) The relation R is called a well-order if R is a partial order and there does not exist an infinite descending chain (with respect to R) within S . An infinite sequence $x_1, x_2, \dots$ of elements of S is called an infinite descending chain if for all i we have $x_{i+1}Rx_i$ and $x_i \neq x_{i+1}$.

Take $S = \mathbb{N} \times \mathbb{N}$ and let the binary relation $\sqsubseteq$ over S be such that $(i_1, j_1) \sqsubseteq (i_2, j_2)$ if and only if either $(i_1 < i_2)$ or $((i_1 = i_2) \wedge (j_1 \leq j_2))$. Which statement is true of $\sqsubseteq$?

- A. $\sqsubseteq$ is an equivalence relation but not a well order.
 B. $\sqsubseteq$ is a partial order but not a well order.
 C. $\sqsubseteq$ is a partial order and a well order.
 D. $\sqsubseteq$ is an equivalence relation and a well order.
 E. $\sqsubseteq$ is neither a partial order nor an equivalence relation.

tifr2012 set-theory&algebra partial-order

Answer key 

9.6.2 Partial Order: TIFR CSE 2013 | Part B | Question: 4



A set S together with partial order $\ll$ is called a well order if it has no infinite descending chains, i.e. there is no infinite sequence $x_1, x_2, \dots$ of elements from S such that $x_{i+1} \ll x_i$ and $x_{i+1} \neq x_i$ for all i .

Consider the set of all words (finite sequence of letters $a - z$), denoted by W , in dictionary order.

- A. Between “ aa ” and “ az ” there are only 24 words.
 B. Between “ aa ” and “ az ” there are only 2^{24} words.
 C. W is not a partial order.
 D. W is a partial order but not a well order.
 E. W is a well order.

tifr2013 set-theory&algebra partial-order

Answer key 

9.7

Relations (1)

9.7.1 Relations: TIFR CSE 2023 | Part A | Question: 4



Let $\mathbb{R}$ be the set of all real numbers. Consider the relation T defined as

$$T = \{(x, y) \in \mathbb{R} \times \mathbb{R} \text{ such that } 4 + xy > 0\}$$

(Recall that a relation $S \subseteq \mathbb{R} \times \mathbb{R}$ is said to be *reflexive* if for all x , $(x, x) \in S$, and *symmetric* if whenever $(x, y) \in S$, it is also the case that $(y, x) \in S$. It is said to be *transitive* if whenever $(x, y) \in S$ and $(y, z) \in S$, it is also the case that $(x, z) \in S$. A relation is said to be an *equivalence relation* if it is symmetric, transitive, and reflexive.)

Then, the relation T is

- A. reflexive and transitive but not symmetric
 B. an equivalence relation
 C. reflexive and symmetric but not transitive
 D. symmetric but not reflexive and not transitive
 E. symmetric and transitive but not reflexive

tifr2023 set-theory&algebra relations

Answer key 

9.8

Set Theory (14)

9.8.1 Set Theory: TIFR CSE 2010 | Part A | Question: 15



Let A, B be sets. Let $\bar{A}$ denote the complement of set A (with respect to some fixed universe), and $(A - B)$ denote the set of elements in A which are not in B . Set $(A - (A - B))$ is equal to:

- A. B
 B. $A \cap \bar{B}$
 C. $A - B$
 D. $A \cap B$
 E. $\bar{B}$

tifr2010 set-theory&algebra set-theory

Answer key 

9.8.2 Set Theory: TIFR CSE 2010 | Part A | Question: 18



Let X be a set of size n . How many pairs of sets (A, B) are there that satisfy the condition $A \subseteq B \subseteq X$?

- A. 2^{n+1}
 B. 2^{2n}
 C. 3^n
 D. $2^n + 1$
 E. 3^{n+1}

Answer key

9.8.3 Set Theory: TIFR CSE 2011 | Part A | Question: 10



Let m, n denote two integers from the set $\{1, 2, \dots, 10\}$. The number of ordered pairs (m, n) such that $2^m + 2^n$ is divisible by 5 is.

- A. 10 B. 14 C. 24 D. 8
- E. None of the above

Answer key

9.8.4 Set Theory: TIFR CSE 2011 | Part B | Question: 23



Suppose $(S_1, S_2, \dots, S_m)$ is a finite collection of non-empty subsets of a universe U . Note that the sets in this collection need not be distinct. Consider the following basic step to be performed on this sequence. While there exist sets S_i and S_j in the sequence, neither of which is a subset of the other, delete them from the sequence, and

- If $S_i \cap S_j \neq \emptyset$, then add the sets $S_i \cup S_j$ and $S_i \cap S_j$ to the sequence;
- If $S_i \cap S_j = \emptyset$, then add only the set $S_i \cup S_j$ to the sequence.

In each step we delete two sets from the sequence and add at most two sets to the sequence. Also, note that empty sets are never added to the sequence. Which of the following statements is TRUE?

- A. The size of the smallest set in the sequence decreases in every step
 B. The size of the largest set in the sequence increases in every step
 C. The process always terminates
 D. The process terminates if U is finite but might not if U is infinite
 E. There is a finite collection of subsets of a finite universe U and a choice of S_i and S_j in each step such that the process does not terminate

Answer key

9.8.5 Set Theory: TIFR CSE 2012 | Part A | Question: 8



How many pairs of sets (A, B) are there that satisfy the condition $A, B \subseteq \{1, 2, \dots, 5\}, A \cap B = \{\}$?

- A. 125 B. 127 C. 130 D. 243
- E. 257

Answer key

9.8.6 Set Theory: TIFR CSE 2016 | Part A | Question: 8



Let A and B be finite sets such that $A \subseteq B$. Then, what is the value of the expression:

$$\sum_{C: A \subseteq C \subseteq B} (-1)^{|C \setminus A|},$$

Where $C \setminus A = \{x \in C : x \notin A\}$?

- A. Always 0
- B. Always 1
- C. 0 if $A = B$ and 1 otherwise
- D. 1 if $A = B$ and 0 otherwise
- E. Depends on the size of the universe

tifr2016 set-theory&algebra set-theory

Answer key

9.8.7 Set Theory: TIFR CSE 2016 | Part B | Question: 14



Consider a family $\mathcal{F}$ of subsets of $\{1, 2, \dots, n\}$ such that for any two distinct sets A and B in $\mathcal{F}$ we have: $A \subset B$ or $B \subset A$ or $A \cap B = \emptyset$. Which of the following statements is TRUE? (Hint: what does the Venn diagram of this family look like?)

- A. $|\mathcal{F}| \leq 2n$ and there exists a family $\mathcal{F}$ such that $|\mathcal{F}| = 2n$
- B. $|\mathcal{F}| \leq n^2$ and there exists a family $\mathcal{F}$ such that $|\mathcal{F}| = n^2$
- C. $|\mathcal{F}| \leq 2n^2$ and there exists a family $\mathcal{F}$ such that $|\mathcal{F}| = 2n^2$
- D. $|\mathcal{F}| \leq 2^{n-1}$ and there exists a family $\mathcal{F}$ such that $|\mathcal{F}| = 2^{n-1}$
- E. None of the above

tifr2016 set-theory&algebra set-theory

Answer key

9.8.8 Set Theory: TIFR CSE 2019 | Part A | Question: 1



Let X be a set with n elements. How many subsets of X have odd cardinality?

- A. n
- B. 2^n
- C. $2^{n/2}$
- D. 2^{n-1}
- E. Can not be determined without knowing whether n is odd or even

tifr2019 engineering-mathematics discrete-mathematics set-theory&algebra set-theory

Answer key

9.8.9 Set Theory: TIFR CSE 2021 | Part A | Question: 9



Fix $n \geq 6$. Consider the set $\mathcal{C}$ of binary strings $x_1, x_2 \dots x_n$ of length n such that the bits satisfy the following set of equalities, all modulo 2: $x_i + x_{i+1} + x_{i+2} = 0$ for all $1 \leq i \leq n-2$, $x_{n-1} + x_n + x_1 = 0$, and $x_n + x_1 + x_2 = 0$. What is the size of the set $\mathcal{C}$?

- A. 1 for all $n \geq 6$
- B. 4 for all $n \geq 6$
- C. 0 for all $n \geq 6$
- D. If $n \geq 6$ is divisible by 3 $|\mathcal{C}| = 1$. If $n \geq 6$ is *not* divisible by 3 then $|\mathcal{C}| = 4$.
- E. If $n \geq 6$ is divisible by 3 $|\mathcal{C}| = 4$. If $n \geq 6$ is not divisible by 3 then $|\mathcal{C}| = 1$.

tifr2021 set-theory&algebra set-theory

Answer key

9.8.10 Set Theory: TIFR CSE 2022 | Part A | Question: 11



Let X be a finite set. A family $\mathcal{F}$ of subsets of X is said to be *upward closed* if the following holds for all sets $A, B \subseteq X$:

$$A \in \mathcal{F} \text{ and } A \subseteq B \Rightarrow B \in \mathcal{F}.$$

For families $\mathcal{F}$ and $\mathcal{G}$ of subsets of X , let

$$\mathcal{F} \sqcup \mathcal{G} = \{A \cup B : A \in \mathcal{F} \text{ and } B \in \mathcal{G}\}.$$

Suppose $\mathcal{F}$ and $\mathcal{G}$ are upward closed families. Then which of the following is true?

- A. $\mathcal{F} \sqcup \mathcal{G} = \mathcal{F} \cap \mathcal{G}$
- B. $\mathcal{F} \sqcup \mathcal{G} = \mathcal{F} \cup \mathcal{G}$
- C. $\mathcal{F} \sqcup \mathcal{G} = \mathcal{F} \setminus \mathcal{G}$
- D. $\mathcal{F} \sqcup \mathcal{G} = \mathcal{G} \setminus \mathcal{F}$
- E. None of the above

tifr2022 set-theory&algebra set-theory

Answer key

9.8.11 Set Theory: TIFR CSE 2022 | Part A | Question: 9



You are given the following properties of sets A, B, X , and Y . For notation, $|A|$ denotes the cardinality of set A (i.e., the number of elements in A), and $A \setminus B$ denotes the set of elements that are in A but not in B .

1. $A \cup B = X \cup Y$
2. $A \cap B = X \cap Y = \emptyset$
3. $|Y \setminus A| = 2$
4. $|A \setminus X| = 4$

Which of the following statements **MUST** then be **FALSE**?

- A. $|X| = 5$
- B. $|Y| = 5$
- C. $|A \cup X| = |B \cup Y|$
- D. $|A \cap X| = |B \cap Y|$
- E. $|A| = |B|$

tifr2022 set-theory&algebra set-theory

Answer key

9.8.12 Set Theory: TIFR CSE 2023 | Part A | Question: 10



Let Q be a set with n elements. Consider a system with states of the form (S, R) where $S \subseteq Q$ and $R \subseteq Q$, with the further restriction that $R \subseteq S$. Note that $(\emptyset, \emptyset)$ is also a valid state of the system (here $\emptyset$ denotes the empty set). How many possible states does the system have?

- A. 3^n
- B. 2^n
- C. 4^n
- D. n^3
- E. $\binom{n}{3}$

tifr2023 set-theory&algebra set-theory

Answer key

9.8.13 Set Theory: TIFR CSE 2023 | Part A | Question: 11



We are given a set $S = \{x_1, \dots, x_n\}$ of distinct positive integers such that $\gcd(x_i, x_j) = 1$ for any $i, j \in \{1, \dots, n\}$ where $i \neq j$. What is the total number of invertible 2×2 matrices whose entries are distinct elements from the set S ?

(Note: For positive integers a and b , $\gcd(a, b)$ denotes the greatest common divisor of a and b).

- A. n^4
 B. $(n-1)^4$
 C. $n^2(n-1)^2/4$
 D. $n(n-1)(n-2)(n-3)$
 E. $n(n-1)(n-2)(n-3)/4!$

tifr2023 set-theory&algebra set-theory

Answer key

9.8.14 Set Theory: TIFR CSE 2025 | Part A | Question: 15



How many subsets of $\{1, 2, \dots, 2n\}$ do not contain two numbers with sum $2n+1$?

- A. $3^n + \binom{2n}{2}$
 B. 3^n
 C. $4^n - \binom{2n}{2}$
 D. 2^n
 E. None of the above

tifr2025 set-theory

Answer key

Answer Keys

9.1.1	D	9.2.1	E	9.3.1	E	9.3.2	D	9.3.3	D
9.3.4	B	9.3.5	B	9.3.6	B	9.3.7	B	9.3.8	C
9.3.9	A	9.4.1	C	9.4.2	D	9.4.3	E	9.5.1	E
9.6.1	C	9.6.2	E	9.7.1	C	9.8.1	D	9.8.2	C
9.8.3	C	9.8.4	C	9.8.5	D	9.8.6	D	9.8.7	A
9.8.8	D	9.8.9	E	9.8.10	A	9.8.11	B	9.8.12	A
9.8.13	D	9.8.14	B						



10.0.1 TIFR CSE 2024 | Part A | Question: 9



Compute $\int_{16}^{\infty} \frac{1}{x} \cdot \frac{1}{\sqrt{\sqrt{x}-1}} dx$.

- A. 0 B. $\frac{\pi}{3}$ C. $\frac{\pi}{2}$ D. $\frac{2\pi}{3}$
E. 2π

tifr2024 calculus

Answer key

10.1

Continuity and Differentiability (1)

10.1.1 Continuity and Differentiability: TIFR CSE 2023 | Part A | Question: 6



For a function $f: \mathbb{R} \rightarrow \mathbb{R}$, consider the following conditions.

- (C1) $|f(x)| \leq |x|$ for all $x \in \mathbb{R}$.
(C2) $|f(x)| \leq |x|^2$ for all $x \in \mathbb{R}$.
(C3) $|f(x)| \leq |x|^3$ for all $x \in \mathbb{R}$.

Which of the above conditions imply that f is differentiable at 0?

- A. Condition (C1) only B. Condition (C2) only
C. Condition (C3) only D. Conditions (C1) and (C2) only
E. Conditions (C2) and (C3) only

tifr2023 calculus continuity-and-differentiability

Answer key

10.2

Convergence (1)

10.2.1 Convergence: TIFR CSE 2014 | Part A | Question: 15



Consider the following statements:

- $b_1 = \sqrt{2}$, series with each $b_i = \sqrt{b_{i-1} + \sqrt{2}}, i \geq 2$, converges.
- $\sum_{i=1}^{\infty} \frac{\cos(i)}{i^2}$ converges.
- $\sum_{i=0}^{\infty} b_i$ converges if $\lim_{i \rightarrow \infty} \frac{|b_{i+1}|}{|b_i|} < 1$

Which of the following is TRUE?

- A. Statements (1) and (2) but not (3). B. Statements (2) and (3) but not (1).
C. Statements (1) and (3) but not (2). D. All the three statements.
E. None of the three statements.

tifr2014 convergence non-gatecse

Answer key

10.3

Definite Integral (3)

10.3.1 Definite Integral: TIFR CSE 2011 | Part A | Question: 11



$$\int_0^1 \log_e(x) dx =$$

- A. 1 B. -1 C. ∞ D. $-\infty$
- E. None of the above

tifr2011 calculus definite-integral

Answer key

10.3.2 Definite Integral: TIFR CSE 2015 | Part A | Question: 10



Let $f(x), x \in [0, 1]$, be any positive real valued continuous function. Then

$$\lim_{n \rightarrow \infty} (n+1) \int_0^1 x^n f(x) dx$$

equals.

- A. $\max_{x \in [0,1]} f(x)$ B. $\min_{x \in [0,1]} f(x)$
- C. $f(0)$ D. $f(1)$
- E. ∞

tifr2015 calculus limits definite-integral

10.3.3 Definite Integral: TIFR CSE 2019 | Part A | Question: 13



Consider the integral

$$\int_0^1 \frac{x^{300}}{1+x^2+x^3} dx$$

What is the value of this integral correct up to two decimal places?

- A. 0.00 B. 0.02 C. 0.10 D. 0.33
- E. 1.00

tifr2019 engineering-mathematics calculus definite-integral

Answer key

10.4

Differential Equation (1)

10.4.1 Differential Equation: TIFR CSE 2012 | Part A | Question: 15



Consider the differential equation $dx/dt = (1-x)(2-x)(3-x)$. Which of its equilibria is unstable?

- A. $x = 0$ B. $x = 1$ C. $x = 2$ D. $x = 3$
- E. None of the above

tifr2012 calculus differential-equation

Answer key

10.5

Differentiation (3)

10.5.1 Differentiation: TIFR CSE 2018 | Part A | Question: 5



Which of the following is the derivative of $f(x) = x^x$ when $x > 0$?

- A. x^x
- B. $x^x \ln x$
- C. $x^x + x^x \ln x$
- D. $(x^x)(x^x \ln x)$
- E. None of the above; function is not differentiable for $x > 0$

tifr2018 calculus differentiation

Answer key

10.5.2 Differentiation: TIFR CSE 2022 | Part A | Question: 14



Suppose $w(t) = 4e^{it}$, $x(t) = 3e^{i(t+\pi/3)}$, $y(t) = 3e^{i(t-\pi/3)}$ and $z(t) = 3e^{i(t+\pi)}$ are points that move in the complex plane as the time t varies in $(-\infty, \infty)$. Let $c(t)$ be the point in the complex plane such that $|w(t) - c(t)|^2 + |x(t) - c(t)|^2 + |y(t) - c(t)|^2 + |z(t) - c(t)|^2$ is minimum. For each value of t , the point $c(t)$ is unique, but $c(t)$ moves at constant speed as t varies. At what speed? That is, what is $|\frac{d}{dt}c(t)|$?

- A. $\frac{1}{2\pi}$
- B. 2π
- C. $\sqrt{3}\pi$
- D. $\frac{1}{\sqrt{3}\pi}$
- E. 1

tifr2022 calculus differentiation

Answer key

10.5.3 Differentiation: TIFR CSE 2023 | Part A | Question: 14



Let $f(x) = ax^3 + bx^2 + cx + d$ be a polynomial, where a, b, c, d are unknown real numbers. It is further given that $f(1) = 1$, $f(2) = 2$, $f(3) = 9$, and $f'(1) = 0$. Then, the value of $f'(2)$ must be

- A. 1
- B. 2
- C. 3
- D. 4
- E. $f'(2)$ cannot be determined uniquely from the information given in the question.

tifr2023 calculus differentiation

Answer key

10.6 Divergence (1)

10.6.1 Divergence: TIFR CSE 2016 | Part A | Question: 5



For a positive integer $N \geq 2$, let

$$A_N := \sum_{n=2}^N \frac{1}{n};$$

$$B_N := \int_{x=1}^N \frac{1}{x} dx$$

Which of the following statements is true?

- A. As $N \rightarrow \infty$, A_N increases to infinity but B_N converges to a finite number
- B. $A_N < B_N$ and the difference decreases as $N \rightarrow \infty$
- C. $A_N < B_N < A_N + 1$
- D. $B_N < A_N < B_N + 1$
- E. As $N \rightarrow \infty$, B_N increases to infinity but A_N converges to a finite number

tifr2016 calculus convergence divergence integration non-gatecse

Answer key 

10.7

Functions (1)

10.7.1 Functions: TIFR CSE 2023 | Part A | Question: 7



Suppose $f(x)$ is a polynomial of the form $ax^2 + bx + c$, with a, b, c unknown real numbers. Suppose you are additionally told that $f(1) = 2$ and $f(-1) = 3$. Consider the following four statements.

- (S1) $f(0)$ cannot be determined from the given data.
- (S2) $f(2)$ cannot be determined from the given data.
- (S3) Both $f(0)$ and $f(2)$ can be determined from the given data.
- (S4) $f(0)$ and $f(2)$ satisfy $3f(0) + f(2) = 9$.

Which of the above statements are true?

- A. Statements (S1) and (S2) only
- B. Statements (S1), (S2), and (S4) only
- C. Statement (S3) only
- D. Statements (S3) and (S4) only
- E. All four statements are true

tifr2023 calculus functions

Answer key 

10.8

Limits (8)

10.8.1 Limits: TIFR CSE 2010 | Part A | Question: 7



The limit of $\frac{10^n}{n!}$ as $n \rightarrow \infty$ is.

- A. 0
- B. 1
- C. e
- D. 10
- E. ∞

tifr2010 calculus limits

Answer key 

10.8.2 Limits: TIFR CSE 2011 | Part A | Question: 14



The limit

$$\lim_{x \rightarrow 0} \frac{d}{dx} \frac{\sin^2 x}{x}$$

is

- A. 0 B. 2 C. 1 D. $\frac{1}{2}$
E. None of the above

tifr2011 calculus limits

Answer key 

10.8.3 Limits: TIFR CSE 2011 | Part A | Question: 17



What is

$$\lim_{x \rightarrow 0} \frac{2^x - 1}{x}$$

- A. 0 B. $\log_2(e)$ C. $\log_e(2)$ D. 1
E. None of the above

tifr2011 limits

Answer key 

10.8.4 Limits: TIFR CSE 2012 | Part A | Question: 14



The limit $\lim_{n \rightarrow \infty} (\sqrt{n^2 + n} - n)$ equals.

- A. ∞ B. 1 C. $1/2$ D. 0
E. None of the above

tifr2012 calculus limits

Answer key 

10.8.5 Limits: TIFR CSE 2014 | Part A | Question: 16



Let $x_0 = 1$ and

$$x_{n+1} = \frac{3+2x_n}{3+x_n}, n \geq 0.$$

$x_\infty = \lim_{n \rightarrow \infty} x_n$ is

- A. $(\sqrt{5} - 1)/2$ B. $(\sqrt{5} + 1)/2$
C. $(\sqrt{13} - 1)/2$ D. $(-\sqrt{13} - 1)/2$
E. None of the above

tifr2014 limits

Answer key 

10.8.6 Limits: TIFR CSE 2014 | Part A | Question: 18



We are given a collection of real numbers where a real number $a_i \neq 0$ occurs n_i times. Let the collection be enumerated as $\{x_1, x_2, \dots, x_n\}$ so that $x_1 = x_2 = \dots = x_{n_1} = a_1$ and so on, and $n = \sum_i n_i$ is finite. What is

$$\lim_{k \rightarrow \infty} \left(\sum_{i=1}^n \frac{1}{|x_i|^k} \right)^{-1/k} ?$$

- A. $\max_i (n_i |a_i|)$ B. $\min_i |a_i|$

- C. $\min_i (n_i |a_i|)$
 E. None of the above

D. $\max_i |a_i|$

tifr2014 limits

Answer key 

10.8.7 Limits: TIFR CSE 2019 | Part A | Question: 15



Consider the matrix

$$A = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{3}{4} & \frac{1}{4} \\ 0 & \frac{1}{4} & \frac{3}{4} \end{bmatrix}$$

What is $\lim_{n \rightarrow \infty} A^n$?

- A. $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$
 B. $\begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$
 C. $\begin{bmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \end{bmatrix}$
 D. $\begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$

E. The limit exists, but it is none of the above

tifr2019 engineering-mathematics calculus limits matrix

Answer key 

10.8.8 Limits: TIFR CSE 2021 | Part A | Question: 8



Consider the sequence

$$y_n = \frac{1}{\int_1^n \frac{1}{(1+x/n)^3} dx}$$

for $n = 2, 3, 4, \dots$. Which of the following is TRUE?

- A. The sequence $\{y_n\}$ does not have a limit as $n \rightarrow \infty$.
 B. $y_n \leq 1$ for all $n = 2, 3, 4, \dots$.
 C. $\lim_{n \rightarrow \infty} y_n$ exists and is equal to $6/\pi^2$.
 D. $\lim_{n \rightarrow \infty} y_n$ exists and is equal to 0.
 E. The sequence $\{y_n\}$ first increases and then decreases as n takes values $2, 3, 4, \dots$.

10.9

Maxima Minima (9)

10.9.1 Maxima Minima: TIFR CSE 2010 | Part A | Question: 3



The function $f(x) = 2.5 \log_e (2 + \exp(x^2 - 4x + 5))$ attains a minimum at $x = ?$

- A. 0 B. 1 C. 2 D. 3
E. 4

tifr2010 calculus maxima-minima

Answer key

10.9.2 Maxima Minima: TIFR CSE 2011 | Part A | Question: 4



Consider the problem of maximizing $x^2 - 2x + 5$ such that $0 < x < 2$. The value of x at which the maximum is achieved is:

- A. 0.5 B. 1 C. 1.5 D. 1.75
E. None of the above

tifr2011 calculus maxima-minima

Answer key

10.9.3 Maxima Minima: TIFR CSE 2012 | Part A | Question: 13



The maximum value of the function

$$f(x, y, z) = (x - 1/3)^2 + (y - 1/3)^2 + (z - 1/3)^2$$

subject to the constraints

$$x + y + z = 1, \quad x \geq 0, y \geq 0, z \geq 0$$

is

- A. $1/3$ B. $2/3$ C. 1 D. $4/3$
E. $4/9$

tifr2012 calculus maxima-minima

Answer key

10.9.4 Maxima Minima: TIFR CSE 2013 | Part A | Question: 16



The minimum of the function $f(x) = x \log_e(x)$ over the interval $[\frac{1}{2}, \infty)$ is

- A. 0 B. $-e$ C. $\frac{-\log_e(2)}{2}$ D. $\frac{-1}{e}$
E. None of the above

tifr2013 calculus maxima-minima

Answer key

10.9.5 Maxima Minima: TIFR CSE 2014 | Part A | Question: 9



Solve $\min x^2 + y^2$ subject to

$$\begin{aligned}x + y &\geq 10, \\2x + 3y &\geq 20, \\x &\geq 4, \\y &\geq 4.\end{aligned}$$

- A. 32 B. 50 C. 52 D. 100
- E. None of the above

tifr2014 calculus maxima-minima

Answer key 

10.9.6 Maxima Minima: TIFR CSE 2015 | Part A | Question: 11



Suppose that $f(x)$ is a continuous function such that $0.4 \leq f(x) \leq 0.6$ for $0 \leq x \leq 1$. Which of the following is always true?

- A. $f(0.5) = 0.5$.
 B. There exists x between 0 and 1 such that $f(x) = 0.8x$.
 C. There exists x between 0 and 0.5 such that $f(x) = x$.
 D. $f(0.5) > 0.5$.
 E. None of the above statements are always true.

tifr2015 maxima-minima calculus

Answer key 

10.9.7 Maxima Minima: TIFR CSE 2020 | Part A | Question: 8



Consider a function $f : [0, 1] \rightarrow [0, 1]$ which is twice differentiable in $(0, 1)$. Suppose it has exactly one global maximum and exactly one global minimum inside $(0, 1)$. What can you say about the behaviour of the first derivative f' and second derivative f'' on $(0, 1)$ (give the most precise answer)?

- A. f' is zero at exactly two points, f'' need not be zero anywhere
 B. f' is zero at exactly two points, f'' is zero at exactly one point
 C. f' is zero at at least two points, f'' is zero at exactly one point
 D. f' is zero at at least two points, f'' is zero at at least one point
 E. f' is zero at at least two points, f'' is zero at at least two points

tifr2020 engineering-mathematics calculus maxima-minima

Answer key 

10.9.8 Maxima Minima: TIFR CSE 2020 | Part B | Question: 4



A *clamp* gate is an analog gate parametrized by two real numbers a and b , and denoted as $\text{clamp}_{a,b}$. It takes as input two non-negative real numbers x and y . Its output is defined as

$$\text{clamp}_{a,b}(x, y) = \begin{cases} ax + by & \text{when } ax + by \geq 0, \text{ and} \\ 0 & \text{when } ax + by < 0. \end{cases}$$

Consider circuits composed only of clamp gates, possibly parametrized by different pairs (a, b) of real numbers. How many clamp gates are needed to construct a circuit that on input non-negative reals x and y outputs the maximum of x and y ?

- A. 1
- B. 2
- C. 3
- D. 4
- E. No circuit composed only of clamp gates can compute the max function

tifr2020 calculus maxima-minima

Answer key

10.9.9 Maxima Minima: TIFR CSE 2022 | Part A | Question: 6



Let f be a polynomial of degree $n \geq 3$ all of whose roots are non-positive real numbers. Suppose that $f(1) = 1$. What is the maximum possible value of $f'(1)$?

- A. 1
- B. n
- C. $n + 1$
- D. $\frac{n(n+1)}{2}$
- E. $f'(1)$ can be arbitrarily large given only the constraints in the question

tifr2022 calculus maxima-minima

Answer key

10.10 Polynomials (1)

10.10.1 Polynomials: TIFR CSE 2012 | Part A | Question: 12



For the polynomial $p(x) = 8x^{10} - 7x^3 + x - 1$ consider the following statements (which may be true or false)

- i. It has a root between $[0, 1]$.
- ii. It has a root between $[0, -1]$.
- iii. It has no roots outside $(-1, 1)$.

Which of the above statements are true?

- A. Only (i).
- B. Only (i) and (ii).
- C. Only (i) and (iii).
- D. Only (ii) and (iii).
- E. All of (i), (ii) and (iii).

tifr2012 calculus polynomials

Answer key

Answer Keys

10.0.1	D	10.1.1	E	10.2.1	D	10.3.1	B	10.3.2	D
10.3.3	A	10.4.1	C	10.5.1	C	10.5.2	E	10.5.3	C
10.6.1	C	10.7.1	B	10.8.1	A	10.8.2	C	10.8.3	C
10.8.4	C	10.8.5	C	10.8.6	B	10.8.7	D	10.8.8	D
10.9.1	C	10.9.2	E	10.9.3	B	10.9.4	C	10.9.5	B

10.9.6

B

10.9.7

D

10.9.8

B

10.9.9

B

10.10.1

E



11.1

Determinant (1)

11.1.1 Determinant: TIFR CSE 2022 | Part A | Question: 4



Consider the polynomial $p(x) = x^3 - x^2 + x - 1$. How many *symmetric* matrices with integer entries are there whose characteristic polynomial is p ? (Recall that the *characteristic polynomial* of a square matrix A in the variable x is defined to be the determinant of the matrix $(A - xI)$ where I is the identity matrix.)

- A. 0 B. 1 C. 2 D. 4
- E. Infinitely many

tifr2022 linear-algebra matrix determinant

Answer key

11.2

Eigen Value (4)

11.2.1 Eigen Value: TIFR CSE 2016 | Part A | Question: 6



Which of the following statements about the eigen values of I_n , the $n \times n$ identity matrix (over complex numbers), is true?

- A. The eigen values are $1, \omega, \omega^2, \dots, \omega^{n-1}$, where ω is a primitive n -th root of unity
- B. The only eigen value is -1
- C. Both 0 and 1 are eigen values, but there are no other eigen values
- D. The eigen values are $1, 1/2, 1/3, \dots, 1/n$
- E. The only eigen value is 1

tifr2016 linear-algebra eigen-value

Answer key

11.2.2 Eigen Value: TIFR CSE 2019 | Part A | Question: 3



A is $n \times n$ square matrix for which the entries in every row sum to 1. Consider the following statements:

- i. The column vector $[1, 1, \dots, 1]^T$ is an eigen vector of A .
- ii. $\det(A - I) = 0$.
- iii. $\det(A) = 0$.

Which of the above statements must be **TRUE**?

- A. Only (i) B. Only (ii) C. Only (i) and (ii) D. Only (i) and (iii)
- E. (i), (ii) and (iii)

tifr2019 engineering-mathematics linear-algebra eigen-value

Answer key

11.2.3 Eigen Value: TIFR CSE 2025 | Part A | Question: 6



Consider the following matrix:

- C. Sum of the diagonal elements is positive.
E. None of the above.

tifr2010 linear-algebra matrix

Answer key

- D. $\det(A)$ is positive.

11.3.3 Matrix: TIFR CSE 2012 | Part B | Question: 12



Let A be a matrix such that $A^k = 0$. What is the inverse of $I - A$?

- A. 0
C. A
E. Inverse is not guaranteed to exist.
B. I
D. $1 + A + A^2 + \dots + A^{k-1}$

tifr2012 linear-algebra matrix

Answer key

11.3.4 Matrix: TIFR CSE 2013 | Part B | Question: 3



How many 4×4 matrices with entries from 0, 1 have odd determinant?

Hint: Use modulo 2 arithmetic.

- A. 20160
B. 32767
C. 49152
D. 57343
E. 65520

tifr2013 linear-algebra matrix

Answer key

11.3.5 Matrix: TIFR CSE 2015 | Part A | Question: 14



Consider the following 3×3 matrices.

$$M_1 = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

$$M_2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

How many 0 – 1 column vectors of the form

$$X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

are there such that $M_1 X = M_2 X$ (modulo 2)? (modulo 2 means all operations are done modulo 2, i.e., $3 = 1$ (modulo 2), $4 = 0$ (modulo 2)).

- A. None
B. Two
C. Three
D. Four
E. Eight

tifr2015 matrix

Answer key

11.3.6 Matrix: TIFR CSE 2018 | Part A | Question: 12



An $n \times n$ matrix M with real entries is said to be positive definite if for every non-zero n -dimensional vector x with real entries, we have $x^T M x > 0$. Let A and B be symmetric, positive definite matrices of size $n \times n$ with real entries.

Consider the following matrices, where I denotes the $n \times n$ identity matrix:

1. $A + B$
2. ABA
3. $A^2 + I$

Which of the above matrices must be positive definite?

- | | |
|--|---|
| A. Only (2) | B. Only (3) |
| C. Only (1) and (3) | D. None of the above matrices are positive definite |
| E. All of the above matrices are positive definite | |

tifr2018 matrix linear-algebra

Answer key

11.3.7 Matrix: TIFR CSE 2018 | Part A | Question: 14



Let A be an $n \times n$ invertible matrix with real entries whose row sums are all equal to c . Consider the following statements:

1. Every row in the matrix $2A$ sums to $2c$.
2. Every row in the matrix A^2 sums to c^2 .
3. Every row in the matrix A^{-1} sums to c^{-1} .

Which of the following is **TRUE**?

- A. none of the statements (1), (2), (3) is correct
- B. statement (1) is correct but not necessarily statements (2) or (3)
- C. statement (2) is correct but not necessarily statements (1) or (3)
- D. statement (1) and (2) are correct but not necessarily statement (3)
- E. all the three statements (1), (2), and (3) are correct

tifr2018 matrix linear-algebra

Answer key

11.3.8 Matrix: TIFR CSE 2020 | Part A | Question: 5



Let A be an $n \times n$ invertible matrix with real entries whose column sums are all equal to 1. Consider the following statements:

1. Every column in the matrix A^2 sums to 2
2. Every column in the matrix A^3 sums to 3
3. Every column in the matrix A^{-1} sums to 1

Which of the following is TRUE?

- A. none of the statements (1), (2), (3) is correct
- B. statement (1) is correct but not statements (2) or (3)
- C. statement (2) is correct but not statements (1) or (3)

- D. statement (3) is correct but not statements (1) or (2)
 E. all the 3 statements (1), (2), and (3) are correct

tifr2020 engineering-mathematics linear-algebra matrix

Answer key

11.3.9 Matrix: TIFR CSE 2021 | Part B | Question: 8



Let A and B be two matrices of size $n \times n$ and with real-valued entries. Consider the following statements.

1. If $AB = B$, then A must be the identity matrix.
2. If A is an idempotent (i.e. $A^2 = A$) nonsingular matrix, then A must be the identity matrix.
3. If $A^{-1} = A$, then A must be the identity matrix.

Which of the above statements **MUST** be true of A ?

- A. 1, 2 and 3 B. Only 2 and 3 C. Only 1 and 2 D. Only 1 and 3
 E. Only 2

tifr2021 linear-algebra matrix

Answer key

11.3.10 Matrix: TIFR CSE 2023 | Part A | Question: 15



Consider the $n \times n$ matrix M defined as follows:

$$M = \begin{pmatrix} 1 & 2 & \dots & n \\ n+1 & n+2 & \dots & 2n \\ 2n+1 & 2n+2 & \dots & 3n \\ \vdots & \vdots & \dots & \vdots \\ (n-1)n+1 & (n-1)n+2 & \dots & n^2 \end{pmatrix}.$$

Let $M_{i,j}$ denote the entry present in the i -th row and the j -th column of M for each $1 \leq i, j \leq n$. Define a set S as

$$S = \{M_{i_1, j_1} + M_{i_2, j_2} + \dots + M_{i_n, j_n} \mid (i_1, \dots, i_n) \text{ and } (j_1, \dots, j_n) \text{ are permutations of } [n]\}.$$

are permutations of $[n]$.

(Here, $[n]$ denotes the set $\{1, 2, \dots, n\}$ of positive integers from 1 to n .) How many elements does S have?

- A. 1 B. $\binom{n}{2}$ C. $n!$ D. $(n!)^2$
 E. n

tifr2023 linear-algebra matrix

Answer key

11.3.11 Matrix: TIFR CSE 2023 | Part A | Question: 5



Suppose A is a 2×2 matrix such that the sum of the principal diagonal entries of A is 10 and the sum of the principal diagonal entries of A^2 is 20. (For any 2×2 matrix B , the principal diagonal entries of B are the entries $B_{1,1}$ and $B_{2,2}$.) What is $\det(A)$?

- A. 0
- B. 40
- C. 80
- D. Nonzero, but cannot be uniquely determined from the above data.
- E. Cannot be uniquely determined from the above data, and could also be zero.

tifr2023 linear-algebra matrix

Answer key

11.3.12 Matrix: TIFR CSE 2024 | Part A | Question: 10



Let M be a 3×3 matrix over the real numbers such that $M^T M = I$. Consider the following statements.

- i. There exists a non-zero vector $x \in \mathbb{R}^3$ such that $Mx = 0$.
- ii. There exist two non-zero vectors $x, y \in \mathbb{R}^3$ such that the angle between them and that between Mx and My are different.
- iii. There exists a non-zero vector x such that its length is different from that of Mx .

Which of the above statements is/are true?

- A. Only (i).
- B. Only (ii).
- C. Only (iii).
- D. All three statements.
- E. None of the three statements.

tifr2024 matrix linear-algebra

11.3.13 Matrix: TIFR CSE 2024 | Part A | Question: 14



Let A and B be two $n \times n$ invertible matrices with real entries such that every row in A sums to 1 and every row in B sums to 2. Consider the following three statements:

- i. Every row in the matrix AB sums to 2.
- ii. Every row in the matrix $A^{-1}B$ sums to 2.
- iii. Every row in the matrix $A^{-1}B^{-1}$ sums to $\frac{1}{2}$.

Which of the above statements is/are true?

- A. None of the statements (i), (ii), (iii) is true.
- B. All the three statements (i), (ii), and (iii) are true.
- C. Statement (i) is true but not necessarily statements (ii) or (iii).
- D. Statements (i) and (ii) are true but not necessarily statement (iii).
- E. Statements (i) and (iii) are true but not necessarily statement (ii).

tifr2024 matrix linear-algebra

11.4.1 Polynomials: TIFR CSE 2024 | Part A | Question: 11



Consider the following sequence of polynomials with real coefficients.

$$\begin{aligned} P_0(x) &= 1 \\ P_1(x) &= 2x \\ P_{n+1}(x) &= 2xP_n(x) - P_{n-1}(x), \text{ for all natural numbers } n \geq 1. \end{aligned}$$

What is the dimension of the linear span of the set

$$\{P_0(x), P_1(x), P_2(x), P_3(x), P_4(x)\}$$

in the vector space of polynomials in variable x with real coefficients?

- A. 1 B. 2 C. 3 D. 4
E. 5

tifr2024 linear-algebra recurrence-relation polynomials

11.5

Rank of Matrix (3)

11.5.1 Rank of Matrix: TIFR CSE 2020 | Part A | Question: 2



Let M be a real $n \times n$ matrix such that for *every* non-zero vector $x \in \mathbb{R}^n$, we have $x^T M x > 0$. Then

- A. Such an M cannot exist B. Such M s exist and their rank is always n
C. Such M s exist, but their eigenvalues are always real D. No eigenvalue of any such M can be real
E. None of the above

tifr2020 engineering-mathematics linear-algebra rank-of-matrix eigen-value

Answer key

11.5.2 Rank of Matrix: TIFR CSE 2021 | Part A | Question: 3



Let M be an $n \times m$ real matrix. Consider the following:

- Let k_1 be the smallest number such that M can be factorized as $A \cdot B$, where A is an $n \times k_1$ and B is a $k_1 \times m$ matrix.
- Let k_2 be the smallest number such that $M = \sum_{i=1}^{k_2} u_i v_i$, where each u_i is an $n \times 1$ matrix and each v_i is an $1 \times m$ matrix.
- Let k_3 be the column-rank of M .

Which of the following statements is TRUE?

- A. $k_1 < k_2 < k_3$ B. $k_1 < k_3 < k_2$
C. $k_2 = k_3 < k_1$ D. $k_1 = k_2 = k_3$
E. No general relationship exists among k_1, k_2 and k_3

tifr2021 linear-algebra matrix rank-of-matrix

11.7.1 Vector Space: TIFR CSE 2010 | Part A | Question: 11



The length of a vector $X = (x_1, \dots, x_n)$ is defined as

$$\|X\| = \sqrt{\sum_{i=1}^n x_i^2}$$

Given two vectors $X = (x_1, \dots, x_n)$ and $Y = (y_1, \dots, y_n)$, which of the following measures of discrepancy between X and Y is insensitive to the length of the vectors?

- A. $\|X - Y\|$
- B. $\|X - Y\| / \|x\| \|y\|$
- C. $\|X\| - \|Y\|$
- D. $\left\| \frac{X}{\|X\|} - \frac{Y}{\|Y\|} \right\|$
- E. None of the above

tifr2010 linear-algebra vector-space

Answer key

11.7.2 Vector Space: TIFR CSE 2016 | Part A | Question: 3



Consider the following set of $3n$ linear equations in $3n$ variables:

$$\begin{array}{llll} x_1 - x_2 = 0 & x_4 - x_5 = 0 & \dots & x_{3n-2} - x_{3n-1} = 0 \\ x_2 - x_3 = 0 & x_5 - x_6 = 0 & & x_{3n-1} - x_{3n} = 0 \\ x_1 - x_3 = 0 & x_4 - x_6 = 0 & & x_{3n-2} = x_{3n} = 0 \end{array}$$

Let $S \subseteq \mathbb{R}^{3n}$ be the set of solutions to this set of equations. Then,

- A. S is empty
- B. S is a subspace of $\mathbb{R}^{3n}$ of dimension 1
- C. S is a subspace of $\mathbb{R}^{3n}$ of dimension n
- D. S is a subspace of $\mathbb{R}^{3n}$ of dimension $n - 1$
- E. S has exactly n elements

tifr2016 linear-algebra vector-space non-gatecse

Answer key

11.7.3 Vector Space: TIFR CSE 2016 | Part B | Question: 6



A subset X of $\mathbb{R}^n$ is convex if for all $x, y \in X$ and all $\lambda \in (0, 1)$, we have $\lambda x + (1 - \lambda)y \in X$. If X is a convex set, which of the following statements is necessarily TRUE?

- A. For every $x \in X$ there exist $y, z \in X - \{x\}$ and $\lambda \in (0, 1)$ so that $x = \lambda y + (1 - \lambda)z$
- B. If $x, y \in X$ and $\lambda \geq 0$, then $\lambda x + (1 - \lambda)y \in X$
- C. If $x_1, \dots, x_n \in X (n \geq 1)$, then $(x_1 + \dots + x_n)/n \in X$
- D. If $x \in X$, then $\lambda x \in X$ for all scalars λ
- E. If $x, y \in X$, then $x - y \in X$

tifr2016 linear-algebra vector-space non-gatecse

11.7.4 Vector Space: TIFR CSE 2017 | Part A | Question: 2



For vectors x, y in $\mathbb{R}^n$, define the inner product $\langle x, y \rangle = \sum_{i=1}^n x_i y_i$, and the length of x to be $\|x\| = \sqrt{\langle x, x \rangle}$. Let a, b be two vectors in $\mathbb{R}^n$ so that $\|b\| = 1$. Consider the following statements:

- i. $\langle a, b \rangle \leq \|b\|$
- ii. $\langle a, b \rangle \leq \|a\|$
- iii. $\langle a, b \rangle = \|a\| \|b\|$
- iv. $\langle a, b \rangle \geq \|b\|$
- v. $\langle a, b \rangle \geq \|a\|$

Which of the above statements must be TRUE of a, b ? Choose from the following options.

- A. ii only
- B. i and ii
- C. iii only
- D. iv only
- E. iv and v

tifr2017 linear-algebra vector-space

Answer key

11.7.5 Vector Space: TIFR CSE 2020 | Part A | Question: 3



Let $d \geq 4$ and fix $w \in \mathbb{R}$. Let

$$S = \{a = (a_0, a_1, \dots, a_d) \in \mathbb{R}^{d+1} \mid f_a(w) = 0 \text{ and } f'_a(w) = 0\},$$

where the polynomial function $f_a(x)$ is defined as $f_a(x) := \sum_{i=0}^d a_i x^i$ and $f'_a(w)$ denotes the derivative of $f_a(x)$ with respect to x , evaluated at w . Then,

- A. S is finite or infinite depending on the value of α
- B. S is a 2-dimensional vector subspace of $\mathbb{R}^{d+1}$
- C. S is a d -dimensional vector subspace of $\mathbb{R}^{d+1}$
- D. S is a $(d-1)$ -dimensional vector subspace of $\mathbb{R}^{d+1}$
- E. None of the other options

tifr2020 engineering-mathematics linear-algebra vector-space

11.7.6 Vector Space: TIFR CSE 2022 | Part B | Question: 15



Let $\mathbb{R}$ denote the set of real numbers. Let $d \geq 4$ and $\alpha \in \mathbb{R}$. Let

$$S = \left\{ (a_0, a_1, \dots, a_d) \in \mathbb{R}^{d+1} : \sum_{i=0}^d a_i \alpha^i = 0 \text{ and } \sum_{i=0}^d i a_i \alpha^{i-1} = 0 \right\}.$$

Then,

- A. S is finite or infinite depending on the value of α
- B. S is a 2-dimensional vector subspace of $\mathbb{R}^{d+1}$
- C. S is a d -dimensional vector subspace of $\mathbb{R}^{d+1}$
- D. S is a $(d-1)$ -dimensional vector subspace of $\mathbb{R}^{d+1}$
- E. For each $(a_0, a_1, \dots, a_d) \in S$, the function

$$x \mapsto \sum_{i=0}^d a_i x^i$$

has a local optimum at α

tifr2022 linear-algebra vector-space

11.7.7 Vector Space: TIFR CSE 2023 | Part A | Question: 1



Let A be a symmetric 3×3 matrix with real entries. Let u and v be non-zero vectors with real entries such that $Au = 2u$ and $Av = 3v$. From the set of values $\{0, 1, -1\}$, which values can the inner product $u^T v$ take?

- A. 0 only
- B. 1 only
- C. -1 only
- D. All of the values 0, 1 and -1
- E. None of the values 0, 1 and -1

tifr2023 linear-algebra vector-space

11.7.8 Vector Space: TIFR CSE 2023 | Part A | Question: 3



A is an $n \times n$ matrix with real-valued entries. Further, there exists a vector $x \neq 0$ such that $Ax = 0$. Now consider a given vector b in $\mathbb{R}^n$. How many possible vectors z exist, so that $Az = b$?

- A. 0
- B. 1
- C. $n - 1$
- D. n
- E. Either 0 or infinite

tifr2023 linear-algebra vector-space

Answer key

11.7.9 Vector Space: TIFR CSE 2023 | Part B | Question: 5



Consider *unit* vectors $\mathbf{a}$ and $\mathbf{b}$ in $\mathbb{R}^n$. Let $\mathbf{w}$ be an arbitrary vector in $\mathbb{R}^n$ and η be a positive real number such that

$$\mathbf{a}^T \mathbf{b} \geq \eta > 0 \geq \mathbf{w}^T \mathbf{b}.$$

Define $\mathbf{z} = \mathbf{w} + \mathbf{b}$. Consider the following statements.

- (S1) $\mathbf{z}^T \mathbf{a} \geq \mathbf{a}^T \mathbf{w} + \eta$
- (S2) $\|\mathbf{z}\|^2 \leq \|\mathbf{w}\|^2 + 1$
- (S3) $\mathbf{z}^T \mathbf{b} \geq 0$

Choose the correct option from those below.

- A. Statements (S1), (S2), and (S3) are all true.
- B. Statements (S1) and (S2) must be true, but statement (S3) must be false.
- C. Statements (S1) and (S2) must be true, but statement (S3) may be either true or false.
- D. Statements (S1) and (S3) must be true, but statement (S2) may be false.
- E. Statement (S1) may be false.

tifr2023 linear-algebra vector-space

Answer key

Answer Keys

11.1.1	A
11.3.1	D
11.3.6	E
11.3.11	B
11.5.2	D
11.7.3	C
11.7.8	E

11.2.1	E
11.3.2	E
11.3.7	E
11.3.12	E
11.5.3	D
11.7.4	A
11.7.9	C

11.2.2	C
11.3.3	D
11.3.8	D
11.3.13	B
11.6.1	C
11.7.5	D

11.2.3	D
11.3.4	A
11.3.9	E
11.4.1	E
11.7.1	D
11.7.6	D

11.2.4	C
11.3.5	B
11.3.10	A
11.5.1	B
11.7.2	C
11.7.7	A



12.1

Bayes Theorem (1)

12.1.1 Bayes Theorem: TIFR CSE 2025 | Part A | Question: 4



There are two bags, one with 10 red balls and 90 blue balls, and another with 90 red balls and 10 blue balls. You pick one of the two bags uniformly at random and draw a random ball from it, and it turns out to be red. You throw that ball away, and draw another from the same bag. Let p be the probability that this ball is also red.

Which of the following is true?

A. $0 \leq p \leq 0.33$

B. $0.33 \leq p < 0.50$

C. $p = 0.5$

D. $0.50 < p < 0.66$

E. $0.66 \leq p \leq 1$

tifr2025 probability bayes-theorem conditional-probability random-variable

Answer key

12.2

Binomial Distribution (7)

12.2.1 Binomial Distribution: TIFR CSE 2010 | Part A | Question: 6



Given 10 tosses of a coin with probability of head = .4 = (1 - the probability of tail), the probability of at least one head is?

A. $(.4)^{10}$

B. $1 - (.4)^{10}$

C. $1 - (.6)^{10}$

D. $(.6)^{10}$

E. $10(.4)(.6)^9$

tifr2010 probability binomial-distribution

Answer key

12.2.2 Binomial Distribution: TIFR CSE 2010 | Part B | Question: 38



Suppose three coins are lying on a table, two of them with heads facing up and one with tails facing up. One coin is chosen at random and flipped. What is the probability that after the flip the majority of the coins (i.e., at least two of them) will have heads facing up?

A. $\left(\frac{1}{3}\right)$

B. $\left(\frac{1}{8}\right)$

C. $\left(\frac{1}{4}\right)$

D. $\left(\frac{1}{4} + \frac{1}{8}\right)$

E. $\left(\frac{2}{3}\right)$

tifr2010 probability binomial-distribution

Answer key

12.2.3 Binomial Distribution: TIFR CSE 2011 | Part A | Question: 3



The probability of three consecutive heads in four tosses of a fair coin is

A. $\left(\frac{1}{4}\right)$

B. $\left(\frac{1}{8}\right)$

C. $\left(\frac{1}{16}\right)$

D. $\left(\frac{3}{16}\right)$

E. None of the above

tifr2011 probability binomial-distribution

Answer key

12.2.4 Binomial Distribution: TIFR CSE 2012 | Part A | Question: 17



A spider is at the bottom of a cliff, and is n inches from the top. Every step it takes brings it one inch closer to the top with probability $1/3$, and one inch away from the top with probability $2/3$, unless it is at the bottom in which case, it always gets one inch closer. What is the expected number of steps for the spider to reach the top as a function of n ?

- A. It will never reach the top.
- B. Linear in n .
- C. Polynomial in n .
- D. Exponential in n .
- E. Double exponential in n .

tifr2012 probability binomial-distribution

Answer key

12.2.5 Binomial Distribution: TIFR CSE 2013 | Part A | Question: 14



An unbiased die is thrown n times. The probability that the product of numbers would be even is

- A. $\frac{1}{(2n)}$
- B. $\frac{1}{[(6n)!]}$
- C. $1 - 6^{-n}$
- D. 6^{-n}
- E. None of the above

tifr2013 probability binomial-distribution

Answer key

12.2.6 Binomial Distribution: TIFR CSE 2013 | Part A | Question: 4



A biased coin is tossed repeatedly. Assume that the outcomes of different tosses are independent and probability of heads is $\frac{2}{3}$ in each toss. What is the probability of obtaining an even number of heads in 5 tosses, zero being treated as an even number?

- A. $\left(\frac{121}{243}\right)$
- B. $\left(\frac{122}{243}\right)$
- C. $\left(\frac{124}{243}\right)$
- D. $\left(\frac{125}{243}\right)$
- E. $\left(\frac{128}{243}\right)$

tifr2013 probability binomial-distribution

Answer key

12.2.7 Binomial Distribution: TIFR CSE 2018 | Part A | Question: 10



Let C be a biased coin such that the probability of a head turning up is p . Let p_n denote the probability that an odd number of heads occurs after n tosses for $n \in \{0, 1, 2, \dots\}$. Then which of the following is TRUE ?

- A. $p_n = \frac{1}{2}$ for all $n \in \{0, 1, 2, \dots\}$.
- B. $p_n = (1 - p)(1 - p_{n-1}) + p \cdot p_{n-1}$ for $n \geq 1$ and $p_0 = 0$.
- C. $p_n = \sum_{i=1}^n p(1 - 2p)^{i-1}$ for $n \geq 1$.
- D. If $p = \frac{1}{2}$, then $p_n = \frac{1}{2}$ for all $n \in \{0, 1, 2, \dots\}$.
- E. $p_n = 1$ if n is odd and 0 otherwise.

tifr2018 probability binomial-distribution

Answer key

12.3.1 Binomial Theorem: TIFR CSE 2021 | Part A | Question: 10



Lavanya and Ketak each flip a fair coin (i.e., both heads and tails have equal probability of appearing) n times. What is the probability that Lavanya sees more heads than Ketak?

In the following, the binomial coefficient $\binom{n}{k}$ counts the number of k -element subsets of an n -element set.

- A. $\frac{1}{2}$
- B. $\frac{1}{2} \left(1 - \sum_{i=0}^n \frac{\binom{n}{i}}{2^{2n}} \right)$
- C. $\frac{1}{2} \left(1 - \sum_{i=0}^n \frac{\binom{n}{i}}{2^{2n}} \right)$
- D. $\frac{1}{2} \left(1 - \frac{1}{2^{2n}} \right)$
- E. $\sum_{i=0}^n \frac{\binom{n}{i}}{2^n}$

tifr2021 probability binomial-theorem

Answer key

12.4 Birthday Problem (1)

12.4.1 Birthday Problem: TIFR CSE 2025 | Part A | Question: 2



Assume that in a room there are k persons. We know that $k = 3$ with probability $\frac{1}{4}$, $k = 5$ with probability $\frac{1}{4}$, and $k = 10$ with probability $\frac{1}{2}$. What is the probability that at least two persons in the room have their birthday on the same day of the week? Assume that all days of the week are equally likely.

- A. $\frac{1}{2} \left(1 + \frac{15 \times 61}{49^2} \right)$
- B. $1 - \frac{1}{2} \left(\frac{30}{49} + \frac{360}{49^2} \right)$
- C. $1 - \frac{15}{98} \left(1 + \frac{12}{49} \right)$
- D. $1 - \frac{15 \times 61}{49^2}$
- E. $\frac{4}{7}$

tifr2025 probability birthday-problem conditional-probability

Answer key

12.5 Conditional Probability (15)

12.5.1 Conditional Probability: TIFR CSE 2010 | Part A | Question: 10



A drawer contains 2 Blue, 4 Red and 2 Yellow balls. No two balls have the same radius. If two balls are randomly selected from the drawer, what is the probability that they will be of the same colour?

- A. $\left(\frac{2}{7} \right)$
- B. $\left(\frac{2}{5} \right)$
- C. $\left(\frac{3}{7} \right)$
- D. $\left(\frac{1}{2} \right)$
- E. $\left(\frac{3}{5} \right)$

tifr2010 probability conditional-probability

Answer key

12.5.2 Conditional Probability: TIFR CSE 2010 | Part A | Question: 19, TIFR CSE 2014 | Part A | Question: 6



Karan tells truth with probability $\frac{1}{3}$ and lies with probability $\frac{2}{3}$. Independently, Arjun tells truth with probability $\frac{3}{4}$ and lies with probability $\frac{1}{4}$. Both watch a cricket match. Arjun tells you that India won, Karan tells you that India lost. What probability will you assign to India's win?

- A. $\left(\frac{1}{2}\right)$ B. $\left(\frac{2}{3}\right)$ C. $\left(\frac{3}{4}\right)$ D. $\left(\frac{5}{6}\right)$
E. $\left(\frac{6}{7}\right)$

tifr2010 probability conditional-probability tifr2014

Answer key

12.5.3 Conditional Probability: TIFR CSE 2011 | Part A | Question: 9



You have to play three games with opponents A and B in a specified sequence. You win the series if you win two consecutive games. A is a stronger player than B . Which sequence maximizes your chance of winning the series?

- A. AAB B. ABA C. BAB D. BAA
E. All are the same.

tifr2011 probability conditional-probability

Answer key

12.5.4 Conditional Probability: TIFR CSE 2012 | Part A | Question: 1



Amar and Akbar both tell the truth with probability $\frac{3}{4}$ and lie with probability $\frac{1}{4}$. Amar watches a test match and talks to Akbar about the outcome. Akbar, in turn, tells Anthony, "Amar told me that India won". What probability should Anthony assign to India's win?

- A. $\left(\frac{9}{16}\right)$ B. $\left(\frac{6}{16}\right)$ C. $\left(\frac{7}{16}\right)$ D. $\left(\frac{10}{16}\right)$
E. None of the above

tifr2012 probability conditional-probability

Answer key

12.5.5 Conditional Probability: TIFR CSE 2012 | Part A | Question: 20



There are 1000 balls in a bag, of which 900 are black and 100 are white. I randomly draw 100 balls from the bag. What is the probability that the 101st ball will be black?

- A. $9/10$ B. More than $9/10$ but less than 1.
C. Less than $9/10$ but more than 0. D. 0
E. 1

tifr2012 probability conditional-probability

Answer key

12.5.6 Conditional Probability: TIFR CSE 2013 | Part A | Question: 6



You are lost in the National park of Kabrastan. The park population consists of tourists and Kabrastanis. Tourists comprise two-thirds of the population the park and give a correct answer to requests for directions with probability $\frac{3}{4}$. The air of Kabrastan has an amnesaic quality, however, and so the answers to repeated questions to tourists are independent, even if the question and the person are the same. If you ask a Kabrastani for directions, the answer is always wrong.

Suppose you ask a randomly chosen passer-by whether the exit from the park is East or West. The answer is East. You then ask the same person again, and the reply is again East. What is the probability of East being correct?

- A. $\left(\frac{1}{4}\right)$ B. $\left(\frac{1}{3}\right)$ C. $\left(\frac{1}{2}\right)$ D. $\left(\frac{2}{3}\right)$
 E. $\left(\frac{3}{4}\right)$

tifr2013 probability conditional-probability

Answer key

12.5.7 Conditional Probability: TIFR CSE 2013 | Part B | Question: 10



Let m, n be positive integers with m a power of 2. Let $s = 100n^2 \log m$. Suppose $S_1, S_2, \dots, S_m$ are subsets of $1, 2, \dots, s$ such that $|S_i| = 10n \log m$ and $|S_i \cap S_j| \leq \log m$ for all $1 \leq i < j \leq m$. Such a collection of sets $S_1, \dots, S_m$ is an example of a so-called Nisan-Wigderson design. We now consider the set membership problem, where we have to store an arbitrary subset $T \subseteq \{1, 2, \dots, m\}$, $|T| = n$ as an array A of s bits so that given any integer x , $1 \leq x \leq m$, we can discover whether $x \in T$ by reading only one bit of A . Consider the following strategy to solve this problem. Array A is initialized to all zeroes. Given the set T to be stored, we put a one in all the locations of A indexed by the union $\bigcup_{t \in T} S_t$. Now, given the integer x , we read a random location in A from S_x and declare that $x \in T$ if the bit in that location is one. This strategy gives the correct answer with probability

- A. 1 if $x \in T$ and at most 0.1 if $x \notin T$.
 B. At least 0.9 if $x \in T$ and at most 0.1 if $x \notin T$.
 C. At least 0.9 if $x \in T$ and at least 0.9 if $x \notin T$.
 D. 1 if $x \in T$ and at least 0.9 if $x \notin T$.
 E. At least 0.9 if $x \in T$ and 1 if $x \notin T$.

tifr2013 probability conditional-probability

Answer key

12.5.8 Conditional Probability: TIFR CSE 2015 | Part A | Question: 1



Consider a 6-sided die with all sides not necessarily equally likely such that probability of an even number is $P(\{2, 4, 6\}) = \frac{1}{2}$, probability of a multiple of 3 is $P(\{3, 6\}) = \frac{1}{3}$ and probability of 1 is $P(\{1\}) = \frac{1}{6}$. Given the above conditions, choose the strongest (most stringent) condition of the following that must always hold about $P(\{5\})$, the probability of 5.

- A. $P(\{5\}) = \frac{1}{6}$ B. $P(\{5\}) \geq \frac{1}{6}$
 C. $P(\{5\}) \leq \frac{1}{6}$ D. $P(\{5\}) \leq \frac{1}{3}$

E. None of the above

tifr2015 probability conditional-probability

Answer key

12.5.9 Conditional Probability: TIFR CSE 2016 | Part A | Question: 12



There are two rocks A and B , located close to each other, in a lily pond. There is a frog that jumps randomly between the two rocks at time $t = 0, 1, 2, \dots$. The location of the frog is determined as follows. Initially, at time $t = 0$, the frog is at A . From then on, the frog's location is determined as follows. If the frog is at A at time t , then at time $t + 1$, with probability $2/3$ it jumps to B and with probability $1/3$, it jumps on the spot and stays at A . If the frog is at B at time t , then at time $t + 1$, with probability $1/2$ it jumps to A and with probability $1/2$ it jumps on the spot and stays at B . What is the probability that the frog is at B at time 3 (just after its third jump)?

A. $\frac{1}{2}$
E. $\frac{2}{3}$

B. $\frac{31}{54}$

C. $\frac{14}{27}$

D. $\frac{61}{108}$

tifr2016 probability conditional-probability

Answer key

12.5.10 Conditional Probability: TIFR CSE 2018 | Part A | Question: 13



A hacker knows that the password to the TIFR server is 10-letter string consisting of lower-case letters from the English alphabet. He guesses a set of 5 distinct 10-letter strings (with lower-case letters) uniformly at random. What is the probability that one of the guesses of the hacker is correct password?

A. $\frac{5}{(26)^{10}}$

B. $1 - \left(1 - \frac{1}{(26)^{10}}\right)^5$

C. $1 - \left\{ \left(\frac{(26)^{10}-1}{(26)^{10}}\right) \left(\frac{(26)^{10}-2}{(26)^{10}}\right) \left(\frac{(26)^{10}-3}{(26)^{10}}\right) \left(\frac{(26)^{10}-4}{(26)^{10}}\right) \left(\frac{(26)^{10}-5}{(26)^{10}}\right) \right\}$

D. $\frac{1}{(26)^{10}}$

E. None of the above

tifr2018 probability conditional-probability

Answer key

12.5.11 Conditional Probability: TIFR CSE 2018 | Part A | Question: 15



Suppose a box contains 20 balls: each ball has a distinct number in $\{1, \dots, 20\}$ written on it. We pick 10 balls (without replacement) uniformly at random and throw them out of the box. Then we check if the ball with number "1" on it is present in the box. If it is present, then we throw it out of the box; else we pick a ball from the box uniformly at random and throw it out of the box.

What is the probability that the ball with number "2" on it is present in the box?

A. $9/20$

B. $9/19$

C. $1/2$

D. $10/19$

E. None of the above

tifr2018 probability conditional-probability

Answer key

12.5.12 Conditional Probability: TIFR CSE 2019 | Part A | Question: 14



A drawer contains 9 pens, of which 3 are red, 3 are blue, and 3 are green. The nine pens are drawn from the drawer one at a time (without replacement) such that each pen is drawn with equal probability from the remaining pens in the drawer. What is the probability that two red pens are drawn in succession?

- A. $7/12$ B. $1/6$ C. $1/12$ D. $1/81$
- E. None of the above

tifr2019 engineering-mathematics probability conditional-probability

Answer key

12.5.13 Conditional Probability: TIFR CSE 2020 | Part A | Question: 11



Suppose we toss $m = 3$ labelled balls into $n = 3$ numbered bins. Let A be the event that the first bin is empty while B be the event that the second bin is empty. $P(A)$ and $P(B)$ denote their respective probabilities. Which of the following is true?

- A. $P(A) > P(B)$ B. $P(A) = \frac{1}{27}$
- C. $P(A) > P(A | B)$ D. $P(A) < P(A | B)$
- E. None of the above

tifr2020 probability conditional-probability

Answer key

12.5.14 Conditional Probability: TIFR CSE 2022 | Part A | Question: 10



Consider a bag containing colored marbles. There are n marbles in the bag such that there is exactly one pair of marbles of color i for each $i \in \{1, \dots, m\}$ and the rest of the marbles are of distinct colors (different from colors $\{1, \dots, m\}$). You draw two marbles uniformly at random (without replacement). What is the probability that both marbles are of same color?

- A. $\frac{m}{n}$ B. $\frac{2m}{n}$ C. $\frac{2m}{n(n-1)}$ D. $\frac{2m}{n^2}$
- E. $\frac{m}{n(n-1)}$

tifr2022 probability conditional-probability

Answer key

12.5.15 Conditional Probability: TIFR CSE 2022 | Part A | Question: 12



Alice plays the following game on a math show. There are 7 boxes and identical prizes are hidden inside 3 of the boxes. Alice is asked to choose a box where a prize might be. She chooses a box uniformly at random. From the unchosen boxes which do not have a prize, the host opens an arbitrary box and shows Alice that there is no prize in it. The host then allows Alice to change her choice if she so wishes. Alice chooses a box uniformly at random from the other 5 boxes (other than the one she chose first and the one opened by the host). Her probability of winning the prize is

- A. $3/7$ B. $1/2$ C. $17/30$ D. $18/35$
- E. $9/19$

tifr2022 probability conditional-probability

Answer key

12.6.1 Expectation: TIFR CSE 2011 | Part A | Question: 6



Assume that you are flipping a fair coin, i.e. probability of heads or tails is equal. Then the expected number of coin flips required to obtain two consecutive heads for the first time is.

- a. 4 b. 3 c. 6 d. 10
e. 5

tifr2011 probability expectation

Answer key

12.6.2 Expectation: TIFR CSE 2012 | Part B | Question: 7



A bag contains 16 balls of the following colors: 8 red, 4 blue, 2 green, 1 black, and 1 white. Anisha picks a ball randomly from the bag, and messages Babu its color using a string of zeros and ones. She replaces the ball in the bag, and repeats this experiment, many times. What is the minimum expected length of the message she has to convey to Babu per experiment?

- A. $\frac{3}{2}$ B. $\log 5$ C. $\frac{15}{8}$ D. $\frac{31}{16}$
E. 2

tifr2012 probability expectation

Answer key

12.6.3 Expectation: TIFR CSE 2014 | Part A | Question: 17



A fair dice (with faces numbered $1, \dots, 6$) is independently rolled repeatedly. Let X denote the number of rolls till an even number is seen and let Y denote the number of rolls till 3 is seen. Evaluate $E(Y|X=2)$.

- A. $6\frac{5}{6}$ B. 6 C. $5\frac{1}{2}$ D. $6\frac{1}{3}$
E. $5\frac{2}{3}$

tifr2014 expectation

Answer key

12.6.4 Expectation: TIFR CSE 2015 | Part A | Question: 6



Ram has a fair coin, i.e., a toss of the coin results in either head or tail and each event happens with probability exactly half ($1/2$). He repeatedly tosses the coin until he gets heads in two consecutive tosses. The expected number of coin tosses that Ram does is.

- A. 2 B. 4 C. 6 D. 8
E. None of the above

tifr2015 expectation

Answer key

12.6.5 Expectation: TIFR CSE 2020 | Part A | Question: 1



Two balls are drawn uniformly at random without replacement from a set of five balls numbered 1, 2, 3, 4, 5. What is the expected value of the larger number on the balls drawn?

- A. 2.5 B. 3 C. 3.5 D. 4
E. None of the above

Answer key

12.6.6 Expectation: TIFR CSE 2021 | Part B | Question: 11



Suppose we toss a fair coin (i.e., both heads and tails have equal probability of appearing) repeatedly until the first time by which at least *two* heads and at least *two* tails have appeared in the sequence of tosses made. What is the expected number of coin tosses that we would have to make?

- A. 8 B. 4 C. 5.5 D. 7.5
E. 4.5

tifr2021 probability expectation

Answer key

12.6.7 Expectation: TIFR CSE 2022 | Part A | Question: 2



We would like to invite a minimum number n of people (their birthdays are independent of each other) to a party such that the expected number of pairs of people that share the same birthday is at least 1. What should n be?

(Ignore leap years, so there are only 365 possible birthdays. Assume that birthdays fall with equal probability on each of the 365 days of the year.)

- A. 23 B. 28 C. 92 D. 183
E. 366

tifr2022 probability expectation

Answer key

12.6.8 Expectation: TIFR CSE 2025 | Part A | Question: 12



Consider a coin which has probability p of coming up heads when tossed. Assume that $0 < p < 1$. A trial is performed by repeatedly tossing the coin until the first time it shows a head: the number of tosses made is then called the *value* of the trial. (Thus, if a trial results in tosses "tails", "tails", "heads", then its value is 3) An *experiment* is a sequence of n independent trials.

Akhil and Bela each perform an experiment, as defined above, with the coin. Akhil defines the value A of his experiment to be the maximum of the values of the n trials in his experiment, while Bela defines the value B of her experiment to be the average of the value of the n trials in her experiment. Which of the following is correct? (Here, for any random variable X , $E[X]$ denotes the expectation of X .)

- A. $E[A]$ is a strictly increasing function of n while $E[B]$ does not depend upon n .
B. $E[A]$ is a non-constant function of n , but it is not strictly increasing, while $E[B]$ does not depend upon n .
C. $E[A]$ and $E[B]$ are both strictly increasing functions of n .
D. $E[A]$ is a non-decreasing function of n while $E[B]$ is a strictly decreasing function of n .
E. Neither $E[A]$ nor $E[B]$ depends upon n .

tifr2025 probability random-variable expectation

Answer key

12.7.1 Independent Events: TIFR CSE 2011 | Part A | Question: 19



Three dice are rolled independently. What is the probability that the highest and the lowest value differ by 4?

- A. $\left(\frac{1}{3}\right)$ B. $\left(\frac{1}{6}\right)$ C. $\left(\frac{1}{9}\right)$ D. $\left(\frac{5}{18}\right)$
 E. $\left(\frac{2}{9}\right)$

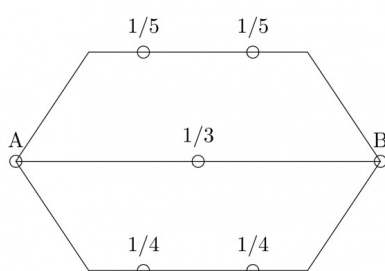
tifr2011 probability independent-events

Answer key

12.7.2 Independent Events: TIFR CSE 2012 | Part A | Question: 19



An electric circuit between two terminals A and B is shown in the figure below, where the numbers indicate the probabilities of failure for the various links, which are all independent.



What is the probability that A and B are connected?

- A. $\left(\frac{6}{25}\right)$ B. $\left(\frac{379}{400}\right)$ C. $\left(\frac{1}{1200}\right)$ D. $\left(\frac{1199}{1200}\right)$
 E. $\left(\frac{59}{60}\right)$

tifr2012 probability independent-events

Answer key

12.7.3 Independent Events: TIFR CSE 2017 | Part A | Question: 9



Consider the *majority* function on three bits, $\mathbf{maj} : \{0, 1\}^3 \rightarrow \{0, 1\}$ where $\mathbf{maj}(x_1, x_2, x_3) = 1$ if and only if $x_1 + x_2 + x_3 \geq 2$. Let $p(\alpha)$ be the probability that the output is 1 when each input is set to 1 independently with probability α . What is $p'(\alpha) = \frac{d}{d\alpha} p(\alpha)$?

- A. 3α B. α^2
 C. $6\alpha(1 - \alpha)$ D. $3\alpha^2(1 - \alpha)$
 E. $6\alpha(1 - \alpha) + \alpha^2$

tifr2017 probability independent-events differentiation

Answer key

12.7.4 Independent Events: TIFR CSE 2020 | Part A | Question: 7



A lottery chooses four random winners. What is the probability that at least three of them are born on the same day of the week? Assume that the pool of candidates is so large that each winner is equally likely to be born on any of the seven days of the week independent of the other winners.

- A. $\frac{17}{2401}$ B. $\frac{48}{2401}$ C. $\frac{105}{2401}$ D. $\frac{175}{2401}$

E. $\frac{294}{2401}$

tifr2020 engineering-mathematics probability independent-events

Answer key

12.8

Permutation and Combination (1)

12.8.1 Permutation and Combination: TIFR CSE 2024 | Part A | Question: 2



Let σ be a uniform random permutation of $\{1, \dots, 100\}$. What is the probability that $\sigma(1) < \sigma(2) < \sigma(3)$ (i.e., what is the probability that the first three elements are in increasing order)?

In the options below, $n! = 1 \times 2 \times \dots \times n$.

- A. $\frac{3}{100!}$ B. $\frac{3!}{100!}$ C. $\frac{6}{100}$ D. $\frac{1}{6}$
 E. $\frac{1}{3}$

tifr2024 probability permutation-and-combination

Answer key

12.9

Poisson Distribution (1)

12.9.1 Poisson Distribution: TIFR CSE 2023 | Part B | Question: 7



Consider the random variables N , H , and T sampled as follows. First, N is sampled from the Poisson(10) distribution. This means that for each integer $n \geq 0$,

$$\Pr[N = n] = \frac{e^{-10} 10^n}{n!}.$$

Then N independent fair coins are tossed. H is then the number of heads, and T the number of tails, obtained in this process. Choose the correct statement from the ones given below.

(**Note:** Recall that for any real number x , e^x is defined by the infinite series

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!},$$

where $0! = 1$ by convention.)

- A. H and T are independent and the variance of H is equal to the variance of N .
 B. H and T are independent and the variance of H is half the variance of N .
 C. H and T are not independent, but they are independent conditioned on N .
 D. H and T are not independent, and $E[HT] < E[H] \times E[T]$.
 E. H and T are not independent, and $E[HT] > E[H] \times E[T]$.

tifr2023 probability poisson-distribution

Answer key

12.10

Polynomials (1)

12.10.1 Polynomials: TIFR CSE 2024 | Part A | Question: 13



Let $n \geq 100$ be a positive integer. Let $X_1, X_2, \dots, X_n$ be independent random variables, each taking values in the set $\{0, 1\}$ such that $\Pr[X_i = 1] = \frac{2}{3}$ for each $1 \leq i \leq n$. Define $S := \sum_{i=1}^n X_i$, and let $p(x)$ be a polynomial such that for each non-negative integer j , the coefficient of x^j in $p(x)$ is $\Pr[S = j]$. What is $p'(1)$?

Note: $p'(x)$ denotes the polynomial obtained by differentiating p with respect to x .

- A. 0 B. n C. $\frac{2n}{3}$ D. $\frac{4n^2+2n}{9}$
E. $\frac{4n^2-4n}{9}$

tifr2024 probability random-variable polynomials

Answer key

12.11

Probability (9)

12.11.1 Probability: TIFR CSE 2010 | Part A | Question: 13



A cube whose faces are colored is split into 1000 small cubes of equal size. The cubes thus obtained are mixed thoroughly. The probability that a cube drawn at random will have exactly two colored faces is:

- A. 0.096 B. 0.12 C. 0.104 D. 0.24
E. None of the above

tifr2010 probability

Answer key

12.11.2 Probability: TIFR CSE 2012 | Part A | Question: 9



The probability of throwing six perfect dices and getting six different faces is

- A. $1 - \frac{6!}{6^6}$ B. $\frac{6!}{6^6}$ C. 6^{-6} D. $1 - 6^{-6}$
E. None of the above

tifr2012 probability

Answer key

12.11.3 Probability: TIFR CSE 2013 | Part A | Question: 13



Doctors A and B perform surgery on patients in stages III and IV of a disease. Doctor A has performed a 100 surgeries (on 80 stage III and 20 stage IV patients) and 80 out of her 100 patients have survived (78 stage III and 2 stage IV survivors). Doctor B has also performed 100 surgeries (on 50 stage III and 50 stage IV patients). Her success rate is $\frac{600}{100}$ (49 stage III survivors and 11 stage IV survivors). A patient has been advised that she is equally likely to be suffering from stage III or stage IV of this disease. Which doctor would you recommend to this patient and why?

- A. Doctor A since she has a higher success rate
B. Doctor A since she specializes in stage III patients and the success of surgery in stage IV patients is anyway too low
C. Doctor B since she has performed more stage IV surgeries
D. Doctor B since she appears to be more successful
E. There is not enough data since the choice depends on the stage of the disease the patient is suffering from.

Answer key

12.11.4 Probability: TIFR CSE 2013 | Part A | Question: 17



A stick of unit length is broken into two at a point chosen at random. Then, the larger part of the stick is further divided into two parts in the ratio $4 : 3$. What is the probability that the three sticks that are left CANNOT form a triangle?

- A. $1/4$ B. $1/3$ C. $5/6$ D. $1/2$
 E. $\log_e(2)/2$

Answer key

12.11.5 Probability: TIFR CSE 2019 | Part A | Question: 4



What is the probability that a point $P = (\alpha, \beta)$ picked uniformly at random from the disk $x^2 + y^2 \leq 1$ satisfies $\alpha + \beta \leq 1$?

- A. $\frac{1}{\pi}$ B. $\frac{3}{4} + \frac{1}{4} \cdot \frac{1}{\pi}$
 C. $\frac{3}{4} + \frac{1}{4} \cdot \frac{2}{\pi}$ D. $\frac{1}{1}$
 E. $\frac{2}{\pi}$

Answer key

12.11.6 Probability: TIFR CSE 2020 | Part A | Question: 10



In a certain year, there were exactly four Fridays and exactly four Mondays in January. On what day of the week did the 20th of the January fall that year (recall that January has 31 days)?

- A. Sunday B. Monday C. Wednesday D. Friday
 E. None of the others

Answer key

12.11.7 Probability: TIFR CSE 2021 | Part A | Question: 4



What is the probability that at least two out of four people have their birthdays in the same month, assuming their birthdays are uniformly distributed over the twelve months?

- A. $\frac{25}{48}$
 B. $\frac{5}{8}$
 C. $\frac{5}{12}$
 D. $\frac{41}{96}$
 E. $\frac{55}{96}$

Answer key

12.11.8 Probability: TIFR CSE 2023 | Part A | Question: 13



We have a coin that is equally likely to land heads (denoted ‘H’) or tails (denoted ‘T’) when tossed. Suppose we keep tossing this coin and stop the game as soon as we see three successive tosses giving either the sequence HTH or the sequence HHT.

What is the probability that when we stop the game, the last three successive tosses are HHT?

- A. $1/4$ B. $2/3$ C. $3/4$ D. $1/3$
E. $1/2$

tifr2023 probability

Answer key

12.11.9 Probability: TIFR CSE 2024 | Part A | Question: 6



For each month in the year (i.e., January, February, March,...), let us assume the probability that a person's birthday falls in that particular month is exactly $1/12$, and let us assume that this is independent for different persons.

What is the smallest value of the natural number n such that, among n independently chosen persons the probability that there is a pair of them born in the same month is at least $1/2$?

- A. 3 B. 4 C. 5 D. 6
E. 7

tifr2024 probability

Answer key

12.12

Probability Density Function (1)

12.12.1 Probability Density Function: TIFR CSE 2023 | Part A | Question: 8



Let μ be a probability distribution on the interval $[0, 1]$ with probability density function $p(x) = c \cdot x^2$ where c is an undetermined constant. Consider an interval $A = [a, b]$, with $0 \leq a < b \leq 1$ such that

$$\Pr_{X \sim \mu}[X \in A] = \int_a^b p(x) dx = \frac{1}{2}.$$

Then, the smallest possible value of $b - a$ is

- A. $1 - \frac{1}{2^{1/3}}$
B. $\frac{1}{2^{1/3}}$
C. $1 - \frac{1}{2^{1/2}}$
D. $\frac{1}{2^{1/2}}$
E. The smallest possible value of $b - a$ cannot be determined uniquely from the information given in the question.

tifr2023 probability probability-density-function

12.13

Random Variable (2)

12.13.1 Random Variable: TIFR CSE 2011 | Part A | Question: 7



Let X and Y be two independent and identically distributed random variables. Then $P(X > Y)$ is.

- A. $\frac{1}{2}$
- B. 1
- C. 0
- D. $\frac{1}{3}$
- E. Information is insufficient.

tifr2011 probability random-variable

Answer key

12.13.2 Random Variable: TIFR CSE 2014 | Part A | Question: 19



Consider the following random function of x

$$F(x) = 1 + Ux + Vx^2 \pmod{5},$$

where U and V are independent random variables uniformly distributed over $\{0, 1, 2, 3, 4\}$. Which of the following is FALSE?

- A. $F(1)$ is uniformly distributed over $\{0, 1, 2, 3, 4\}$.
- B. $F(1), F(2)$ are independent random variables and both are uniformly distributed over $\{0, 1, 2, 3, 4\}$.
- C. $F(1), F(2), F(3)$ are independent and identically distributed random variables.
- D. All of the above.
- E. None of the above.

tifr2014 probability random-variable

Answer key

12.14

Uniform Distribution (8)

12.14.1 Uniform Distribution: TIFR CSE 2013 | Part A | Question: 18



Consider three independent uniformly distributed (taking values between 0 and 1) random variables. What is the probability that the middle of the three values (between the lowest and the highest value) lies between a and b where $0 \leq a < b \leq 1$?

- A. $3(1-b)a(b-a)$
- B. $3((b-a) - (b^2 - a^2)/2)$
- C. $6(1-b)a(b-a)$
- D. $(1-b)a(b-a)$
- E. $6((b^2 - a^2)/2 - (b^3 - a^3)/3)$.

tifr2013 probability random-variable uniform-distribution

Answer key

12.14.2 Uniform Distribution: TIFR CSE 2015 | Part A | Question: 12



Consider two independent and identically distributed random variables X and Y uniformly distributed in $[0, 1]$. For $\alpha \in [0, 1]$, the probability that $\alpha \max(X, Y) < XY$ is

- A. $1/(2\alpha)$
- B. $\exp(1 - \alpha)$
- C. $1 - \alpha$
- D. $(1 - \alpha)^2$
- E. $1 - \alpha^2$

tifr2015 probability random-variable uniform-distribution

Answer key

12.14.3 Uniform Distribution: TIFR CSE 2016 | Part A | Question: 11



In one of the islands that his travels took him to, Gulliver noticed that the probability that a (uniformly) randomly chosen inhabitant has height at least 2 meters is 0.2. Also, 0.2 is the probability that a (uniformly) randomly chosen inhabitant has height at most 1.5 meters. What can we conclude about the average height h in meters of the inhabitants of the island?

- i. $1.5 \leq h \leq 2$
- ii. $h \geq 1.3$
- iii. $h \leq 2.2$

Which of the above statements is necessarily true?

- A. ii only
- B. iii only
- C. i, ii and iii
- D. ii and iii only
- E. None of the above

tifr2016 probability uniform-distribution

Answer key

12.14.4 Uniform Distribution: TIFR CSE 2016 | Part A | Question: 4



There are n balls $b_1, \dots, b_n$ and n boxes. Each ball is placed in box chosen independently and uniformly at random. We say that (b_i, b_j) is a *colliding pair* if $i < j$, and b_i and b_j are placed in the same box. What is the expected number of *colliding pairs*?

- A. $\frac{n-1}{2}$
- B. 0
- C. 1
- D. $\frac{n}{4}$
- E. $\binom{n}{2}$

tifr2016 probability random-variable uniform-distribution

Answer key

12.14.5 Uniform Distribution: TIFR CSE 2020 | Part A | Question: 4



Fix $n \geq 4$. Suppose there is a particle that moves randomly on the number line, but never leaves the set $\{1, 2, \dots, n\}$. Let the initial probability distribution of the particle be denoted by $\vec{\pi}$. In the first step, if the particle is at position i , it moves to one of the positions in $\{1, 2, \dots, i\}$ with uniform distribution; in the second step, if the particle is in location j , then it moves to one of the locations in $\{j, j+1, \dots, n\}$ with uniform distribution. Suppose after two steps, the final distribution of the particle is uniform. What is the initial distribution $\vec{\pi}$?

- A. $\vec{\pi}$ is not unique
- B. $\vec{\pi}$ is uniform
- C. $\pi(i)$ is non-zero for all even i and zero otherwise
- D. $\vec{\pi}(1) = 1$ and $\vec{\pi}(i) = 0$ for $i \neq 1$
- E. $\vec{\pi}(n) = 1$ and $\vec{\pi}(i) = 0$ for $i \neq n$

tifr2020 engineering-mathematics probability uniform-distribution

12.14.6 Uniform Distribution: TIFR CSE 2022 | Part A | Question: 15



Fix $n \geq 4$. Suppose there is a particle that moves randomly on the number line, but never leaves the set $\{1, 2, \dots, n\}$. The initial probability distribution of the particle is π i.e., the probability

that particle is in location i is given by $\pi(i)$. In the first step, if the particle is at position i , it moves to one of the positions in $\{1, 2, \dots, i\}$ with uniform distribution; in the second step, if the particle is in location j , then it moves to one of the locations in $\{j, j+1, \dots, n\}$ with uniform distribution. Suppose after two steps, the final distribution of the particle is uniform. What is the initial distribution π ?

- A. π is not unique
 B. π is uniform
 C. $\pi(i)$ is non-zero for all even i and zero otherwise
 D. $\pi(1) = 1$ and $\pi(i) = 0$ for $i \neq 1$
 E. $\pi(n) = 1$ and $\pi(i) = 0$ for $i \neq n$

tifr2022 probability uniform-distribution

Answer key

12.14.7 Uniform Distribution: TIFR CSE 2022 | Part A | Question: 5



Let $\mathcal{F}$ be the set of all functions mapping $\{1, \dots, n\}$ to $\{1, \dots, m\}$. Let f be a function that is chosen uniformly at random from $\mathcal{F}$. Let x, y be distinct elements from the set $\{1, \dots, n\}$. Let p denote the probability that $f(x) = f(y)$. Then,

- A. $p = 0$
 B. $p = \frac{1}{n^m}$
 C. $0 < p \leq \frac{1}{m^n}$
 D. $p = \frac{1}{m}$
 E. $p = \frac{1}{n}$

tifr2022 probability uniform-distribution functions

Answer key

12.14.8 Uniform Distribution: TIFR CSE 2022 | Part A | Question: 7



Initially, N white beads are arranged in a circle. A number k is chosen uniformly at random from $\{1, \dots, N-1\}$. Then a set of k beads is chosen uniformly from the white beads, and these k beads are coloured black. The position of the beads remains unchanged. What is the probability that the black beads occur sequentially in the circle, i.e., at most two black beads have white beads next to them?

- A. $\frac{2N}{2N+1}$
 B. $\frac{N^2}{(N-1)(N-1)!}$
 C. $\frac{N}{N-1} \sum_{k=1}^{N-1} \frac{1}{\binom{N}{k}}$
 D. $\frac{1}{N} + \sum_{k=1}^{N-1} \frac{1}{\binom{N}{k}}$
 E. None of the above

tifr2022 probability uniform-distribution

Answer key

Answer Keys

12.1.1	E	12.2.1	C	12.2.2	E	12.2.3	D	12.2.4	D
12.2.5	E	12.2.6	B	12.2.7	C	12.3.1	B	12.4.1	C
12.5.1	A	12.5.2	E	12.5.3	B	12.5.4	D	12.5.5	A
12.5.6	C	12.5.7	D	12.5.8	D	12.5.9	B	12.5.10	A
12.5.11	B	12.5.12	A	12.5.13	C	12.5.14	C	12.5.15	D
12.6.1	C	12.6.2	C	12.6.3	E	12.6.4	C	12.6.5	D
12.6.6	C	12.6.7	B	12.6.8	A	12.7.1	E	12.7.2	B

12.7.3	C
12.11.1	A
12.11.6	A
12.13.1	E
12.14.4	A

12.7.4	D
12.11.2	B
12.11.7	D
12.13.2	C
12.14.5	D

12.8.1	D
12.11.3	D
12.11.8	B
12.14.1	E
12.14.6	D

12.9.1	B
12.11.4	A
12.11.9	C
12.14.2	D
12.14.7	D

12.10.1	C
12.11.5	C
12.12.1	A
12.14.3	A
12.14.8	C



13.1

Logical Reasoning (10)

13.1.1 Logical Reasoning: TIFR CSE 2010 | Part A | Question: 1



A box contains 731 black balls and 2000 white balls. The following process is to be repeated as long as possible. Arbitrarily select two balls from the box. If they are of the same color, throw them out and put a black ball into the box (enough extra black balls are available to do this). If they are of different color, place the white ball back into the box and throw the black ball away. Which of the following is correct?

- A. The process can be applied indefinitely without any prior bound
- B. The process will stop with a single white ball in the box
- C. The process will stop with a single black ball in the box
- D. The process will stop with the box empty
- E. None of the above

tifr2010 analytical-aptitude logical-reasoning

Answer key

13.1.2 Logical Reasoning: TIFR CSE 2013 | Part A | Question: 10



Three men and three rakhsasas arrive together at a ferry crossing to find a boat with an oar, but no boatman. The boat can carry one or at the most two persons, for example, one man and one rakhsasas, and each man or rakhsasas can row. But if at any time, on any bank, (including those who maybe are in the boat as it touches the bank) rakhsasas outnumber men, the former will eat up the latter. If all have to go to the other side without any mishap, what is the minimum number of times that the boat must cross the river?

- A. 7
- B. 9
- C. 11
- D. 13
- E. 15

tifr2013 analytical-aptitude logical-reasoning

Answer key

13.1.3 Logical Reasoning: TIFR CSE 2013 | Part A | Question: 11



Let there be a pack of 100 cards numbered 1 to 100. The i^{th} card states: "There are at most $i - 1$ true cards in this pack". Then how many cards of the pack contain TRUE statements?

- A. 0
- B. 1
- C. 100
- D. 50
- E. None of the above

tifr2013 logical-reasoning

Answer key

13.1.4 Logical Reasoning: TIFR CSE 2013 | Part A | Question: 2



Consider the following two types of elections to determine which of two parties A and B forms the next government in the 2014 Indian elections. Assume for simplicity an Indian population of size 545545 ($= 545 * 1001$). There are only two parties A and B and every citizen votes.

TYPE C: The country is divided into 545 constituencies and each constituency has 1001 voters. Elections are held for each constituency and a party is said to win a constituency if it receive a majority of the vote in that constituency. The party that wins the most constituencies forms the next government.

TYPE P: There are no constituencies in this model. Elections are held throughout the country and the party that wins the most votes (among 545545 voters forms the government.

Which of the following is true?

- A. If the party forms the govt. by election TYPE C winning at least two-third of the constituencies, then it will also forms the govt. by election TYPE P.
- B. If a party forms govt. by election TYPE C, then it will also form the govt. by election TYPE P.
- C. If a party forms govt. by election TYPE P, then it will also form the govt. by election TYPE C.
- D. All of the above
- E. None of the above

tifr2013 logical-reasoning

Answer key

13.1.5 Logical Reasoning: TIFR CSE 2016 | Part A | Question: 1



Suppose the following statements about three persons in a room are true.

Chandni, Sooraj and Tara are in a room. Nobody else is in the room. Chandni is looking at Sooraj. Sooraj is looking at Tara. Chandni is married. Tara is not married. A married person in the room is looking at an unmarried person.

Then, Which of the following is necessarily true?

- A. Sooraj is married
- B. Sooraj is unmarried
- C. The situation described is impossible
- D. There is insufficient information to conclude if Sooraj is married or unmarried
- E. None of the above

tifr2016 logical-reasoning

Answer key

13.1.6 Logical Reasoning: TIFR CSE 2017 | Part A | Question: 14



Consider the following game with two players, Aditi and Bharat. There are n tokens in a bag. The two players know n , and take turns removing tokens from the bag. In each turn, a player can either remove one token or two tokens. The player that removes the last token from the bag loses. Assume that Aditi always goes first. Further, we say that a player has a winning strategy if she or he can win the game, no matter what other player does. Which of the following statements is TRUE?

- A. For $n = 3$, Bharath has a winning strategy. For $n = 4$, Aditi has a winning strategy.
- B. For $n = 7$, Bharath has a winning strategy. For $n = 8$, Aditi has a winning strategy.
- C. For both $n = 3$ and $n = 4$, Aditi has a winning strategy.
- D. For both $n = 7$ and $n = 8$, Bharat has a winning strategy.
- E. Bharat never has a winning strategy.

tifr2017 analytical-aptitude logical-reasoning

Answer key

13.1.7 Logical Reasoning: TIFR CSE 2018 | Part A | Question: 11



We are given a (possibly empty) set of objects. Each object in the set is colored either black or white, is shaped either circular or rectangular, and has a profile that is either fat or thin, Those properties obey the following principles:

1. Each white object is also circular.
2. Not all thin objects are black.
3. Each rectangular object is also either thin or white or both thin and white.

Consider the following statements:

- i. If there is a thin object in the set, then there is also a white object.
- ii. If there is a rectangular object in the set, then there are at least two objects.
- iii. Every fat object in the set is circular.

Which of the above statements must be TRUE for the set?

- A. (i) only
- B. (i) and (ii) only
- C. (i) and (iii) only
- D. None of the statements must be TRUE
- E. All of the statements must be TRUE

tifr2018 analytical-aptitude logical-reasoning

Answer key 

13.1.8 Logical Reasoning: TIFR CSE 2018 | Part A | Question: 8



A crime has been committed with four people at the scene of the crime. You are responsible for finding out who did it. You have recorded the following statements from the four witnesses, and you know one of them has committed the crime.

1. Anuj says that Binky did it.
2. Binky says that Anuj did it.
3. Chacko says that Binky is telling the truth.
4. Desmond says that Chacko is not lying.

You know that exactly three of the statements recorded are FALSE. Who committed the crime?

- A. Anuj
- B. Binky
- C. Chacko
- D. Desmond
- E. Either Anuj or Binky; the information is insufficient to pinpoint the criminal

tifr2018 analytical-aptitude logical-reasoning

Answer key 

13.1.9 Logical Reasoning: TIFR CSE 2019 | Part A | Question: 10



Avni and Badal alternately choose numbers from the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ without replacement (starting with Avni). The first person to choose numbers of which any 3 sum to 15 wins the game (for example, Avni wins if she chooses the numbers 8, 3, 5, 2 since $8 + 5 + 2 = 15$). A player is said to have a winning strategy if the player can always win the game, no matter what the other player does. Which of the following statements is TRUE?

As a hint, there are exactly 8 ways in which 3 numbers from the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ can sum up to 15, shown as the three rows, the three columns, and the two diagonals in the following square:

8	1	6
3	5	7
4	9	2

- A. Avni has a winning strategy
- B. Badal has a winning strategy
- C. Both of them have a winning strategy
- D. Neither of them has a winning strategy
- E. The Player that picks 9 has a winning strategy

tifr2019 general-aptitude analytical-aptitude logical-reasoning

Answer key 

13.1.10 Logical Reasoning: TIFR CSE 2019 | Part A | Question: 11



Suppose there are n guests at a party (and no hosts). As the night progresses, the guests meet each other and shake hands. The same pair of guests might shake hands multiple times. for some parties stretch late into the night , and it is hard to keep track. Still, they don't shake hands with themselves. Let Odd be the set of guests who have shaken an odd number of hands, and let even be the set of guests who have shaken an even number of hands. Which of the following stays invariant throughout the night?

- A. $|\text{Odd}| \bmod 2$
- B. $|\text{Even}|$
- C. $|\text{Even}| - |\text{Odd}|$
- D. $2 \mid |\text{Even}| - |\text{Odd}|$
- E. $2 \mid |\text{Odd}| - |\text{Even}|$

tifr2019 general-aptitude analytical-aptitude logical-reasoning

Answer key 

Answer Keys

13.1.1	C	13.1.2	C	13.1.3	D	13.1.4	E	13.1.5	D
13.1.6	B	13.1.7	E	13.1.8	B	13.1.9	D	13.1.10	A



14.1

Alligation Mixture (1)

14.1.1 Alligation Mixture: TIFR CSE 2019 | Part A | Question: 9



Let A and B be two containers. Container A contains 50 litres of liquid X and container B contains 100 litres of liquid Y . Liquids X and Y are soluble in each other.

We now take 30 ml of liquid X from container A and put it into container B . The mixture in container B is then thoroughly mixed and 20 ml of the resulting mixture is put back into container A . At the end of this process let V_{AY} be the volume of liquid Y and V_{BX} be the volume of liquid X in container B . Which of the following must be TRUE ?

- A. $V_{AY} < V_{BX}$
 C. $V_{AY} = V_{BX}$
 E. $V_{AY} + V_{BX} = 20$

- B. $V_{AY} > V_{BX}$
 D. $V_{AY} + V_{BX} = 30$

tifr2019 quantitative-aptitude alligation-mixture

Answer key

14.2

Cartesian Coordinates (4)

14.2.1 Cartesian Coordinates: TIFR CSE 2013 | Part B | Question: 9



Suppose n straight lines are drawn on a plane. When these lines are removed, the plane falls apart into several connected components called regions. A region R is said to be convex if it has the following property: whenever two points are in R , then the entire line segment joining them is in R . Suppose no two of the n lines are parallel. Which of the following is true?

- A. $O(n)$ regions are produced, and each region is convex.
 B. $O(n^2)$ regions are produced but they need not all be convex.
 C. $O(n^2)$ regions are produced, and each region is convex.
 D. $O(n \log n)$ regions are produced, but they need not all be convex.
 E. All regions are convex but there may be exponentially many of them.

tifr2013 quantitative-aptitude geometry cartesian-coordinates

Answer key

14.2.2 Cartesian Coordinates: TIFR CSE 2014 | Part A | Question: 13



Let L be a line on the two dimensional plane. L 's intercepts with the X and Y axes are respectively a and b . After rotating the co-ordinate system (and leaving L untouched), the new intercepts are a' and b' respectively. Which of the following is TRUE?

- A. $\frac{1}{a} + \frac{1}{b} = \frac{1}{a'} + \frac{1}{b'}$.
 B. $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{a'^2} + \frac{1}{b'^2}$.
 C. $\frac{b}{a^2} + \frac{a}{b^2} = \frac{b'}{a'^2} + \frac{a'}{b'^2}$.
 D. $\frac{b}{a} + \frac{a}{b} = \frac{b'}{a'} + \frac{a'}{b'}$.
 E. None of the above.

tifr2014 geometry cartesian-coordinates

Answer key

14.2.3 Cartesian Coordinates: TIFR CSE 2015 | Part A | Question: 13



Imagine the first quadrant of the real plane as consisting of unit squares. A typical square has 4 corners: (i, j) , $(i+1, j)$, $(i+1, j+1)$, and $(i, j+1)$, where (i, j) is a pair of non-negative integers. Suppose a line segment l connecting $(0, 0)$ to $(90, 1100)$ is drawn. We say that l passes

through a unit square if it passes through a point in the interior of the square. How many unit squares does l pass through?

- A. 98,990 B. 9,900 C. 1,190 D. 1,180
E. 1,010

tifr2015 quantitative-aptitude cartesian-coordinates

Answer key 

14.2.4 Cartesian Coordinates: TIFR CSE 2020 | Part B | Question: 5



Let u be a point on the unit circle in the first quadrant (i.e., both coordinates of u are positive). Let θ be the angle subtended by u and the x axis at the origin. Let ℓ_u denote the infinite line passing through the origin and u . Consider the following operation O_u on points in the plane.

Operation O_u

INPUT: a point v on the plane

1. Reflect v in the x axis, obtaining $\tilde{v}$.
2. Reflect $\tilde{v}$ in ℓ_u , obtaining $\hat{v}$.
3. Output $\hat{v}$.

If $\hat{v}$ is the output of applying O_u on v , we write $O_u(v) = \hat{v}$. Further, we denote by O_u^k the iterates of O_u , i.e., $O_u^1(v) := O_u(v)$ and $O_u^k(v) := O_u(O_u^{k-1}(v))$ for all integers $k > 1$.

Consider a point v in the first quadrant such that v and the x -axis subtend an angle ϕ at the origin. Define $w = O_u^8(v)$. Assuming $\theta = 5^\circ$ and $\phi = 10^\circ$, what is the angle subtended by w and the x -axis at the origin?

- A. 50° B. 85° C. 90° D. 145°
E. 165°

tifr2020 quantitative-aptitude cartesian-coordinates

Answer key 

14.3 Circle (4)

14.3.1 Circle: TIFR CSE 2011 | Part A | Question: 18



The equation of the tangent to the unit circle at point $(\cos \alpha, \sin \alpha)$ is

- A. $x \cos \alpha - y \sin \alpha = 1$ B. $x \sin \alpha - y \cos \alpha = 1$
C. $x \cos \alpha + y \sin \alpha = 1$ D. $x \sin \alpha - y \cos \alpha = 1$
E. None of the above

tifr2011 quantitative-aptitude geometry circle

Answer key 

14.3.2 Circle: TIFR CSE 2018 | Part A | Question: 1



Consider a point A inside a circle C that is at distance 9 from the centre of a circle. Suppose you told that there is a chord of length 24 passing through A with A as its midpoint. How many distinct chords of C have integer length and pass through A ?

- A. 2 B. 6 C. 7 D. 12
E. 14

tifr2018 quantitative-aptitude geometry circle

Answer key

14.3.3 Circle: TIFR CSE 2020 | Part A | Question: 13



What is the area of the largest rectangle that can be inscribed in a circle of radius R ?

- A. $R^2/2$ B. $\pi \times R^2/2$ C. R^2 D. $2R^2$
- E. None of the above

tifr2020 quantitative-aptitude geometry circle

Answer key

14.3.4 Circle: TIFR CSE 2021 | Part A | Question: 2



What is the area of a rectangle with the largest perimeter that can be inscribed in the unit circle (i.e., all the vertices of the rectangle are on the circle with radius 1)?

- A. 1 B. 2 C. 3 D. 4
- E. 5

tifr2021 quantitative-aptitude geometry circle

Answer key

14.4 Clock Time (5)

14.4.1 Clock Time: TIFR CSE 2010 | Part A | Question: 2



The hour hand and the minute hands of a clock meet at noon and again at mid-night. In between they meet N times, where N is.:

- A. 6 B. 11 C. 12 D. 13
- E. None of the above

tifr2010 quantitative-aptitude clock-time

Answer key

14.4.2 Clock Time: TIFR CSE 2013 | Part A | Question: 20



Consider a well functioning clock where the hour, minute and the seconds needles are exactly at zero. How much time later will the minutes needle be exactly one minute ahead ($1/60$ th of the circumference) of the hours needle and the seconds needle again exactly at zero?

Hint: When the desired event happens both the hour needle and the minute needle have moved an integer multiple of $1/60$ th of the circumference.

- A. 144 minutes B. 66 minutes C. 96 minutes D. 72 minutes
- E. 132 minutes

tifr2013 quantitative-aptitude clock-time

Answer key

14.4.3 Clock Time: TIFR CSE 2014 | Part A | Question: 10



A person went out between 4pm and 5pm to chat with her friend and returned between 5pm and 6pm. On her return, she found that the hour-hand and the minute-hand of her (well-functioning) clock had just exchanged their positions with respect to their earlier positions at the time of her leaving. The person must have gone out to chat at

A. Twenty five minutes past 4pm.

B. Twenty six and $\frac{122}{143}$ minutes past 4pm.

C. Twenty seven and $\frac{1}{3}$ minutes past 4pm.

D. Twenty eight minutes past 4pm.

E. None of the above.

tifr2014 quantitative-aptitude clock-time

Answer key

14.4.4 Clock Time: TIFR CSE 2020 | Part A | Question: 12



The hour needle of a clock is malfunctioning and travels in the anti-clockwise direction, i.e., opposite to the usual direction, at the same speed it would have if it was working correctly. The minute needle is working correctly. Suppose the two needles show the correct time at 12 noon, thus both needles are together at the 12 mark. After how much time do the two needles meet again?

A. $\frac{10}{11}$ hour

B. $\frac{11}{12}$ hour

C. $\frac{12}{13}$ hour

D. $\frac{19}{22}$ hour

E. One hour

tifr2020 quantitative-aptitude clock-time

Answer key

14.4.5 Clock Time: TIFR CSE 2025 | Part A | Question: 10



A clock shows 3 : 00. After 10^{1024} minutes, what minute does the minute hand point to?

A. 0

B. 20

C. 40

D. 50

E. 10

tifr2025 quantitative-aptitude clock-time

Answer key

14.5 Complex Number (3)

14.5.1 Complex Number: TIFR CSE 2011 | Part A | Question: 13



If $z = \frac{\sqrt{3} - i}{2}$ and $(z^{95} + i^{67})^{97} = z^n$, then the smallest value of n is

A. 1

B. 10

C. 11

D. 12

E. None of the above

tifr2011 quantitative-aptitude complex-number

Answer key

14.5.2 Complex Number: TIFR CSE 2011 | Part A | Question: 5



Three distinct points x, y, z lie on a unit circle of the complex plane and satisfy $x + y + z = 0$. Then x, y, z form the vertices of .

A. An isosceles but not equilateral triangle.

B. An equilateral triangle.

C. A triangle of any shape.

D. A triangle whose shape can't be determined.

E. None of the above.

tifr2011 quantitative-aptitude geometry complex-number non-gatecse

Answer key

14.5.3 Complex Number: TIFR CSE 2013 | Part A | Question: 7



For any complex number z , $\arg z$ defines its phase, chosen to be in the interval $0 \leq \arg z < 360^\circ$. If z_1, z_2 and z_3 are three complex numbers with the same modulus but different phases ($\arg z_3 < \arg z_2 < \arg z_1 < 180^\circ$), then the quantity

$$\frac{\arg(z_1/z_2)}{\arg[(z_1 - z_3)/(z_2 - z_3)]}$$

is a constant, and has the value

A. 2

B. $\frac{1}{3}$

C. 1

D. 3

E. $\frac{1}{2}$

tifr2013 quantitative-aptitude complex-number non-gatecse

Answer key

14.6 Convex Sets Functions (1)

14.6.1 Convex Sets Functions: TIFR CSE 2014 | Part A | Question: 12



Let $f(x) = 2^x$. Consider the following inequality for real numbers a, b and $0 < \lambda < 1$:

$$f(\lambda a + b) \leq \lambda f(a) + (1 - \lambda)f\left(\frac{b}{1 - \lambda}\right).$$

Consider the following 3 conditions:

1. $\lambda = 0.5$

2. $0 < a \leq 2, b > 0$

3. $a/\lambda > 2, 0 < b \leq 1 - \lambda$

Which of the following statements is TRUE?

A. The above inequality holds under conditions (1) and (2) but not under condition (3).

B. The above inequality holds under conditions (2) and (3) but not under condition (1).

C. The above inequality holds under conditions (1) and (3) but not under condition (2).

D. The above inequality holds under all the three conditions.

E. The above inequality holds under none of the three conditions.

tifr2014 quantitative-aptitude convex-sets-functions non-gatecse

14.7 Cost Market Price (1)

14.7.1 Cost Market Price: TIFR CSE 2012 | Part A | Question: 6



A certain pair of used shoes can be repaired for $Rs.1250$ and will last for 1 year. A pair of the same kind of shoes can be purchased new for $Rs.2800$ and will last for 2 years. The average cost per year of the new shoes is what percent greater than the cost of repairing the used shoes?

A. 5

B. 12

C. 15

D. 3

E. 24

tifr2012 cost-market-price

Answer key 

14.8 Factors (3)

14.8.1 Factors: TIFR CSE 2010 | Part A | Question: 20



How many integers from 1 to 1000 are divisible by 30 but not by 16?

- A. 29 B. 31 C. 32 D. 33
E. 25

tifr2010 quantitative-aptitude factors

Answer key 

14.8.2 Factors: TIFR CSE 2011 | Part A | Question: 15



The exponent of 3 in the product $100!$ is

- A. 27 B. 33 C. 44 D. 48
E. None of the above

tifr2011 quantitative-aptitude factors tricky

Answer key 

14.8.3 Factors: TIFR CSE 2013 | Part A | Question: 12



Among numbers 1 to 1000 how many are divisible by 3 or 7?

- A. 333 B. 142 C. 475 D. 428
E. None of the above

tifr2013 quantitative-aptitude factors normal

Answer key 

14.9 Fractions (2)

14.9.1 Fractions: TIFR CSE 2014 | Part A | Question: 11



A large community practices birth control in the following peculiar fashion. Each set of parents continues having children until a son is born; then they stop. What is the ratio of boys to girls in the community if, in the absence of birth control, 51% of the babies are born male?

- A. 51 : 49 B. 1 : 1 C. 49 : 51 D. 51 : 98
E. 98 : 51

tifr2014 quantitative-aptitude fractions tricky

Answer key 

14.9.2 Fractions: TIFR CSE 2017 | Part A | Question: 1



A suitcase weighs one kilogram plus half of its weight. How much does the suitcase weigh?

- A. 1.3333... kilograms B. 1.5 kilograms
C. 1.666... kilograms D. 2 kilograms

E. cannot be determined from the given data

tifr2017 quantitative-aptitude fractions normal

Answer key

14.10

Geometry (9)

14.10.1 Geometry: TIFR CSE 2010 | Part A | Question: 17



Suppose there is a sphere with diameter **at least 6** inches. Through this sphere we drill a hole along a diameter. The part of the sphere lost in the process of drilling the hole looks like two caps joined to a cylinder, where the cylindrical part has length 6 inches. It turns out that the volume of the remaining portion of the sphere does not depend on the diameter of the sphere. Using this fact, determine the volume of the remaining part.

- A. 24π cu. inches
B. 36π cu. inches
C. 27π cu. inches
D. 32π cu. inches
E. 35π cu. inches

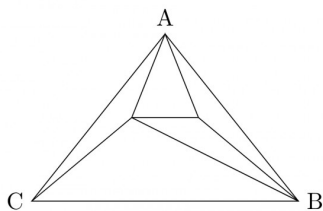
tifr2010 quantitative-aptitude geometry

Answer key

14.10.2 Geometry: TIFR CSE 2012 | Part A | Question: 4



Let ABC be a triangle with n distinct points inside. A triangulation of ABC with respect to the n points is obtained by connecting as many points as possible, such that no more line segments can be added without intersecting other line segments. In other words ABC has been partitioned into triangles with end points at the n points or at the vertices A, B, C . For example, the following figure gives one possible triangulation of ABC with two points inside it.



Although there are many different ways to triangulate ABC with the n points inside, the number of triangles depends only on n . In the above figure it is five. How many triangles are there in a triangulation of ABC with n points inside it?

- A. $3n - 1$
B. $n^2 + 1$
C. $n + 3$
D. $2n + 1$
E. $4n - 3$

tifr2012 quantitative-aptitude geometry

Answer key

14.10.3 Geometry: TIFR CSE 2012 | Part A | Question: 5



What is the maximum number of points of intersection between the diagonals of a convex octagon (8-vertex planar polygon)? Note that a polygon is said to be convex if the line segment joining any two points in its interior lies wholly in the interior of the polygon. Only points of intersection between diagonals that lie in the interior of the octagon are to be considered for this problem.

- A. 55
B. 60
C. 65
D. 70
E. 75

tifr2012 quantitative-aptitude geometry

Answer key

14.10.4 Geometry: TIFR CSE 2013 | Part A | Question: 5



The late painter Maqbool Fida Husain once coloured the surface of a huge hollow steel sphere, of radius 1 metre, using just two colours, Red and Blue. As was his style however, both the red and blue areas were a bunch of highly irregular disconnected regions. The late sculptor Ramkinkar Baij then tried to fit in a cube inside the sphere, the eight vertices of the cube touching only red coloured parts of the surface of the sphere. Assume $\pi = 3.14$ for solving this problem. Which of the following is true?

- A. Baij is bound to succeed if the area of the red part is 10sq. metres ;
- B. Baij is bound to fail if the area of the red part is 10sq. metres ;
- C. Baij is bound to fail if the area of the red part is 11sq. metres ;
- D. Baij is bound to succeed if the area of the red part is 11sq. metres ;
- E. None of the above.

tifr2013 geometry quantitative-aptitude

Answer key

14.10.5 Geometry: TIFR CSE 2015 | Part A | Question: 2



Consider a circle with a circumference of one unit length. Let $d < \frac{1}{6}$. Suppose that we independently throw two arcs, each of length d , randomly on this circumference so that each arc is uniformly distributed along the circle circumference. The arc attaches itself exactly to the circumference so that arc of length d exactly covers length d of the circumference. What can be said about the probability that the two arcs do not intersect each other?

- A. It equals $(1 - d)$
- B. It equals $(1 - 3d)$
- C. It equals $(1 - 2d)$
- D. It equals 1
- E. It equals $(1 - d)(1 - d)$

tifr2015 geometry

Answer key

14.10.6 Geometry: TIFR CSE 2015 | Part A | Question: 9



Consider a square of side length 2. We throw five points into the square. Consider the following statements:

- i. There will always be three points that lie on a straight line.
- ii. There will always be a line connecting a pair of points such that two points lie on one side of the line and one point on the other.
- iii. There will always be a pair of points which are at distance at most $\sqrt{2}$ from each other.

Which of the above is true:

- A. (i) only
- B. (ii) only
- C. (iii) only
- D. (ii) and (iii)
- E. None of the above

tifr2015 geometry quantitative-aptitude easy

Answer key

14.10.7 Geometry: TIFR CSE 2016 | Part A | Question: 9



Suppose a rectangular farm has area 100 square meters. The lengths of its sides are not known. It is known, however, that all the edges are at least 2 meters in length. Which of the following statements about the rectangle's perimeter p (in meters) is FALSE?

- A. p can take all values between 45 and 50
- B. p can be 52 for some configuration
- C. p can take all values between 55 and 60
- D. p can be 70 for some configuration
- E. p can be 39 for some configuration

tifr2016 quantitative-aptitude geometry

Answer key

14.10.8 Geometry: TIFR CSE 2017 | Part A | Question: 13



A set of points $S \subseteq \mathbb{R}^2$ is convex if for any points $x, y \in S$, every point on the straight line joining x and y is also in S . For two sets of points $S, T \subset \mathbb{R}^2$, define the sum $S + T$ as the set of points obtained by adding a point in S to a point in T . That is, $S + T := \{(x_1, x_2) \in \mathbb{R}^2 : x_1 = y_1 + z_1, x_2 = y_2 + z_2, (y_1, y_2) \in S, (z_1, z_2) \in T\}$. Similarly, $S - T := \{(x_1, x_2) \in \mathbb{R}^2 : x_1 = y_1 - z_1, x_2 = y_2 - z_2, (y_1, y_2) \in S, (z_1, z_2) \in T\}$ is the set of points obtained by subtracting a point in T from a point in S . Which of the following statements is TRUE for all convex sets S, T ?

- A. $S + T$ is convex but not $S - T$
- B. $S - T$ is convex but not $S + T$
- C. exactly one of $S + T$ and $S - T$ is convex, but it depends on S and T which one
- D. neither $S + T$ nor $S - T$ is convex
- E. both $S + T$ and $S - T$ are convex

tifr2017 quantitative-aptitude geometry

Answer key

14.10.9 Geometry: TIFR CSE 2017 | Part A | Question: 8



In a tutorial on geometrical constructions, the teacher asks a student to construct a right-angled triangle ABC where the hypotenuse BC is 8 inches and the length of the perpendicular dropped from A onto the hypotenuse is h inches, and offers various choices for the value of h . For which value of h can such a triangle NOT exist?

- A. 3.90 inches
- B. $2\sqrt{2}$ inches
- C. $2\sqrt{3}$ inches
- D. 4.1 inches
- E. none of the above

tifr2017 quantitative-aptitude geometry

Answer key

14.11

Modular Arithmetic (7)

14.11.1 Modular Arithmetic: TIFR CSE 2011 | Part A | Question: 20



Let $n > 1$ be an odd integer. The number of zeros at the end of the number $99^n + 1$ is

- A. 1
- B. 2
- C. 3
- D. 4
- E. None of the above

Answer key 

14.11.2 Modular Arithmetic: TIFR CSE 2018 | Part B | Question: 1

What is the remainder when 4444^{4444} is divided by 9?

- A. 1 B. 2 C. 5 D. 7
E. 8

Answer key 

14.11.3 Modular Arithmetic: TIFR CSE 2019 | Part A | Question: 2



How many proper divisors (that is, divisors other than 1 or 7200) does 7200 have ?

- A. 18 B. 20 C. 52 D. 54
E. 60

Answer key 

14.11.4 Modular Arithmetic: TIFR CSE 2019 | Part A | Question: 7

What are the last two digits of $1! + 2! + \dots + 100!$?

- A. 00 B. 13 C. 30 D. 33
E. 73

Answer key 

14.11.5 Modular Arithmetic: TIFR CSE 2019 | Part B | Question: 14

Let m and n be two positive integers. Which of the following is NOT always true?

- A. If m and n are co-prime, there exist integers a and b such that $am + bn = 1$
B. $m^{n-1} \equiv 1 \pmod{n}$
C. The rational number $\frac{n}{m} \cdot \frac{n-1}{m-1} \cdot \frac{n-2}{m-2} \dots \frac{n-(m-2)}{m-(m-2)} \cdot \frac{n-(m-1)}{m-(m-1)}$ is an integer
D. $m+1$ is a factor of $m^{n(n+1)} - 1$
E. If $2^n - 1$ is prime, then n is prime

Answer key 

14.11.6 Modular Arithmetic: TIFR CSE 2021 | Part A | Question: 13

What are the last two digits of 7^{2021} ?

- A. 67 B. 07 C. 27 D. 01
E. 77

Answer key

14.11.7 Modular Arithmetic: TIFR CSE 2023 | Part B | Question: 11



Let $m = 2877426671$. It is known that $p = 5754853343 = 2m + 1$ is a 10 -digit prime number. What is $16^m \pmod{p}$?

- A. 1
C. 16
E. $5754853342 \pmod{p}$ (which is actually $-1 \pmod{p}$)
- B. 4
D. 2877426671

tifr2023 quantitative-aptitude modular-arithmetic

Answer key

14.12

Number Series (4)

14.12.1 Number Series: TIFR CSE 2011 | Part A | Question: 8



The sum of the first n terms of the series $1, 11, 111, 1111, \dots$, is.

- A. $\frac{1}{81}(10^{n+1} - 9n - 10)$
C. $\frac{1}{9}(10^{n+1} - 1)$
E. None of the above
- B. $\frac{1}{81}(10^n - 9n)$
D. $\frac{1}{9}(10^{n+1} - n10^n)$

tifr2011 quantitative-aptitude number-series

Answer key

14.12.2 Number Series: TIFR CSE 2013 | Part A | Question: 15



$$\text{Let } \text{sgn}(x) = \begin{cases} +1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}$$

What is the value of the following summation?

$$\sum_{i=0}^{50} \text{sgn}((2i-1)(2i-3)\dots(2i-99))$$

- A. 0
E. 50
- B. -1
C. +1
D. 25

tifr2013 quantitative-aptitude number-series

Answer key

14.12.3 Number Series: TIFR CSE 2013 | Part A | Question: 8



Find the sum of the infinite series

$$\frac{1}{1 \times 3 \times 5} + \frac{1}{3 \times 5 \times 7} + \frac{1}{5 \times 7 \times 9} + \frac{1}{7 \times 9 \times 11} + \dots$$

- A. ∞
E. $\left(\frac{1}{14}\right)$
- B. $\left(\frac{1}{2}\right)$
C. $\left(\frac{1}{6}\right)$
D. $\left(\frac{1}{12}\right)$

tifr2013 quantitative-aptitude number-series

Answer key

14.12.4 Number Series: TIFR CSE 2014 | Part A | Question: 7



Consider a sequence of non-negative numbers $x_n : n = 1, 2, \dots$. Which of the following statements cannot be true?

- A. $\sum_{n=1}^{\infty} x_n = \infty$ and $\sum_{n=1}^{\infty} x_n^2 = \infty$.
- B. $\sum_{n=1}^{\infty} x_n = \infty$ and $\sum_{n=1}^{\infty} x_n^2 < \infty$.
- C. $\sum_{n=1}^{\infty} x_n < \infty$ and $\sum_{n=1}^{\infty} x_n^2 < \infty$.
- D. $\sum_{n=1}^{\infty} x_n \leq 5$ and $\sum_{n=1}^{\infty} x_n^2 \geq 25$.
- E. $\sum_{n=1}^{\infty} x_n < \infty$ and $\sum_{n=1}^{\infty} x_n^2 = \infty$.

tifr2014 quantitative-aptitude number-series

Answer key

14.13

Number System (1)

14.13.1 Number System: TIFR CSE 2023 | Part B | Question: 9



Fix a positive integer n , and let $n = p_1^{e_1} p_2^{e_2} \dots p_k^{e_k}$ be the prime factorization of n . Here, $p_1, \dots, p_k$ are prime numbers and $e_i \geq 1$ for all $1 \leq i \leq k$. Call a sequence $(n_0, n_1, \dots, n_t)$ where $n_0 = n$ and $n_t = 1$ *relevant* if for every $0 \leq i \leq t-1$, the number n_i/n_{i+1} is a prime number. What is the total number of relevant sequences?

- A. $(p_1 + 1) \times (p_2 + 1) \times \dots \times (p_k + 1)$
- B. $(p_1 + 1)^{e_1} \times (p_2 + 1)^{e_2} \times \dots \times (p_k + 1)^{e_k}$
- C. $(e_1 + e_2 + \dots + e_k)!$
- D. $\frac{(e_1 + e_2 + \dots + e_k)!}{(e_1! \times e_2! \times \dots \times e_k!)}$
- E. None of the above

tifr2023 quantitative-aptitude number-system

Answer key

14.14

Number Theory (5)

14.14.1 Number Theory: TIFR CSE 2014 | Part A | Question: 20



Consider the equation $x^2 + y^2 - 3z^2 - 3t^2 = 0$. The total number of integral solutions of this equation in the range of the first 10000 numbers, i.e., $1 \leq x, y, z, t \leq 10000$, is

- A. 200
- B. 55
- C. 100
- D. 1
- E. None of the above

tifr2014 number-theory quantitative-aptitude

Answer key

14.14.2 Number Theory: TIFR CSE 2020 | Part A | Question: 15



The sequence $s_0, s_1, \dots, s_9$ is defined as follows:

- $s_0 = s_1 + 1$
- $2s_i = s_{i-1} + s_{i+1} + 2$ for $1 \leq i \leq 8$
- $2s_9 = s_8 + 2$

What is s_0 ?

- A. 81 B. 95 C. 100 D. 121
E. 190

tifr2020 quantitative-aptitude number-theory

Answer key 

14.14.3 Number Theory: TIFR CSE 2020 | Part A | Question: 6



What is the maximum number of regions that the plane $\mathbb{R}^2$ can be partitioned into using 10 lines?

- A. 25 B. 50 C. 55 D. 56
E. 1024

Hint: Let $A(n)$ be the maximum number of partitions that can be made by n lines. Observe that $A(0) = 1$, $A(2) = 2$, $A(2) = 4$ etc. Come up with a recurrence equation for $A(n)$.

tifr2020 general-aptitude quantitative-aptitude number-theory

Answer key 

14.14.4 Number Theory: TIFR CSE 2020 | Part A | Question: 9



A contiguous part, i.e., a set of adjacent sheets, is missing from Tharoor's GRE preparation book. The number on the first missing page is 183, and it is known that the number on the last missing page has the same three digits, but in a different order. Note that every sheet has two pages, one at the front and one at the back. How many pages are missing from Tharoor's book?

- A. 45 B. 135 C. 136 D. 198
E. 450

tifr2020 quantitative-aptitude number-theory

Answer key 

14.14.5 Number Theory: TIFR CSE 2021 | Part A | Question: 7



Let d be the positive square integers (that is, it is a square of some integer) that are factors of $20^5 \times 21^5$. Which of the following is true about d ?

- A. $50 \leq d < 100$ B. $100 \leq d < 150$
C. $150 \leq d < 200$ D. $200 \leq d < 300$
E. $300 \leq d$

tifr2021 quantitative-aptitude number-theory

Answer key 

14.15 Numerical Computation (5)

14.15.1 Numerical Computation: TIFR CSE 2010 | Part A | Question: 14



A marine biologist wanted to estimate the number of fish in a large lake. He threw a net and found 30 fish in the net. He marked all these fish and released them into the lake. The next morning he again threw the net and this time caught 40 fish, of which two were found to be marked. The (approximate) number of fish in the lake is:

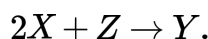
- A. 600 B. 1200 C. 68 D. 800
E. 120

Answer key 

14.15.2 Numerical Computation: TIFR CSE 2014 | Part A | Question: 1



Consider the reactions



Let n_X , n_Y , n_Z denote the numbers of molecules of chemicals X, Y, Z in the reaction chamber. Then which of the following is conserved by both reactions?

- A. $n_X + n_Y + n_Z$.
 B. $n_X + 7n_Y + 5n_Z$.
 C. $2n_X + 9n_Y - 3n_Z$.
 D. $3n_X - 3n_Y + 13n_Z$.
 E. None of the above.

tifr2014 quantitative-aptitude numerical-computation

Answer key 

14.15.3 Numerical Computation: TIFR CSE 2014 | Part A | Question: 4



Consider numbers greater than one that satisfy the following properties:

- They have no repeated prime factors;
- For all primes $p \geq 2$, p divides the number if and only if $p - 1$ divides the number.

The number of such numbers is

- A. 0
 B. 5
 C. 100
 D. Infinite
 E. None of the above

tifr2014 quantitative-aptitude difficult numerical-computation

Answer key 

14.15.4 Numerical Computation: TIFR CSE 2015 | Part A | Question: 3



Let $|z| < 1$. Define $M_n(z) = \sum_{i=1}^{10^n} z^{10^i(i-1)}$? what is

$$\prod_{i=0}^{\infty} M_i(z) = M_0(z) \times M_1(z) \times M_2(z) \times \dots?$$

- A. Can't be determined
 B. $1/(1 - z)$
 C. $1/(1 + z)$
 D. $1 - z^9$
 E. None of the above

tifr2015 quantitative-aptitude numerical-computation number-series

Answer key 

14.15.5 Numerical Computation: TIFR CSE 2015 | Part B | Question: 12



Let t_n be the sum of the first n natural numbers, for $n > 0$. A number is called triangular if it is equal to t_n for some n . Which of the following statements are true:

- There exists three successive triangular numbers whose product is a perfect square.
- If the triangular number t_n is a perfect square, then so is $t_{4n(n+1)}$.
- The sum of the reciprocals of the first n triangular numbers is less than 2, i.e.

$$\frac{1}{1} + \frac{1}{3} + \frac{1}{6} + \dots + \frac{1}{t_n} < 2$$

- A. (i) only
 C. (iii) only
 E. None of the above
- B. (ii) only
 D. All of the above

tifr2015 quantitative-aptitude normal numerical-computation

Answer key 

14.16 Permutation and Combination (1)

14.16.1 Permutation and Combination: TIFR CSE 2024 | Part B | Question: 2



Let S be the set of all 4-digit numbers created using just the digits 1, 2, 3, 4, 5 such that no two successive digits are the same. If the numbers in S are arranged in ascending order, what is the 100th number in this sequence?

- A. 2135
 B. 2324
 C. 2315
 D. 2352
 E. 2415

tifr2024 combinatory counting permutation-and-combination quantitative-aptitude

Answer key 

14.17 Polynomials (1)

14.17.1 Polynomials: TIFR CSE 2013 | Part B | Question: 2



Consider polynomials in a single variable x of degree d . Suppose $d < n/2$. For such a polynomial $p(x)$, let C_p denote the n -tuple $(P(i))_{1 \leq i \leq n}$. For any two such distinct polynomials p, q , the number of coordinates where the tuples C_p, C_q differ is.

- A. At most d
 C. Between d and $n - d$
 E. None of the above.
- B. At most $n - d$
 D. At least $n - d$

tifr2013 polynomials non-gatecse

Answer key 

14.18 Ratio Proportion (2)

14.18.1 Ratio Proportion: TIFR CSE 2012 | Part A | Question: 18



A large community practices birth control in the following peculiar fashion. Each set of parents continues having children until a son is born; then they stop. What is the ratio of boys to girls in the community if, in the absence of birth control, 51% of the babies are born male?

- A. 51 : 49
 B. 1 : 1
 C. 49 : 51
 D. 51 : 98
 E. 98 : 51

tifr2012 quantitative-aptitude ratio-proportion

Answer key 

14.18.2 Ratio Proportion: TIFR CSE 2014 | Part A | Question: 2



A body at a temperature of 30 Celsius is immersed into a heat bath at 0 Celsius at time $t = 0$. The body starts cooling at a rate proportional to the temperature difference. Assuming that the

heat bath does not change in temperature throughout the process, calculate the ratio of the time taken for the body to reach 1 Celsius divided by the time taken for the body to reach 5 Celsius.

- A. $\log 5$ B. $\frac{\log 29}{\log 25}$ C. e^5 D. $1 + \log_6 5$
E. None of the above

tifr2014 quantitative-aptitude ratio-proportion

Answer key 

14.19

Sequence Series (3)

14.19.1 Sequence Series: TIFR CSE 2013 | Part A | Question: 19



Consider a sequence of numbers $(\epsilon_n : n = 1, 2, \dots)$, such that $\epsilon_1 = 10$ and

$$\epsilon_{n+1} = \frac{20\epsilon_n}{20 + \epsilon_n}$$

for $n \geq 1$. Which of the following statements is true?

Hint: Consider the sequence of reciprocals.

- A. The sequence $(\epsilon_n : n = 1, 2, \dots)$ converges to zero.
B. $\epsilon_n \geq 1$ for all n
C. The sequence $(\epsilon_n : n = 1, 2, \dots)$ is decreasing and converges to 1.
D. The sequence $(\epsilon_n : n = 1, 2, \dots)$ is decreasing and then increasing. Finally it converges to 1.
E. None of the above.

tifr2013 quantitative-aptitude sequence-series

Answer key 

14.19.2 Sequence Series: TIFR CSE 2016 | Part A | Question: 10



Consider the sequence $\langle s_n : n \geq 0 \rangle$ defined as follows: $s_0 = 0, s_1 = 1, s_2 = 1$, and $s_n = s_{n-1} + s_{n-2} + s_{n-3}$, for $n \geq 3$. Which of the following statements is FALSE?

- A. s_{4k} is even, for any $k \geq 0$
B. s_{4k+1} is odd, for any $k \geq 0$
C. s_{4k+2} is odd, for any $k \geq 0$
D. s_n is a multiple of 3, for only finitely many values of n
E. s_{4k+3} is even, for any $k \geq 0$

tifr2016 quantitative-aptitude sequence-series

Answer key 

14.19.3 Sequence Series: TIFR CSE 2023 | Part A | Question: 9



Let S be the value of following infinite series:

$$\sum_{n=1}^{\infty} \frac{1}{n^4}.$$

In which of the following intervals must S lie?

- A. $\left[\frac{\pi^4}{900}, \frac{\pi^4}{450} \right]$
 C. $\left[\frac{\pi^4}{100}, \frac{\pi^4}{80} \right]$
 E. The series diverges, so S must be infinity.

- B. $[0.95, 1.05]$
 D. $\left[\frac{\pi^4}{2}, \pi^4 \right]$

tifr2023 quantitative-aptitude sequence-series

14.20

Speed Time Distance (4)

14.20.1 Speed Time Distance: TIFR CSE 2012 | Part A | Question: 16



Walking at $\frac{4}{5}$ is normal speed a man is 10 minute too late. Find his usual time in minutes.

- A. 81
 B. 64
 C. 52
 D. 40
 E. It is not possible to determine the usual time from given data.

tifr2012 quantitative-aptitude speed-time-distance

Answer key

14.20.2 Speed Time Distance: TIFR CSE 2017 | Part A | Question: 3



On planet TIFR, the acceleration of an object due to gravity is half that on planet earth. An object on planet earth dropped from a height h takes time t to reach the ground. On planet TIFR, how much time would an object dropped from height h take to reach the ground?

- A. $\left(\frac{t}{\sqrt{2}} \right)$ B. $\sqrt{2}t$ C. $2t$ D. $\left(\frac{h}{t} \right)$
 E. $\left(\frac{h}{2t} \right)$

tifr2017 quantitative-aptitude speed-time-distance

Answer key

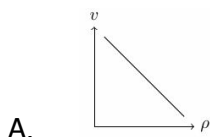
14.20.3 Speed Time Distance: TIFR CSE 2019 | Part A | Question: 8



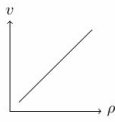
Consider the following toy model of traffic on a straight , single lane, highway. We think of cars as points, which move at the maximum speed v that satisfies the following constraints:

1. The speed is no more than the speed limit v_{max} mandated for the highway.
2. The speed is such that when traveling at this speed, it takes at least time t_0 (where t_0 is a fixed time representing the reaction time of drivers) to reach the car ahead, in case the car ahead stops suddenly.

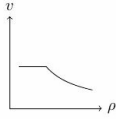
Let us assume that in the steady state, all cars on the highway move at the same speed v satisfying both the above constraints, and the distance between any two successive cars is the same. Let ρ denote the “density”, that is, the number of cars per unit length of the highway. Which of the following graphs most accurately captures the relationship between the speed v and the density ρ in this model ?



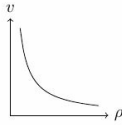
B.



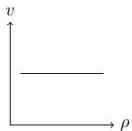
C.



D.



E.



tifr2019 quantitative-aptitude speed-time-distance non-gatecse

Answer key

14.20.4 Speed Time Distance: TIFR CSE 2020 | Part A | Question: 14



A ball is thrown directly upwards from the ground at a speed of 10 ms^{-1} , on a planet where the gravitational acceleration is 10 ms^{-2} . Consider the following statements:

1. The ball reaches the ground exactly 2 seconds after it is thrown up
2. The ball travels a total distance of 10 metres before it reaches the ground
3. The velocity of the ball when it hits the ground is 10 ms^{-1}

What can you say now?

- A. Only Statement 1 is correct
 B. Only Statement 2 is correct
 C. Only Statement 3 is correct
 D. None of the Statements 1,2 or 3 is correct
 E. All of the Statements 1,2 and 3 are correct

tifr2020 quantitative-aptitude speed-time-distance

Answer key

14.21

Statistics (1)

14.21.1 Statistics: TIFR CSE 2015 | Part A | Question: 15



Let A and B be non-empty disjoint sets of real numbers. Suppose that the average of the numbers in the first set is μ_A and the average of the numbers in the second set is μ_B ; let the corresponding variances be v_A and v_B respectively. If the average of the elements in $A \cup B$ is $\mu = p \cdot \mu_A + (1 - p) \cdot \mu_B$, what is the variance of the elements in $A \cup B$?

- A. $p \cdot v_A + (1 - p) \cdot v_B$
 B. $(1 - p) \cdot v_A + p \cdot v_B$
 C. $p \cdot [v_A + (\mu_A - \mu)^2] + (1 - p) \cdot [v_B + (\mu_B - \mu)^2]$
 D. $(1 - p) \cdot [v_A + (\mu_A - \mu)^2] + p \cdot [v_B + (\mu_B - \mu)^2]$
 E. $p \cdot v_A + (1 - p) \cdot v_B + (\mu_A - \mu_B)^2$

Answer key

14.22

Three Dimensional Geometry (1)

14.22.1 Three Dimensional Geometry: TIFR CSE 2018 | Part A | Question: 2



Consider the following subset of $\mathbb{R}^3$ (the first two are cylinder, the third is a plane):

- $C_1 = \{(x, y, z) : y^2 + z^2 \leq 1\}$;
- $C_2 = \{(x, y, z) : x^2 + z^2 \leq 1\}$;
- $H = \{(x, y, z) : z = 0.2\}$;

Let $A = C_1 \cap C_2 \cap H$. Which of the following best describe the shape of set A ?

- A. Circle
- B. Ellipse
- C. Triangle
- D. Square
- E. An octagonal convex figure with curved sides

tifr2018 quantitative-aptitude geometry three-dimensional-geometry non-gatecse

Answer key

14.23

Work Time (1)

14.23.1 Work Time: TIFR CSE 2018 | Part A | Question: 4



The distance from your home to your office is 4 kilometers and your normal walking speed is 4 Km/hr. On the first day, you walk at your normal walking speed and take time T_1 to reach office.

On the second day, you walk at a speed of 3 Km/hr from 2 Kilometers, and at a speed of 5 Km/hr for the remaining 2 Kilometers and you take time T_2 to reach office.

On the third day, you walk at a speed of 3 Km/hr for 30 minutes, and at 5 Km/hr for the remaining time and take time T_3 to reach office.

What can you say about the ordering of T_1, T_2 and T_3

- A. $T_1 > T_2$ and $T_1 < T_3$
- B. $T_1 = T_2 = T_3$
- C. $T_1 < T_2$ and $T_1 > T_3$
- D. $T_1 = T_2$ and $T_1 < T_3$
- E. $T_1 < T_2$ and $T_1 = T_3$

tifr2018 quantitative-aptitude work-time

Answer key

Answer Keys

14.1.1	A	14.2.1	C	14.2.2	B	14.2.3	D	14.2.4	C
14.3.1	C	14.3.2	D	14.3.3	D	14.3.4	B	14.4.1	E
14.4.2	E	14.4.3	B	14.4.4	C	14.4.5	C	14.5.1	A
14.5.2	B	14.5.3	A	14.6.1	D	14.7.1	B	14.8.1	A
14.8.2	D	14.8.3	D	14.9.1	A	14.9.2	D	14.10.1	B
14.10.2	D	14.10.3	D	14.10.4	D	14.10.5	C	14.10.6	C

14.10.7	E
14.11.3	C
14.12.1	A
14.14.1	E
14.15.1	A
14.16.1	E
14.19.2	D
14.20.4	E

14.10.8	E
14.11.4	B
14.12.2	C
14.14.2	C
14.15.2	B
14.17.1	D
14.19.3	C
14.21.1	C

14.10.9	D
14.11.5	B
14.12.3	D
14.14.3	D
14.15.3	E
14.18.1	A
14.20.1	D
14.22.1	D

14.11.1	B
14.11.6	B
14.12.4	E
14.14.4	C
14.15.4	B
14.18.2	D
14.20.2	B
14.23.1	E

14.11.2	D
14.11.7	A
14.13.1	D
14.14.5	C
14.15.5	D
14.19.1	A
14.20.3	C

15.1.1 Factors: TIFR CSE 2025 | Part A | Question: 3



How many distinct factors does $2^2 3^3 4^4 5^5$ have? For example, 12 has 6 distinct factors: 1, 2, 3, 4, 6, and 12.

- A. 240 B. 264 C. 324 D. 360
- E. None of the above

tifr2025 factors number-theory

Answer key 

15.2.1 Perfect Square: TIFR CSE 2025 | Part A | Question: 1



What is the smallest positive integer that when multiplied with $2^3 3^2 4^3 5^7 6^5$ results in a perfect square?

- A. 6 B. 10 C. 15 D. 60
- E. 210

tifr2025 number-theory perfect-square

Answer key 

Answer Keys

15.1.1

B

15.2.1

C

16.1

Area (1)

16.1.1 Area: TIFR CSE 2024 | Part A | Question: 7



Let $S := \{(a, b) \mid 0 \leq a \leq 1, 0 \leq b \leq 1\}$, a unit square, in $\mathbb{R}^2$. Let $B := \{(x, y) \mid x^2 + y^2 \leq 1\}$, a unit disk, in $\mathbb{R}^2$. Define the set $S + B$ as follows:

$$S + B := \{(u, v) \mid \exists (a, b) \in S, (x, y) \in B \text{ such that } u = a + x, v = b + y\}.$$

What is the area of $S + B$?

- A. $\pi + 4$
- B. $\pi + 5$
- C. $\pi + 3$
- D. $\pi + 2$
- E. None of the above.

tifr2024 geometry area

16.2

Complex Number (1)

16.2.1 Complex Number: TIFR CSE 2025 | Part A | Question: 14



Let $c_1, c_2, \dots, c_n$ and z be complex numbers such that

$$\frac{1}{z - c_1} + \frac{1}{z - c_2} + \dots + \frac{1}{z - c_n} = 0$$

Assume that the numbers $c_1, c_2, \dots, c_n$ are represented in the complex plane by the vertices of a convex n -gon C . Then

- A. the number z always lies strictly outside C .
- B. the number z always lies inside or on C .
- C. the number z may lie inside or outside C depending on $c_1, c_2, \dots, c_n$.
- D. if z lies inside C then z must be the centroid, i.e., $z = \frac{1}{n} \sum_{k=1}^n c_k$.
- E. None of the above

tifr2025 complex-number geometry

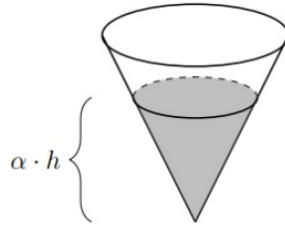
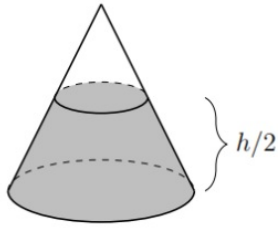
16.3

Cone (1)

16.3.1 Cone: TIFR CSE 2025 | Part A | Question: 7



Suppose we had a conical container with some liquid filled in that reaches until half its height when placed with the circular face downwards. If we were to turn the cone upside-down, suppose that the liquid reaches until $\alpha \cdot h$ height when measured from the bottom (the pointy end). (Diagram below not drawn to scale.)



Which of the following is true about α ?

- A. $\alpha \leq 0.5$
 B. $0.5 < \alpha < 0.6$
 C. $0.6 < \alpha \leq 0.7$
 D. $0.7 < \alpha \leq 0.8$
 E. $0.8 < \alpha \leq 1$

tifr2025 geometry cone

Answer key

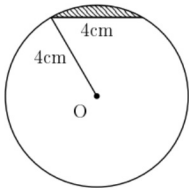
16.4

Geometry (1)

16.4.1 Geometry: TIFR CSE 2024 | Part A | Question: 1



If O is the center of the circle, what is the area of the shaded portion in square cm?



- A. $4\pi - 4\sqrt{2}$
 B. $\frac{7}{2}\pi - 4\sqrt{2}$
 C. $\frac{7}{2}\pi - 4\sqrt{3}$
 D. $\frac{8}{3}\pi - 4\sqrt{3}$
 E. $\frac{8}{3}\pi - 4\sqrt{2}$

tifr2024 geometry

Answer key

16.5

Variance (1)

16.5.1 Variance: TIFR CSE 2025 | Part A | Question: 13



Let $(X, Y, Z) \in \mathbb{R}^3$ be a point chosen uniformly at random from the unit sphere $S^2 := \{(a, b, c) \in \mathbb{R}^3 \mid a^2 + b^2 + c^2 = 1\}$ in three dimensions. What is the variance of the random variable Z ?

- A. $\frac{1}{2}$
 B. $\frac{1}{3}$
 C. $\frac{1}{4}$
 D. $\frac{1}{5}$
 E. 1

tifr2025 geometry uniform-distribution random-variable variance

Answer key

Answer Keys

16.1.1

B

16.2.1

B

16.3.1

E

16.4.1

D

16.5.1

B

17.1.1 Object Oriented Programming: TIFR CSE 2011 | Part B | Question: 40



Consider the class of object oriented languages. Which of the following is true?

- A. Pascal is an object oriented language.
- B. Object oriented languages require heap management.
- C. Object oriented languages cannot be implemented in language C.
- D. Object oriented languages are more powerful than declarative programming languages.
- E. Parallelism cannot be realized in object oriented languages.

tifr2011 programming object-oriented-programming non-gatecse

Answer key 

Answer Keys

17.1.1	B
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18.1

Page Replacement (1)

18.1.1 Page Replacement: TIFR CSE 2013 | Part B | Question: 14



Assume a demand paged memory system where ONLY THREE pages can reside in the memory at a time. The following sequence gives the order in which the program references the pages.

1, 3, 1, 3, 4, 2, 2, 4

Assume that least frequently used page is replaced when necessary. If there is more than one least frequently used pages then the least recently used page among them is replaced. During the program's execution, how many times will the pages 1, 2, 3 and 4 be brought to the memory?

- | | |
|-----------------------------------|-----------------------------------|
| A. 2, 2, 2, 2 times, respectively | B. 1, 1, 1, 2 times, respectively |
| C. 1, 1, 1, 1 times, respectively | D. 2, 1, 2, 2 times, respectively |
| E. None of the above | |

tifr2013 operating-system page-replacement

Answer key

18.2

Process Scheduling (1)

18.2.1 Process Scheduling: TIFR CSE 2020 | Part B | Question: 8



Jobs keep arriving at a processor. A job can have an associated time length as well as a priority tag. New jobs may arrive while some earlier jobs are running. Some jobs may keep running indefinitely. A **starvation free** job-scheduling policy guarantees that no job waits indefinitely for service. Which of the following job-scheduling policies is starvation free?

- | | |
|-----------------------|-----------------------|
| A. Round – robin | B. Shortest job first |
| C. Priority queuing | D. Latest job first |
| E. None of the others | |

tifr2020 operating-system process-scheduling round-robin-scheduling

Answer key

18.3

Process Synchronization (8)

18.3.1 Process Synchronization: TIFR CSE 2010 | Part B | Question: 28



Consider the **concurrent** program:

```
x: 1;
cobegin
  x := x + 3 || x := x + x + 2
coend
```

Reading and writing of variables is atomic, but the evaluation of an expression is not atomic. The set of possible values of variable x at the end of the execution of the program is:

- | | | | |
|--------------|--------|---------------|------------------|
| A. {4} | B. {8} | C. {4, 7, 10} | D. {4, 7, 8, 10} |
| E. {4, 7, 8} | | | |

tifr2010 process-synchronization

Answer key

18.3.2 Process Synchronization: TIFR CSE 2010 | Part B | Question: 32



Consider the following solution (expressed in Dijkstra's guarded command notation) to the mutual exclusion problem.

```
process P1 is
begin
  loop
    Non_critical_section;
    while not (Turn=1) do skip od;
    Critical_section_1;
    Turn:=2;
  end loop
end
```

```
process P2 is
begin
  loop
    Non_critical_section;
    while not (Turn=2) do skip od;
    Critical_section_2;
    Turn:=1;
  end loop
end
```

Initially, $\text{Turn} = 1$. Assume that the two processes run forever and that no process stays in its critical and non-critical section infinitely. A mutual exclusion program is correct if it satisfies the following requirements.

1. Only one process can be in a critical region at a time.
2. Program is a dead-lock free, i.e., if both processes are trying to enter the critical region then at least one of them does enter the critical region.
3. Program is starvation-free; i.e., a process trying to enter the critical region eventually manages to do so.

The above mutual exclusion solution.

- A. Does not satisfy the requirement (1).
- B. Satisfies the requirement (1) but does not satisfy the requirement (2).
- C. Satisfies the requirements (1) and (2), but does not satisfy the requirement (3).
- D. Satisfies the requirement (1) and (3), but does not satisfy the requirement (2).
- E. Satisfies all the requirements (1), (2), and (3).

tifr2010 operating-system process-synchronization

Answer key

18.3.3 Process Synchronization: TIFR CSE 2011 | Part B | Question: 22



Consider the program

```
P:: x:=1; y:=1; z:=1; u:=0
```

And the program

```
Q:: x, y, z, u := 1, 1, 1, 1; u:= 0
```

Which of the following is true?

- A. P and Q are equivalent for sequential processors.
- B. P and Q are equivalent for all multi-processor models.
- C. P and Q are equivalent for all multi-core machines.

- D. P and Q are equivalent for all networks of computers.
E. None of the above

tifr2011 operating-system process-synchronization

Answer key

18.3.4 Process Synchronization: TIFR CSE 2011 | Part B | Question: 26



Consider the following two scenarios in the dining philosophers problem:

- First a philosopher has to enter a room with the table that restricts the number of philosophers to four.
- There is no restriction on the number of philosophers entering the room.

Which of the following is true?

- | | |
|--|--------------------------------------|
| A. Deadlock is possible in (i) and (ii). | B. Deadlock is possible in (i). |
| C. Starvation is possible in (i). | D. Deadlock is not possible in (ii). |
| E. Starvation is not possible in (ii) | |

tifr2011 operating-system process-synchronization

Answer key

18.3.5 Process Synchronization: TIFR CSE 2011 | Part B | Question: 28



Consider a basic block:

```
x:= a[i]; a[j]:= y; z:= a[j]
```

optimized by removing common sub expression $a[i]$ as follows:

```
x:= a[i]; z:= x; a[j]:= y.
```

Which of the following is true?

- Both are equivalent.
- The values computed by both are exactly the same.
- Both give exactly the same values only if i is not equal to j .
- They will be equivalent in concurrent programming languages with shared memory.
- None of the above.

tifr2011 process-synchronization operating-system normal

Answer key

18.3.6 Process Synchronization: TIFR CSE 2011 | Part B | Question: 34



Consider the class of synchronization primitives. Which of the following is false?

- Test and set primitives are as powerful as semaphores.
- There are various synchronizations that can be implemented using an array of semaphores but not by binary semaphores.
- Split binary semaphores and binary semaphores are equivalent.
- All statements a - c are false.
- Petri nets with and without inhibitor arcs have the same power.

tifr2011 operating-system process-synchronization

Answer key

18.3.7 Process Synchronization: TIFR CSE 2012 | Part B | Question: 9



Consider the concurrent program

```
x := 1;
cobegin
  x := x + x + 1 || x := x + 2
coend;
```

Reading and writing of a variable is atomic, but evaluation of an expression is not atomic. The set of possible values of variable x at the end of execution of the program is

- A. {3} B. {7} C. {3,5,7} D. {3,7}
- E. {3,5}

tifr2012 process-synchronization operating-system

Answer key

18.3.8 Process Synchronization: TIFR CSE 2015 | Part B | Question: 14



Consider the following concurrent program (where statements separated by $||$ with-in cobegin-coend are executed concurrently).

```
x:=1
cobegin
  x:=x+1 || x:=x+1 || x:=x+1
coend
```

Reading and writing of variables is atomic but evaluation of expressions is not atomic. The set of possible values of x at the end of execution of the program is

- A. {4} B. {2,3,4}
- C. {2,4} D. {2,3}
- E. {2}

tifr2015 process-synchronization operating-system normal

Answer key

18.4

Semaphore (1)

18.4.1 Semaphore: TIFR CSE 2012 | Part B | Question: 10



Consider the blocked-set semaphore where the signaling process awakens any one of the suspended process; i.e.,

Wait (S): If $S > 0$ then $S \leftarrow S - 1$, else suspend the execution of this process.

Signal (S): If there are processes that have been suspended on semaphore S , then wake any one of them, else $S \leftarrow S + 1$

Consider the following solution of mutual exclusion problem using blocked-set semaphores.

```
s := 1;
cobegin
  P(1) || P(2) || ..... || P(N)
coend
```

Where the task body $P(i)$ is

```
begin
while true do
begin
  < non critical section >
  Wait (S)
  <critical section>
  Signal (S)
```

end
end

Here N is the number of concurrent processors. Which of the following is true?

- A. The program fails to achieve mutual exclusion of critical regions.
- B. The program achieves mutual exclusion, but starvation freedom is ensured only for $N \leq 2$
- C. The program does not ensure mutual exclusion if $N \geq 3$
- D. The program achieves mutual exclusion, but allows starvation for any $N \geq 2$
- E. The program achieves mutual exclusion and starvation freedom for any $N \geq 1$

tifr2012 operating-system process-synchronization semaphore

Answer key 

Answer Keys

18.1.1	B	18.2.1	A	18.3.1	C	18.3.2	E	18.3.3	D
18.3.4	C	18.3.5	E	18.3.6	TBA	18.3.7	C	18.3.8	B
18.4.1	B								



19.1

Array (1)

19.1.1 Array: TIFR CSE 2011 | Part B | Question: 30



Consider an array $A[1\dots n]$. It consists of a permutation of numbers $1\dots n$. Now compute another array $B[1\dots n]$ as follows: $B[A[i]] := i$ for all i . Which of the following is true?

- A. B will be a sorted array.
- B. B is a permutation of array A .
- C. Doing the same transformation twice will not give the same array.
- D. B is not a permutation of array A .
- E. None of the above.

tifr2011 data-structures array

Answer key

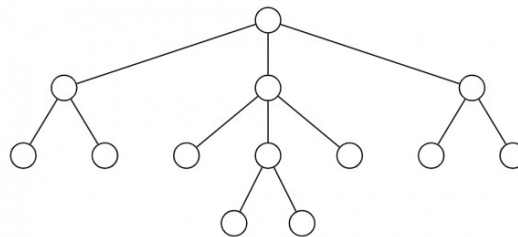
19.2

Binary Heap (1)

19.2.1 Binary Heap: TIFR CSE 2014 | Part B | Question: 19



Consider the following tree with 13 nodes.



Suppose the nodes of the tree are randomly assigned distinct labels from $\{1, 2, \dots, 13\}$, each permutation being equally likely. What is the probability that the labels form a min-heap (i.e., every node receives the minimum label in its subtree)?

- A. $\left(\frac{1}{6!}\right) \left(\frac{1}{3!}\right)^2$
- B. $\left(\frac{1}{3!}\right)^2 \left(\frac{1}{2!}\right)^3$
- C. $\left(\frac{1}{13}\right) \left(\frac{1}{6}\right) \left(\frac{1}{3}\right)^3$
- D. $\frac{2}{13}$
- E. $\frac{1}{2^{13}}$

tifr2014 binary-heap

Answer key

19.3

Binary Search Tree (1)

19.3.1 Binary Search Tree: TIFR CSE 2010 | Part B | Question: 26



Suppose there is a balanced binary search tree with n nodes, where at each node, in addition to the key, we store the number of elements in the sub tree rooted at that node.

Now, given two elements a and b , such that $a < b$, we want to find the number of elements x in the tree that lie between a and b , that is, $a \leq x \leq b$. This can be done with (choose the best solution).

- A. $O(\log n)$ comparisons and $O(\log n)$ additions.
- B. $O(\log n)$ comparisons but no further additions.
- C. $O(\sqrt{n})$ comparisons but $O(\log n)$ additions.

- D. $O(\log n)$ comparisons but a constant number of additions.
E. $O(n)$ comparisons and $O(n)$ additions, using depth-first-search.

tifr2010 binary-search-tree

Answer key 

19.4

Binary Tree (5)

19.4.1 Binary Tree: TIFR CSE 2012 | Part B | Question: 16



Consider a complete binary tree of height n , where each edge is one Ohm resistor. Suppose all the leaves of the tree are tied together. Approximately how much is the effective resistance from the root to this bunch of leaves for very large n ?

- A. Exponential in n .
B. Cubic in n .
C. Linear in n .
D. Logarithmic in n .
E. Of the order square root of n .

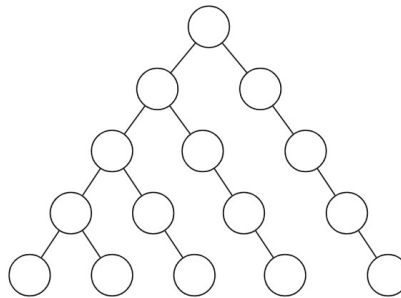
tifr2012 binary-tree

Answer key

19.4.2 Binary Tree: TIFR CSE 2013 | Part B | Question: 13



Given a binary tree of the following form and having n nodes, the height of the tree is



- A. $\Theta(\log n)$
 B. $\Theta(n)$
 C. $\Theta(\sqrt{n})$
 D. $\Theta(n/\log n)$
 E. None of the above.

tifr2013 binary-tree data-structures

Answer key

19.4.3 Binary Tree: TIFR CSE 2014 | Part B | Question: 1



Let T be a rooted binary tree whose vertices are labelled with symbols $a, b, c, d, e, f, g, h, i, j, k$. Suppose the in-order (visit left subtree, visit root, visit right subtree) and post-order (visit left subtree, visit right subtree, visit root) traversals of T produce the following sequences.

in-order: $a, b, c, d, e, f, g, h, i, j, k$

post-order: $a, c, b, e, f, h, j, k, i, g, d$

How many leaves does the tree have?

- A. THREE.
C. FIVE.
E. Cannot be determined uniquely from the given information.
- B. FOUR.
D. SIX.

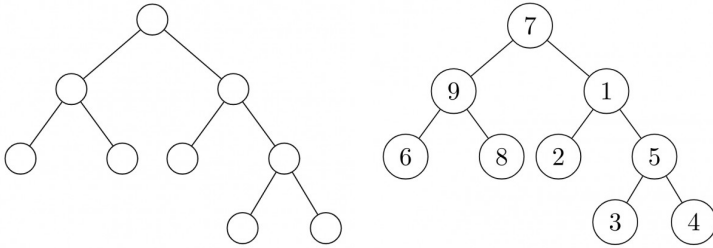
tifr2014 binary-tree data-structures easy

Answer key

19.4.4 Binary Tree: TIFR CSE 2015 | Part B | Question: 4



First, consider the tree on the left.



On the right, the nine nodes of the tree have been assigned numbers from the set $\{1, 2, \dots, 9\}$ so that for every node, the numbers in its left subtree and right subtree lie in disjoint intervals (that is, all numbers in one subtree are less than all numbers in the other subtree). How many such assignments are possible? Hint: Fix a value for the root and ask what values can then appear in its left and right subtrees.

- A. $2^9 = 512$ B. $2^4 \cdot 3^2 \cdot 5 \cdot 9 = 6480$ C. $2^3 \cdot 3 \cdot 5 \cdot 9 = 1080$ D. $2^4 = 16$
 E. $2^3 \cdot 3^3 = 216$

tifr2015 binary-tree combinatory

Answer key

19.4.5 Binary Tree: TIFR CSE 2018 | Part B | Question: 6



Consider the following implementation of a binary tree data structure. The operator $+$ denotes list-concatenation.

That is, $[a, b, c] + [d, e] = [a, b, c, d, e]$.

```
struct TreeNode:
    int value
    TreeNode leftChild
    TreeNode rightChild

function preOrder(T):
    if T == null:
        return []
    else:
        return [T.value] + preOrder(T.leftChild) + preOrder(T.rightChild)

function inOrder(T):
    if T == null:
        return []
    else:
        return inOrder(T.leftChild) + [T.value] + inOrder(T.rightChild)

function postOrder(T):
    if T == null:
        return []
    else:
        return postOrder(T.leftChild) + postOrder(T.rightChild) + [T.value]
```

For some T the functions $\text{inOrder}(T)$ and $\text{preOrder}(T)$ return the following:

$\text{inOrder}(T) : [12, 10, 6, 9, 7, 2, 15, 5, 1, 13, 4, 3, 8, 14, 11]$

$\text{preOrder}(T) : [5, 2, 10, 12, 9, 6, 7, 15, 13, 1, 3, 4, 14, 8, 11]$

What does $\text{postOrder}(T)$ return ?

- A. $[12, 6, 10, 7, 15, 2, 9, 1, 4, 13, 8, 11, 14, 3, 5]$

- B. [11, 8, 14, 4, 3, 1, 13, 15, 7, 6, 9, 12, 10, 2, 5]
- C. [11, 14, 8, 3, 4, 13, 1, 5, 15, 2, 7, 9, 6, 10, 12]
- D. [12, 6, 7, 9, 10, 15, 2, 1, 4, 8, 11, 14, 3, 13, 5]
- E. Cannot be uniquely determined from given information.

tifr2018 data-structures binary-tree

Answer key

19.5 Infix Prefix (1)

19.5.1 Infix Prefix: TIFR CSE 2021 | Part B | Question: 3



What is the prefix expression corresponding to the expression:

$$((9 + 8) * 7 + (6 * (5 + 4)) * 3) + 2?$$

You may assume that $*$ has precedence over $+$?

- A. $* + + 987 * * 6 + + 5432$
- B. $* + + + 987 * * 6 + 5432$
- C. $+ * + + 987 * * 6 + 5432$
- D. $+ + * + 987 * * 6 + 5432$
- E. $+ * + * 987 + + 6 * 5432$

tifr2021 data-structures stack infix-prefix

Answer key

19.6 Linked List (2)

19.6.1 Linked List: TIFR CSE 2021 | Part B | Question: 2



Let L be a singly-linked list X and Y be additional pointer variables such that X points to the first element of L and Y points to the last element of L . Which of the following operations cannot be done in time that is bound above by a constant?

- A. Delete the first element of L .
- B. Delete the last element of L .
- C. Add an element after the last element of L .
- D. Add an element before the first element of L .
- E. Interchange the first two elements of L .

tifr2021 data-structures linked-list

Answer key

19.6.2 Linked List: TIFR CSE 2022 | Part B | Question: 5



There is an unsorted list of n integers. You are given 3 distinct integers and you have to check if all 3 integers are present in the list or not. The only operation that you are allowed to perform is a comparison. Let A be an algorithm for this task that performs the least number of comparisons. Let c be the number of comparisons done by A . Then,

- A. $c = 3n$
- B. $c = 2n + 5$
- C. $c \geq 3n - 1$
- D. $c \leq n$
- E. $c \leq 2n + 3$

tifr2022 data-structures linked-list

Answer key

19.7

Queue (1)

19.7.1 Queue: TIFR CSE 2022 | Part B | Question: 1



Which data structure is commonly used to implement breadth first search in a graph?

- A. A queue
- B. A stack
- C. A heap
- D. A hash table
- E. A splay tree

tifr2022 data-structures queue easy

Answer key

19.8

Stack (1)

19.8.1 Stack: TIFR CSE 2017 | Part B | Question: 3



We have an implementation that supports the following operations on a stack (in the instructions below, **s** is the name of the stack).

- **isempty(s)** : returns **True** if **s** is empty, and **False** otherwise.
- **top(s)** : returns the top element of the stack, but does not pop the stack; returns **null** if the stack is empty.
- **push(s, x)** : places **x** on top of the stack.
- **pop(s)** : pops the stack; does nothing if **s** is empty.

Consider the following code:

```
pop_ray_pop(x):
    s=empty
    for i=1 to length(x):
        if (x[i] == '('):
            push(s, x[i])
        else:
            while (top(s)=='('):
                pop(s)
            end while
            push(s, ')')
        end if
    end for
    while not isempty(s):
        print top(s)
        pop(s)
    end while
```

What is the output of this program when

pop_ray_pop("(((())((()(((("))

is executed?

- A. (((((
- B.))) (((
- C.)))
- D. (((()))
- E. ())

tifr2017 data-structures stack

Answer key

19.9

Tree (2)

19.9.1 Tree: TIFR CSE 2012 | Part B | Question: 15



Let T be a tree of n nodes. Consider the following algorithm, that constructs a sequence of leaves $u_1, u_2, \dots$. Let u_1 be some leaf of tree. Let u_2 be a leaf that is farthest from u_1 . Let u_3 be the leaf that is farthest from u_2 , and, in general, let u_{i+1} be a leaf of T that is farthest from u_i (if there are many choices for u_{i+1} , pick one arbitrarily). The algorithm stops when some u_i is visited again. What can you say about the distance between u_i and u_{i+1} , as $i = 1, 2, \dots$?

- A. For some trees, the distance strictly reduces in each step.
- B. For some trees, the distance increases initially and then decreases.
- C. For all trees, the path connecting u_2 and u_3 is a longest path in the tree.
- D. For some trees, the distance reduces initially, but then stays constant.
- E. For the same tree, the distance between the last two vertices visited can be different, based on the choice of the first leaf u_1 .

tifr2012 data-structures tree

Answer key

19.9.2 Tree: TIFR CSE 2023 | Part B | Question: 8



Let U be a finite set and let h be a function mapping $U \times U$ to U . Consider the following process that assigns values to all nodes of a complete binary tree with 128 leaves. Initially, the leaf nodes are assigned arbitrary values from U . Then, the value that is assigned to an internal node is the output of h on the values assigned to this node's children. The value of the tree is given by the value assigned to the root.

Once you have computed the value of the tree, you realize that the initial value that was assigned to the 53^{rd} leaf was incorrect and you therefore need to use a different value for the 53^{rd} leaf. Given this incorrect tree as input, at most how many times would an optimal algorithm need to recompute h in order to obtain the correct value of the tree?

- A. 7
- B. 8
- C. 53
- D. 127
- E. 255

tifr2023 data-structures tree

Answer key

Answer Keys

19.1.1	B	19.2.1	C	19.3.1	A	19.4.1	A	19.4.2	C
19.4.3	C	19.4.4	B	19.4.5	D	19.5.1	D	19.6.1	B
19.6.2	E	19.7.1	A	19.8.1	D	19.9.1	C	19.9.2	A

20.0.1 TIFR CSE 2011 | Part B | Question: 24



Consider the program

```
x:=0; y:=0; (r1:=x; r2:=x; y:= if r1 = r2 then 1 || r3:= y; x:= r3)
```

Note that $\parallel$ denotes the parallel operator. In which of the following cases can the program possibly result in a final state with $r1 = 0; r2 = r3 = 1$.

- A. Such a transformation is not possible in Java.
- B. Such a program transformation is possible in Java.
- C. Possible in Pascal when the compiler appropriately translates the $\parallel$ operator to interleaved pascal statements.
- D. Possible in all sequential programming languages when the compiler appropriately translates the $\parallel$ operator to interleaved statements in the sequential language.
- E. None of the above.

tifr2011 programming non-gatecse

20.1

Functions (1)

20.1.1 Functions: TIFR CSE 2020 | Part B | Question: 12



Given the pseudocode below for the function **remains()**, which of the following statements is true about the output, if we pass it a positive integer $n > 2$?

```
int remains(int n)
{
    int x = n;
    for (i=(n-1); i>1; i--) {
        x = x % i;
    }
    return x;
}
```

- A. Output is always 0
- B. Output is always 1
- C. Output is 0 only if n is NOT a prime number
- D. Output is 1 only if n is a prime number
- E. None of the above

tifr2020 programming programming-in-c functions

Answer key

20.2

Loop Invariants (4)

20.2.1 Loop Invariants: TIFR CSE 2010 | Part B | Question: 30



Consider the following program for summing the entries of the array b : array $[0..N-1]$ of integers, where N is a positive integer. (The symbol ' $\neq$ ' denotes 'not equal to').

```
var
    i, s: integer;
Program
    i := 0;
    s := 0;
    [*] while i <> N do
        s := s + b[i];
        i := i + 1;
    od
```

Which of the following gives the invariant that holds at the beginning of each loop, that is, each time the program arrives at point $[*]$?

- A. $s = \sum_{j=0}^N b[j] \ \& \ 0 \leq i \leq N$
- B. $s = \sum_{j=0}^{i-1} b[j] \ \& \ 0 \leq i < N$
- C. $s = \sum_{j=0}^i b[j] \ \& \ 0 < i \leq N$
- D. $s = \sum_{j=1}^N b[j] \ \& \ 0 \leq i < N$
- E. $s = \sum_{j=0}^{i-1} b[j] \ \& \ 0 \leq i \leq N$

tifr2010 programming loop-invariants

Answer key

20.2.2 Loop Invariants: TIFR CSE 2010 | Part B | Question: 37



Consider the program where a, b are integers with $b > 0$.

```
x:=a; y:=b; z:=0;
while y > 0 do
  if odd (x) then
    z:= z + x;
    y:= y - 1;
  else y:= y % 2;
    x:= 2 * x;
  fi
```

Invariant of the loop is a condition which is true before and after every iteration of the loop. In the above program the loop invariant is given by

$$0 \leq y \text{ and } z + x * y = a * b$$

Which of the following is true of the program?

- A. The program will not terminate for some values of a, b .
- B. The program will terminate with $z = 2^b$
- C. The program will terminate with $z = a * b$.
- D. The program will not terminate for some values of a, b but when it does terminate, the condition $z = a * b$ will hold.
- E. The program will terminate with $z = a^b$

tifr2010 programming loop-invariants

Answer key

20.2.3 Loop Invariants: TIFR CSE 2017 | Part B | Question: 5



Consider the following psuedocode fragment, where y is an integer that has been initialized.

```
int i=1
int j=1
while (i<10):
  j=j*i
  i=i+1
  if (i==y):
```

```
break
end if
end while
```

Consider the following statements:

- i. $(i == 10)$ or $(i == y)$
- ii. If $y > 10$, then $i == 10$
- iii. If $j = 6$, then $y == 4$

Which of the above statements is/are TRUE at the end of the while loop? Choose from the following options.

- A. i only
- B. iii only
- C. ii and iii only
- D. i, ii, and iii
- E. None of the above

tifr2017 programming loop-invariants

Answer key

20.2.4 Loop Invariants: TIFR CSE 2019 | Part B | Question: 9



Consider the following program fragment:

```
var x, y: integer;
x := 1; y := 0;
while y < x do
begin
  x := 2*x;
  y := y+1
end;
```

For the above fragment, which of the following is a loop invariant ?

- A. $x = y + 1$
- B. $x = (y + 1)^2$
- C. $x = (y + 1)2^y$
- D. $x = 2^y$
- E. None of the above, since the loop does not terminate

tifr2019 programming loop-invariants

Answer key

20.3

Parameter Passing (2)

20.3.1 Parameter Passing: TIFR CSE 2011 | Part B | Question: 32



Various parameter passing mechanisms have been in used in different programming languages. Which of the following statements is true?

- A. Call by value result is used in language Ada.
- B. Call by value result is the same as call by name.
- C. Call by value is the most robust.
- D. Call by reference is the same as call by name.
- E. Call by name is the most efficient.

tifr2011 programming parameter-passing

Answer key

20.3.2 Parameter Passing: TIFR CSE 2019 | Part B | Question: 8



Consider the following program fragment:

```

var a,b : integer;
procedure G(c,d: integer);
begin
  c:=c-d;
  d:=c+d;
  c:=d-c
end;
a:=2;
b:=3;
G(a,b);

```

If both parameters to G are passed by reference, what are the values of a and b at the end of the above program fragment ?

- A. $a = 0$ and $b = 2$ B. $a = 3$ and $b = 2$ C. $a = 2$ and $b = 3$ D. $a = 1$ and $b = 5$
 E. None of the above

tifr2019 programming parameter-passing

Answer key 

20.4

Programming In C (1)

20.4.1 Programming In C: TIFR CSE 2019 | Part B | Question: 6



Given the following pseudocode for function `printx()` below, how many times is x printed if we execute `printx(5)`?

```

void printx(int n) {
  if(n==0){
    printf("x");
  }
  for(int i=0;i<=n-1;++i){
    printx(n-1);
  }
}

```

- A. 625 B. 256 C. 120 D. 24
 E. 5

tifr2019 programming programming-in-c

Answer key 

20.5

Recursion (5)

20.5.1 Recursion: TIFR CSE 2010 | Part B | Question: 31



Consider the following computation rules. **Parallel-outermost rule**: Replace all the outermost occurrences of F (i.e., all occurrences of F which do not occur as arguments of other F 's) simultaneously. **Parallel - innermost rule**: Replace all the innermost occurrences of F (i.e., all occurrences of F with all arguments free of F 's) simultaneously. Now consider the evaluations of the recursive program over the integers.

```

F(x, y) <== if x = 0 then 0 else
  [ F(x + 1, F(x, y)) * F(x - 1, F(x, y))]

```

where the multiplication functions $*$ is extended as follows:

```

0 * w & w * 0 are 0
a * w & w * a are w (for any non-zero integer a)
w * w is w

```

We say that $F(x, y) = w$ when the evaluation of $F(x, y)$ does not terminate. Computing $F(1, 0)$ using

the parallel - innermost and parallel - outermost rule yields

- A. w and 0 respectively
- B. 0 and 0 respectively
- C. w and w respectively
- D. w and 1 respectively
- E. none of the above

tifr2010 programming recursion

Answer key

20.5.2 Recursion: TIFR CSE 2011 | Part B | Question: 38



Consider the class of recursive and iterative programs. Which of the following is false?

- A. Recursive programs are more powerful than iterative programs.
- B. For every iterative program there is an equivalent recursive program.
- C. Recursive programs require dynamic memory management.
- D. Recursive programs do not terminate sometimes.
- E. Iterative programs and recursive programs are equally expressive.

tifr2011 recursion programming

Answer key

20.5.3 Recursion: TIFR CSE 2016 | Part B | Question: 5



Consider the recursive function **mc91**.

```
int mc91(int n)
{
    print n
    if (n > 100) {
        return n-10;
    }
    else {
        return mc91(mc91(n+11));
    }
}
```

Let

Out = $\{n : \text{there is an } x \in \{0, 1, \dots, 100\} \text{ such that } n \text{ is one of the integers printed by mc91}(x)\}$

Then which of the following is **Out**?

- A. $\{n : -\infty < n \leq 100\}$
- B. $\{n : 0 \leq n \leq 101\}$
- C. $\{n : 0 \leq n \leq 110\}$
- D. $\{n : 0 \leq n \leq 111\}$
- E. $\{n : 0 \leq n < +\infty\}$

tifr2016 programming-in-c recursion

Answer key

20.5.4 Recursion: TIFR CSE 2018 | Part A | Question: 7



Consider the following function definition.

```
void greet(int n)
{
    if(n>0)
    {
        printf("hello");
        greet(n-1);
    }
    printf("world");
}
```

If you run `greet(n)` for some non-negative integer `n`, what would it print?

- A. `n` times "hello", followed by `n+1` times "world"
- B. `n` times "hello", followed by `n` times "world"
- C. `n` times "helloworld"
- D. `n+1` times "helloworld"
- E. `n` times "helloworld", followed by "world"

tifr2018 programming programming-in-c recursion

Answer key

20.5.5 Recursion: TIFR CSE 2021 | Part B | Question: 6



Consider the following pseudocode:

```
procedure HowManyDash(n)
  if n=0 then
    print '-'
  else if n=1 then
    print '-'
  else
    HowManyDash(n-1)
    HowManyDash(n-2)
  end if
end procedure
```

How many '-' does `HowManyDash(10)` print?

- A. 9
- B. 10
- C. 55
- D. 89
- E. 1024

tifr2021 programming recursion

Answer key

Answer Keys

20.0.1	B	20.1.1	B	20.2.1	E	20.2.2	A	20.2.3	D
20.2.4	D	20.3.1	E	20.3.2	B	20.4.1	C	20.5.1	A
20.5.2	E	20.5.3	D	20.5.4	A	20.5.5	D		



21.1.1 Closure Property: TIFR CSE 2013 | Part B | Question: 11



Which of the following statements is FALSE?

- A. The intersection of a context free language with a regular language is context free.
- B. The intersection of two regular languages is regular.
- C. The intersection of two context free languages is context free
- D. The intersection of a context free language and the complement of a regular language is context free.
- E. The intersection of a regular language and the complement of a regular language is regular.

tifr2013 theory-of-computation easy closure-property

Answer key

21.1.2 Closure Property: TIFR CSE 2014 | Part B | Question: 14



Which the following is FALSE?

- A. Complement of a recursive language is recursive.
- B. A language recognized by a non-deterministic Turing machine can also be recognized by a deterministic Turing machine.
- C. Complement of a context free language can be recognized by a Turing machine.
- D. If a language and its complement are both recursively enumerable then it is recursive.
- E. Complement of a non-recursive language can never be recognized by any Turing machine.

tifr2014 theory-of-computation closure-property

Answer key

21.1.3 Closure Property: TIFR CSE 2016 | Part B | Question: 2



Which language class has the following properties?

It is closed under union and intersection but not complement.

- | | |
|--|------------------------------------|
| A. Regular language | B. Context-free language |
| C. Recursive language | D. Recursively enumerable language |
| E. Languages that are not recursively enumerable | |

tifr2016 theory-of-computation closure-property

Answer key

21.2.1 Complex Number: TIFR CSE 2024 | Part B | Question: 11



Let $\mathbb{C}$ denote the set of complex numbers and let k be a positive integer. Given a non-zero univariate polynomial $f(x)$ with coefficients in $\mathbb{C}$ and an $a \in \mathbb{C}$, we say that a is a zero of f of multiplicity k if $f(a) = 0$, $\frac{d^k f}{dx^k}(a) \neq 0$, and for all $i \in \{1, \dots, k-1\}$, $\frac{d^i f}{dx^i}(a) = 0$.

Which of the following is true for every polynomial f of degree d and every positive integer k ?

- A. The number of distinct zeroes in $\mathbb{C}$ of f of multiplicity k is at most d/k , and can be smaller than d/k

as well.

- B. The number of distinct zeroes in $\mathbb{C}$ of f of multiplicity k is at least d/k , and can be larger than d/k as well.
- C. The number of distinct zeroes in $\mathbb{C}$ of f of multiplicity k is equal to d/k .
- D. The number of distinct zeroes in $\mathbb{C}$ of f of multiplicity k is at least d , and can be larger than d as well.
- E. The number of distinct zeroes in $\mathbb{C}$ of f of multiplicity k is equal to d .

tifr2024 polynomials calculus complex-number

21.3

Complexity (1)

21.3.1 Complexity: TIFR CSE 2024 | Part B | Question: 12



In the n -queens completion problem, the input is an $n \times n$ chess board with queens on some squares, and the goal is to determine if there is a way to place more queens so that the total number of queens is n and no two queens attack each other (two queens are said to attack each other if they are on the same row, or they are on the same column, or they are on the same diagonal).

Consider the following statements:

- The n -queens completion problem is decidable.
- The n -queens completion problem is decidable in time $O(n^n)$.
- The problem of "checking whether a given program solves the n -queens completion problem" is decidable.
- The problem of "checking whether a given program solves the n -queens completion problem in time $O(n^n)$ " is decidable.
- The problem of "checking whether a given program solves the n -queens completion problem" is decidable in time $O(n^n)$.

Which of the above is true?

- A. Only (i) and (ii). B. Only (i) and (iii).
- C. Only (i), (ii) and (iv). D. Only (iii), (iv) and (v).
- E. Only (i), (iii) and (iv).

tifr2024 theory-of-computation decidability complexity analytical-aptitude

Answer key

21.4

Context Free Grammar (2)

21.4.1 Context Free Grammar: TIFR CSE 2020 | Part B | Question: 6



Consider the context-free grammar below (ϵ denotes the empty string, alphabet is $\{a, b\}$):

$$S \rightarrow \epsilon \mid aSb \mid bSa \mid SS.$$

What language does it generate?

- A. $(ab)^* + (ba)^*$ B. $(abba)^* + (baab)^*$
- C. $(aabb)^* + (bbaa)^*$ D. Strings of the form $a^n b^n$ or $b^n a^n$, n any positive integer
- E. Strings with equal numbers of a and b

tifr2020 theory-of-computation context-free-grammar

Answer key

21.4.2 Context Free Grammar: TIFR CSE 2021 | Part B | Question: 9



Let L be a context-free language generated by the context-free grammar $G = (V, \Sigma, R, S)$ where V is the finite set of variables, Σ the finite set of terminals (disjoint from V), R the finite set of rules and $S \in V$ the start variable. Consider the context-free grammar G' obtained by adding $S \rightarrow SS$ to the set of rules in G . What must be true for the language L' generated by G' ?

- A. $L' = LL$
- B. $L' = L$
- C. $L' = L^*$
- D. $L' = \{xx \mid x \in L\}$
- E. None of the above

tifr2021 theory-of-computation context-free-grammar

Answer key

21.5

Context Free Language (1)

21.5.1 Context Free Language: TIFR CSE 2021 | Part B | Question: 5



For a language L over the alphabet $\{a, b\}$, let $\bar{L}$ denote the complement of L and let L^* denote the Kleene-closure of L . Consider the following sentences.

- i. $\bar{L}$ and L^* are both context-free.
- ii. $\bar{L}$ is not context-free but L^* is context-free.
- iii. $\bar{L}$ is context-free but L^* is regular.

Which of the above sentence(s) is/are true if $L = \{a^n b^n \mid n \geq 0\}$?

- A. Both (i) and (iii)
- B. Only (i)
- C. Only (iii)
- D. Only (ii)
- E. None of the above

tifr2021 theory-of-computation context-free-language

Answer key

21.6

Countable Uncountable Set (1)

21.6.1 Countable Uncountable Set: TIFR CSE 2022 | Part A | Question: 3



A binary string is a sequence of 0's and 1's. A binary string is *finite* if the sequence is finite, otherwise it is *infinite*. Examples of finite binary strings include 00010100, and 1111101010.

Which of the following is **TRUE** about the set of all finite binary strings and the set of all infinite binary strings?

- A. The set of all finite binary strings is countable while the set of all infinite binary strings is uncountable
- B. The set of all finite binary strings is uncountable while the set of all infinite binary strings is countable
- C. The set of all finite binary strings and the set of all infinite binary strings are both countable
- D. The set of all finite binary strings and the set of all infinite binary strings are both uncountable
- E. The set of all finite binary strings is countable while whether the set of all infinite binary strings is countable or not is not known

tifr2022 theory-of-computation countable-uncountable-set

Answer key

21.7

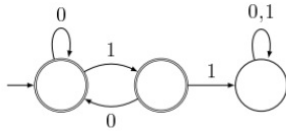
Counting (1)

21.7.1 Counting: TIFR CSE 2024 | Part B | Question: 15



Consider the following automata:

Let N be the number of 0/1-strings of length exactly 6 accepted by this automata. Which of the following is true about N ?



- A. $N \leq 4$
- B. $4 < N \leq 8$
- C. $8 < N \leq 16$
- D. $16 < N \leq 32$
- E. $32 < N \leq 64$

tifr2024 finite-automata counting theory-of-computation

Answer key

21.8 Decidability (4)

21.8.1 Decidability: TIFR CSE 2010 | Part B | Question: 25



Which of the following problems is decidable? (Here, CFG means context free grammar and CFL means context free language.)

- A. Given a CFG G , find whether $L(G) = R$, where R is regular set.
- B. Given a CFG G , find whether $L(G) = \{\}$.
- C. Find whether the intersection of two CFLs is empty.
- D. Find whether the complement of CFL is a CFL.
- E. Find whether CFG G_1 and CFG G_2 generate the same language, i.e, $L(G_1) = L(G_2)$.

tifr2010 theory-of-computation context-free-language decidability

Answer key

21.8.2 Decidability: TIFR CSE 2011 | Part B | Question: 25



Let A_{TM} be defined as follows:

$A_{TM} = \{\langle M, w \rangle \mid \text{The Turing machine } M \text{ accepts the word } w\}$

And let L be some **NP** – complete language. Which of the following statements is FALSE?

- A. $L \in \mathbf{NP}$
- B. Every problem in **NP** is polynomial time reducible to L .
- C. Every problem in **NP** is polynomial time reducible to A_{TM} .
- D. Since L is **NP** – complete, A_{TM} is polynomial time reducible to L .
- E. $A_{TM} \notin \mathbf{NP}$.

tifr2011 theory-of-computation decidability

Answer key

21.8.3 Decidability: TIFR CSE 2020 | Part B | Question: 2



Consider the following statements.

1. The intersection of two context-free languages is always context-free
2. The super-set of a context-free languages is never regular
3. The subset of a decidable language is always decidable

4. Let $\Sigma = \{a, b, c\}$. Let $L \subseteq \Sigma$ be the language of all strings in which either the number of occurrences of a is the same as the number of occurrences of b OR the number occurrences of b is the same as the number of occurrences of c . Then, L is not context-free.

Which of the above statements are true?

- A. Only (1)
 B. Only (1) and (2)
 C. Only (1), (2) and (3)
 D. Only (4)
 E. None of (1), (2), (3), (4) are true.

tifr2020 theory-of-computation context-free-language decidability

Answer key

21.8.4 Decidability: TIFR CSE 2024 | Part B | Question: 5



For two languages A, B over the alphabet Σ , let the perfect shuffle of A and B be the language

$$\left\{ w = a_1 b_1 a_2 b_2 \cdots a_k b_k \text{ where } a_1 a_2 \cdots a_k \in A \text{ and } b_1 b_2 \cdots b_k \in B. \right. \\ \left. \text{and } k \in \mathbb{N} \right\}$$

Consider the following statements.

- If A and B are regular, then perfect shuffle of A and B is regular.
- If A and B are regular, then perfect shuffle of A and B is context-free.
- If A and B are decidable, then perfect shuffle of A and B is decidable.

Which of above statements is/are true?

- A. All of (i), (ii) and (iii).
 B. Only (i) and (ii).
 C. Only (ii).
 D. Only (ii) and (iii).
 E. None of (i), (ii), (iii) is true.

tifr2024 theory-of-computation regular-language decidability context-free-language

Answer key

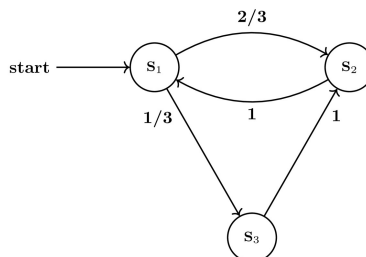
21.9

Finite Automata (2)

21.9.1 Finite Automata: TIFR CSE 2022 | Part A | Question: 13



Consider the transition system shown in the figure below with the initial state s_1 . A token is initially placed at s_1 , and it moves to s_2 with probability $\frac{2}{3}$, and to s_3 with probability $\frac{1}{3}$. From s_2 and s_3 , the token always moves to s_1 and s_2 respectively. A *run* of the system consists of an infinite sequence of states constructed by moving the token from one state to another following the transitions forever. Assuming such a run is chosen randomly, what is the fraction of times that the state s_2 is expected to appear in the run?



A. $\frac{1}{7}$

B. $\frac{2}{7}$

C. $\frac{3}{7}$

D. $\frac{5}{7}$

E. None of the above

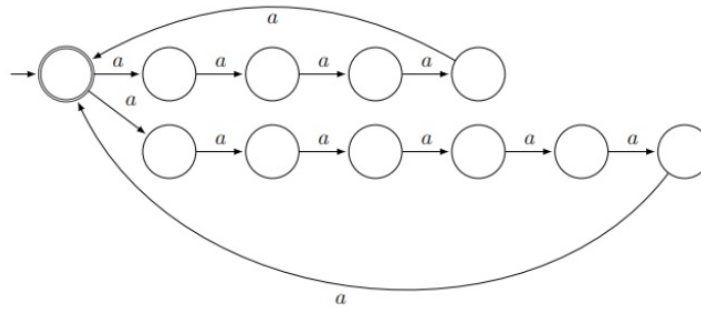
tifr2022 theory-of-computation finite-automata probability

Answer key

21.9.2 Finite Automata: TIFR CSE 2025 | Part B | Question: 7



Consider the following non-deterministic automata for a unary language $L \subseteq \{a\}^*$ consisting of 11 states with the start state as the unique final state. All transitions shown are on the symbol a .



Let ℓ be the length of the longest string that is NOT accepted by the above automata.

Which of the following is TRUE about ℓ ?

A. $\ell = 18$

B. $\ell = 22$

C. $\ell = 23$

D. $\ell = 26$

E. ℓ is not well-defined as there are arbitrary long strings that the automata does not accept.

tifr2025 theory-of-computation finite-automata

Answer key

21.10

Formal Languages (1)

21.10.1 Formal Languages: TIFR CSE 2024 | Part B | Question: 6



The four nucleotides in DNA are called A, C, G, and T. Consider the following languages over the alphabet $\{A, C, G, \text{ and } T\}$.

$$L_1 = \{(AC)^n(GT)^n \mid n \geq 0\}$$

$$L_2 = \{A^n C^m G^n T^n \mid n \geq 0\}$$

$$L_3 = \{(AT)^n(CG)^m \mid m \neq n \text{ and } m, n \geq 0\}$$

Which of the above languages is/are context-free?

A. Only L_1 .B. Only L_3 .C. Only L_1 and L_3 .D. Only L_1 and L_2 .E. All three of L_1, L_2, L_3 .

tifr2024 theory-of-computation context-free-language formal-languages

21.11

Identify Class Language (12)

21.11.1 Identify Class Language: TIFR CSE 2010 | Part B | Question: 22

Let L consist of all binary strings beginning with a 1 such that its value when converted to decimal is divisible by 5. Which of the following is true?

- A. L can be recognized by a deterministic finite state automaton.
- B. L can be recognized by a non-deterministic finite state automaton but not by a deterministic finite state automaton.
- C. L can be recognized by a deterministic push-down automaton but not by a non-deterministic finite state automaton.
- D. L can be recognized by a non-deterministic push-down automaton but not by a deterministic push-down automaton.
- E. L cannot be recognized by any push-down automaton.

tifr2010 theory-of-computation identify-class-language

Answer key

21.11.2 Identify Class Language: TIFR CSE 2010 | Part B | Question: 35

Consider the following languages over the alphabet $\{0, 1\}$.

$$L1 = \{x.x^R \mid x \in \{0, 1\}^*\}$$

$$L2 = \{x.x \mid x \in \{0, 1\}^*\}$$

Where x^R is the reverse of string x ; e.g. $011^R = 110$. Which of the following is true?

- A. Both $L1$ and $L2$ are regular.
- B. $L1$ is context-free but not regular where as $L2$ is regular.
- C. Both $L1$ and $L2$ are context free and neither is regular.
- D. $L1$ is context free but $L2$ is not context free.
- E. Both $L1$ and $L2$ are not context free.

tifr2010 theory-of-computation identify-class-language

Answer key

21.11.3 Identify Class Language: TIFR CSE 2012 | Part B | Question: 18

Let a^i denote a sequence $a.a \dots a$ with i letters and let $\mathbb{N}$ be the set of natural numbers $1, 2, \dots$. Let $L_1 = \{a^i b^{2i} \mid i \in \mathbb{N}\}$ and $L_2 = \{a^i b^{i^2} \mid i \in \mathbb{N}\}$ be two languages. Which of the following is correct?

- A. Both L_1 and L_2 are context-free languages.
- B. L_1 is context-free and L_2 is recursive but not context-free.
- C. Both L_1 and L_2 are recursive but not context-free.
- D. L_1 is regular and L_2 is context-free.
- E. Complement of L_2 is context-free.

tifr2012 theory-of-computation identify-class-language

Answer key

21.11.4 Identify Class Language: TIFR CSE 2014 | Part B | Question: 13

Let L be a given context-free language over the alphabet $\{a, b\}$. Construct L_1, L_2 as follows.

Let $L_1 = L - \{xyx \mid x, y \in \{a, b\}^*\}$, and $L_2 = L \cdot L$. Then,

- A. Both L_1 and L_2 are regular.
- B. Both L_1 and L_2 are context free but not necessarily regular.
- C. L_1 is regular and L_2 is context free.
- D. L_1 and L_2 both may not be context free.
- E. L_1 is regular but L_2 may not be context free.

tifr2014 theory-of-computation identify-class-language

Answer key 

21.11.5 Identify Class Language: TIFR CSE 2015 | Part B | Question: 8



Let $\Sigma_1 = \{a\}$ be a one letter alphabet and $\Sigma_2 = \{a, b\}$ be a two letter alphabet. A language over an alphabet is a set of finite length words comprising letters of the alphabet. Let L_1 and L_2 be the set of languages over Σ_1 and Σ_2 respectively. Which of the following is true about L_1 and L_2 :

- A. Both are finite.
- B. Both are countably infinite.
- C. L_1 is countable but L_2 is not.
- D. L_2 is countable but L_1 is not.
- E. Neither of them is countable.

tifr2015 identify-class-language

Answer key 

21.11.6 Identify Class Language: TIFR CSE 2017 | Part B | Question: 14



Consider the following grammar G with terminals $\{[,]\}$, start symbol S , and non-terminals $\{A, B, C\}$:

$$S \rightarrow AC \mid SS \mid AB$$

$$C \rightarrow SB$$

$$A \rightarrow [$$

$$B \rightarrow]$$

A language L is called prefix-closed if for every $x \in L$, every prefix of x is also in L . Which of the following is FALSE?

- A. $L(G)$ is context free
- B. $L(G)$ is infinite
- C. $L(G)$ can be recognized by a deterministic push down automaton
- D. $L(G)$ is prefix-closed
- E. $L(G)$ is recursive

tifr2017 theory-of-computation identify-class-language

Answer key 

21.11.7 Identify Class Language: TIFR CSE 2017 | Part B | Question: 4



Let L be the language over the alphabet $\{1, 2, 3, (,)\}$ generated by the following grammar (with start symbol S , and non-terminals $\{A, B, C\}$):

$$\begin{aligned}
 S &\rightarrow ABC \\
 A &\rightarrow (\\
 B &\rightarrow 1B \mid 2B \mid 3B \\
 B &\rightarrow 1 \mid 2 \mid 3 \\
 C &\rightarrow)
 \end{aligned}$$

Then, which of the following is TRUE?

- A. L is finite
- B. L is not recursively enumerable
- C. L is regular
- D. L contains only strings of even length
- E. L is context-free but not regular

tifr2017 theory-of-computation identify-class-language

Answer key 

21.11.8 Identify Class Language: TIFR CSE 2018 | Part B | Question: 11



Consider the language $L \subseteq \{a, b, c\}^*$ defined as

$$L = \{a^p b^q c^r : p = q \text{ or } q = r \text{ or } r = p\}.$$

Which of the following answer is TRUE about the complexity of this language?

- A. L is regular but not context-free
- B. L is context-free but not regular
- C. L is decidable but not context free
- D. The complement of L , defined as $\overline{L} = \{a, b, c\}^* \setminus L$, is regular.
- E. L is regular, context-free and decidable

tifr2018 identify-class-language theory-of-computation

Answer key 

21.11.9 Identify Class Language: TIFR CSE 2018 | Part B | Question: 14



Define the language $\text{INFINITE}_{DFA} \equiv \{(A) \mid A \text{ is a DFA and } L(A) \text{ is an infinite language}\}$, where (A) denotes the description of the deterministic finite automata (DFA). Then which of the following about INFINITE_{DFA} is TRUE:

- A. It is regular.
- B. It is context-free but not regular.
- C. It is Turing decidable (recursive).
- D. It is Turing recognizable but not decidable.
- E. Its complement is Turing recognizable but it is not decidable.

tifr2018 theory-of-computation identify-class-language

Answer key 

21.11.10 Identify Class Language: TIFR CSE 2019 | Part B | Question: 10



Let the language D be defined in the binary alphabet $\{0, 1\}$ as follows:

$D := \{w \in \{0,1\}^* \mid \text{substrings } 01 \text{ and } 10 \text{ occur an equal number of times in } w\}$

For example, $101 \in D$ while $1010 \notin D$. Which of the following must be TRUE of the language D ?

- A. D is regular
- B. D is context-free but not regular
- C. D is decidable but not context-free
- D. D is decidable but not in NP
- E. D is undecidable

tifr2019 theory-of-computation identify-class-language

Answer key

21.11.11 Identify Class Language: TIFR CSE 2023 | Part B | Question: 15



Consider the language $L = \{a^i \$ a^j \$ b^k \$ \mid k \leq \max(i, j), i, j, k \geq 0\}$ over the alphabet $\Sigma = \{a, b, \$\}$. The complement of the language L , that is, $\Sigma^* \setminus L$ is denoted by $\bar{L}$. Which of the following is true?

- A. Both L and $\bar{L}$ are regular languages.
- B. L is a context-free language and $\bar{L}$ is a regular language.
- C. Both L and $\bar{L}$ are context-free languages.
- D. L is a context-free language and $\bar{L}$ is not a context-free language.
- E. Neither is L a context-free language nor is $\bar{L}$ a context-free language.

tifr2023 theory-of-computation identify-class-language

21.11.12 Identify Class Language: TIFR CSE 2023 | Part B | Question: 2



Which of the following is true about the set of regular languages and the set of recursively enumerable languages over a finite alphabet Σ ?

- A. The set of regular languages is countable while the set of recursively enumerable languages is uncountable.
- B. The set of regular languages is uncountable while the set of recursively enumerable languages is countable.
- C. The set of regular languages and the set of recursively enumerable languages are both countable.
- D. The set of regular languages and the set of recursively enumerable languages are both uncountable.
- E. The set of regular languages is countable while whether the set of recursively enumerable languages is countable or not is not known and is a longstanding open problem.

tifr2023 theory-of-computation identify-class-language

Answer key

21.12

Number Theory (1)

21.12.1 Number Theory: TIFR CSE 2024 | Part B | Question: 3



For any positive integer N , let $p(N)$ be the probability that a uniformly random number $a \in \{1, \dots, N\}$ has an odd number of factors (including 1 and the number itself). Which of the following is true about the function $p : \mathbb{N} \rightarrow \mathbb{R}$?

- A. $\lim_{N \rightarrow \infty} p(N) = \frac{1}{2}$.
- B. $p(N) = \Theta\left(\frac{\log N}{N}\right)$.
- C. $p(N) = \Theta\left(\frac{1}{N}\right)$.
- D. $p(N) = \Theta\left(\frac{1}{\sqrt{N}}\right)$.

$$E. p(N) = \Theta\left(\frac{1}{\log N}\right).$$

tifr2024 probability number-theory limits

Answer key 

21.13

P NP NPC NPH (5)

21.13.1 P NP NPC NPH: TIFR CSE 2016 | Part B | Question: 8



Consider the following language

$$\text{PRIMES} = \left\{ \underbrace{111 \dots 11}_{p \text{ times}} : p \text{ is prime} \right\}$$

Then, which of the following is TRUE?

- A. PRIMES is regular
- B. PRIMES is undecidable
- C. PRIMES is decidable in polynomial time
- D. PRIMES is context free but not regular
- E. PRIMES is NP-complete and $P \neq NP$

tifr2016 theory-of-computation decidability p-np-npc-nph

Answer key 

21.13.2 P NP NPC NPH: TIFR CSE 2019 | Part B | Question: 7



A formula is said to be a 3-CF-formula if it is a conjunction (i.e., an AND) of clauses, and each clause has at most 3 literals. Analogously, a formula is said to be a 3-DF-formula if it is a disjunction (i.e., an OR) of clauses of at most 3 literals each.

Define the languages 3-CF-SAT and 3-DF-SAT as follows:

$$3\text{-CF-SAT} = \{\Phi \mid \Phi \text{ is a satisfiable 3-CF formula}\}$$

$$3\text{-DF-SAT} = \{\Phi \mid \Phi \text{ is a satisfiable 3-DF formula}\}$$

Which of the following best represents our current knowledge of these languages ?

- A. Both 3-CF-SAT and 3-DF-SAT are in NP but only 3-CF-SAT is NP-complete
- B. Both 3-CF-SAT and 3-DF-SAT are in NP-complete
- C. Both 3-CF-SAT and 3-DF-SAT are in P
- D. Both 3-CF-SAT and 3-DF-SAT are in NP but only 3-DF-SAT is NP-complete
- E. Neither 3-CF-SAT nor 3-DF-SAT are in P

tifr2019 theory-of-computation p-np-npc-nph

Answer key 

21.13.3 P NP NPC NPH: TIFR CSE 2021 | Part B | Question: 4



Consider the following two languages.

$$\text{PRIME} = \{1^n \mid n \text{ is a prime number}\},$$

$$\text{FACTOR} = \{1^n 01^a 01^b \mid n \text{ has a factor in the range } [a, b]\}.$$

What can you say about the languages **PRIME** and **FACTOR**?

- A. **PRIME** is in **P**, but **FACTOR** is not in **P**.
- B. Neither **PRIME** nor **FACTOR** are in **P**.
- C. Both **PRIME** and **FACTOR** are in **P**.
- D. **PRIME** is not in **P**, but **FACTOR** is in **P**.
- E. None of the above since we can answer this question only if we resolve the status of the **NP** vs. **P** question.

tifr2021 theory-of-computation p-np-npc-nph

Answer key

21.13.4 P NP NPC NPH: TIFR CSE 2023 | Part B | Question: 14



A k -term $T = \ell_1 \wedge \ell_2 \wedge \dots \wedge \ell_r$ is defined to be a conjunction of at most k literals, where each literal ℓ_i is a Boolean variable or its negation and $r \leq k$. A formula Φ is said to be a k -DNF if it is of the form $T_1 \vee T_2 \vee \dots \vee T_m$ where each T_i is a k -term. An assignment of TRUE/FALSE values to the variables appearing in a formula Φ is said to be a *satisfying* assignment for Φ if, under the assignment, Φ evaluates to TRUE. Otherwise, the assignment is said to be a *violating* assignment for Φ . Consider the following statements

- (S1) Given a 2-DNF Φ , one can find a violating assignment for Φ in polynomial time.
- (S2) It is NP-hard to find a violating assignment for a given 3-DNF Φ .
- (S3) Given a 3-DNF Φ , one can find a satisfying assignment for Φ in polynomial time.

Which of the following statements are correct?

- A. Statements (S1), (S2), and (S3) are all true.
- B. Statements (S1) and (S2) are true, but statement (S3) is not known to be true.
- C. Statement (S2) is true, but statements (S1) and (S3) are not known to be true.
- D. Statements (S1), (S2), and (S3) are false.
- E. Statements (S1) and (S3) are true, but (S2) is not known to be true.

tifr2023 theory-of-computation p-np-npc-nph

21.13.5 P NP NPC NPH: TIFR CSE 2025 | Part B | Question: 6



The complexity class **NP** corresponds to the class of languages which can be accepted by some nondeterministic Turing machine in polynomial time.

The complexity class **coNP** consists of all languages whose complement language is in the class **NP**.

A complexity class $\mathcal{C}$ is said to be closed under union if for any two languages L_1 and L_2 in $\mathcal{C}$, we also have that $L_1 \cup L_2 \in \mathcal{C}$. Similarly, we also say that a complexity class $\mathcal{C}$ is *closed under intersection* if for any two languages L_1 and L_2 in $\mathcal{C}$, we also have that $L_1 \cap L_2 \in \mathcal{C}$.

Consider the following statements.

- i. The class **NP** is closed under union.
- ii. The class **coNP** is closed under union.
- iii. The class **NP** is closed under intersection.
- iv. The class **coNP** is closed under intersection.

Which of the above statements are TRUE?

- A. Only (i) and (iv)
- B. Only (ii) and (iii)
- C. Only (i) and (iii)
- D. Only (ii) and (iv)

E. All of (i), (ii), (iii), and (iv)

tifr2025 theory-of-computation p-np-npc-nph turing-machine

Answer key

21.14

Polynomials (1)

21.14.1 Polynomials: TIFR CSE 2024 | Part A | Question: 5



Let $p(x)$ be a polynomial with real coefficients which satisfies $p(r) = p(-r)$ for every real number r . Let $n \geq 5$ be a positive integer. Suppose that $p(i) = i$ for all $1 \leq i \leq n$. What is the maximum possible value of the absolute value of the coefficient of x^5 in $p(x)$?

- A. 0 B. 5 C. 10 D. n
- E. $n + 1$

tifr2024 polynomials

Answer key

21.15

Pushdown Automata (1)

21.15.1 Pushdown Automata: TIFR CSE 2025 | Part B | Question: 4



Consider the following languages:

- L_1 is the set of languages recognised by a deterministic pushdown automaton.
- L_2 is the set of languages recognised by a nondeterministic pushdown automaton.
- L_3 is the set of languages recognised by a nondeterministic Turing machine.
- L_4 is the set of languages recognised by a nondeterministic Turing machine with two tapes.

Choose the correct option below.

- A. $L_1 \subsetneq L_2 \subsetneq L_3 \subsetneq L_4$ B. $L_1 \subsetneq L_2 \subsetneq L_3 = L_4$.
- C. $L_1 \subsetneq L_2 = L_3 = L_4$ D. $L_1 = L_2 \subsetneq L_3 = L_4$
- E. $L_1 \subseteq L_2 = L_3 \subsetneq L_4$

tifr2025 theory-of-computation pushdown-automata turing-machine recursive-and-recursively-enumerable-languages

Answer key

21.16

Recursive and Recursively Enumerable Languages (2)

21.16.1 Recursive and Recursively Enumerable Languages: TIFR CSE 2010 | Part B | Question: 40



Which of the following statement is FALSE?

- A. All recursive sets are recursively enumerable.
- B. The complement of every recursively enumerable sets is recursively enumerable.
- C. Every Non-empty recursively enumerable set is the range of some totally recursive function.
- D. All finite sets are recursive.
- E. The complement of every recursive set is recursive.

tifr2010 theory-of-computation recursive-and-recursively-enumerable-languages

Answer key

21.16.2 Recursive and Recursively Enumerable Languages: TIFR CSE 2012 | Part B | Question: 19

Which of the following statements is TRUE?



- A. Every turning machine recognizable language is recursive.
- B. The complement of every recursively enumerable language is recursively enumerable.
- C. The complement of a recursive language is recursively enumerable.
- D. The complement of a context-free language is context-free.
- E. The set of turning machines which do not halt on empty input forms a recursively enumerable set.

tifr2012 theory-of-computation recursive-and-recursively-enumerable-languages

Answer key

21.17

Reduction (2)

21.17.1 Reduction: TIFR CSE 2016 | Part B | Question: 3



Assume $P \neq NP$. Which of the following is not TRUE?

- A. 2-SAT in NP
- B. 2-SAT in coNP
- C. 3-SAT is polynomial-time reducible to 2-SAT
- D. 4-SAT is polynomial-time reducible to 3-SAT
- E. 2-SAT in P

tifr2016 theory-of-computation reduction p-np-npc-nph

Answer key

21.17.2 Reduction: TIFR CSE 2018 | Part B | Question: 15



G represents an undirected graph and a cycle refers to a simple cycle (no repeated edges or vertices).

Define the following two languages.

$SCYCLE = \{(G, k) \mid G \text{ contains a cycle of length at most } k\}$
and

$LCYCLE = \{(G, k) \mid G \text{ contains a cycle of length at least } k\}$

Which of the following is NOT known to be TRUE (to the best of our current knowledge) ?

- A. $SCYCLE \in P$
- B. $LCYCLE \in NP$.
- C. $LCYCLE \leq_p SCYCLE$ (i.e, there is a polynomial time many-to-one reduction from $LCYCLE$ to $SCYCLE$).
- D. $LCYCLE$ is NP-complete.
- E. $SCYCLE \leq_p LCYCLE$ (i.e, there is a polynomial time many-to-one reduction from $SCYCLE$ to $LCYCLE$).

tifr2018 theory-of-computation reduction p-np-npc-nph non-gatecse

Answer key

21.18

Regular Expression (7)

21.18.1 Regular Expression: TIFR CSE 2010 | Part B | Question: 34



Let r, s, t be regular expressions. Which of the following identities is correct?

- A. $(r + s)^* = r^* s^*$
- B. $r(s + t) = rs + t$
- C. $(r + s)^* = r^* + s^*$
- D. $(rs + r)^* r = r(sr + r)^*$

E. $(r^*s)^* = (rs)^*$

tifr2010 theory-of-computation regular-expression

Answer key

21.18.2 Regular Expression: TIFR CSE 2015 | Part B | Question: 7



Let a, b, c be regular expressions. Which of the following identities is correct?

A. $(a + b)^* = a^*b^*$

B. $a(b + c) = ab + c$

C. $(a + b)^* = a^* + b^*$

D. $(ab + a)^*a = a(ba + a)^*$

E. None of the above.

tifr2015 theory-of-computation regular-expression

Answer key

21.18.3 Regular Expression: TIFR CSE 2017 | Part B | Question: 9



Which of the following regular expressions correctly accepts the set of all 0/1-strings with an even (possibly zero) number of 1s?

A. $(10^*10^*)^*$

B. $(0^*10^*1)^*$

C. $0^*1(10^*1)^*10^*$

D. $0^*1(0^*10^*10^*)^*10^*$

E. $(0^*10^*1)^*0^*$

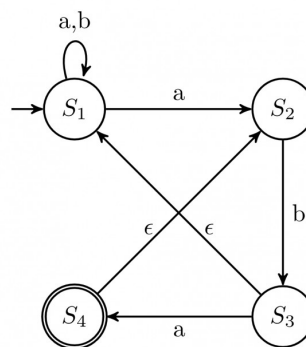
tifr2017 theory-of-computation regular-expression

Answer key

21.18.4 Regular Expression: TIFR CSE 2018 | Part B | Question: 2



Consider the following non-deterministic automation, where s_1 is the start state and s_4 is the final (accepting) state. The alphabet is $\{a, b\}$. A transition with label ϵ can be taken without consuming any symbol from the input.



Which of the following regular expressions corresponds to the language accepted by this automation ?

A. $(a + b)^*aba$

B. $aba(a + b)^*aba$

C. $(a + b)aba(b + a)^*$

D. $aba(a + b)^*$

E. $(ab)^*aba$

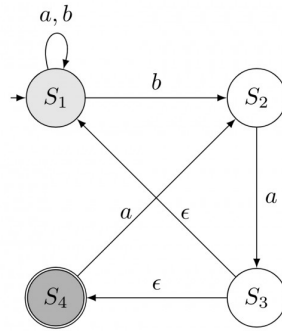
tifr2018 regular-expression finite-automata

Answer key

21.18.5 Regular Expression: TIFR CSE 2019 | Part B | Question: 11



Consider the following non-deterministic automaton, where s_1 is the start state and s_4 is the final (accepting) state. The alphabet is $\{a, b\}$. A transition with label ϵ can be taken without consuming any symbol from the input.



Which of the following regular expressions correspond to the language accepted by this automaton ?

- A. $(a + b)^*aba$ B. $(a + b)^*ba^*$ C. $(a + b)^*ba(aa)^*$ D. $(a + b)^*$
 E. $(a + b)^*baa^*$

tifr2019 theory-of-computation regular-expression

Answer key

21.18.6 Regular Expression: TIFR CSE 2021 | Part B | Question: 7



Which of the following regular expressions defines a language that is different from the other choices?

- A. $b^*(a + b)^*a(a + b)^*ab^*(a + b)^*$
 B. $a^*(a + b)^*ab^*(a + b)^*a(a + b)^*$
 C. $(a + b)^*ab^*(a + b)^*a(a + b)^*b^*$
 D. $(a + b)^*a(a + b)^*b^*a(a + b)^*a^*$
 E. $(a + b)^*b^*a(a + b)^*b^*(a + b)^*$

tifr2021 theory-of-computation regular-expression

Answer key

21.18.7 Regular Expression: TIFR CSE 2022 | Part B | Question: 8



Let r_1 and r_2 be two regular expressions. They symbol $\equiv$ stands for equivalence of two regular expressions in the sense that if $r_1 \equiv r_2$, then both regular expressions describe the same language. Which of the following is/are FALSE?

- i. $(r_1 r_2)^* r_1 \equiv r_1 (r_2 r_1)^*$
 ii. $(r_1^* r_2)^* r_1^* \equiv (r_1 + r_2)^*$
 iii. $(r_1^* r_2^*)^* \equiv (r_1 + r_2)^*$

- A. Only (i) is false B. Only (ii) is false
 C. Only (iii) is false D. Both (i) and (iii) are false
 E. None of the above

tifr2022 theory-of-computation regular-expression

Answer key

21.19.1 Regular Language: TIFR CSE 2013 | Part B | Question: 6



Let L and L' be languages over the alphabet Σ . The left quotient of L by L' is

$$L/L' \stackrel{\text{def}}{=} \{w \in \Sigma^* : wx \in L \text{ for some } x \in L'\}$$

Which of the following is true?

- A. If L/L' is regular then L' is regular.
- B. If L is regular then L/L' is regular.
- C. If L/L' is regular then L is regular.
- D. L/L' is a subset of L .
- E. If L/L' and L' are regular, then L is regular.

tifr2013 theory-of-computation regular-language

Answer key

21.19.2 Regular Language: TIFR CSE 2013 | Part B | Question: 8



Which one of the following languages over the alphabet $0, 1$ is regular?

- A. The language of balanced parentheses where $0, 1$ are thought of as $(,)$ respectively.
- B. The language of palindromes, i.e. bit strings x that read the same from left to right as well as right to left.
- C. $L = \{0^{m^2} : 3 \leq m\}$
- D. The Kleene closure L^* , where L is the language in (c) above.
- E. $\{0^m 1^n | 1 \leq m \leq n\}$

tifr2013 theory-of-computation regular-language

Answer key

21.19.3 Regular Language: TIFR CSE 2014 | Part B | Question: 12



Consider the following three statements:

- i. Intersection of infinitely many regular languages must be regular.
- ii. Every subset of a regular language is regular.
- iii. If L is regular and M is not regular then $L \bullet M$ is necessarily not regular.

Which of the following gives the correct true/false evaluation of the above?

- A. true, false, true.
- B. false, false, true.
- C. true, false, true.
- D. false, false, false.
- E. true, true, true.

tifr2014 theory-of-computation regular-language

Answer key

21.19.4 Regular Language: TIFR CSE 2015 | Part B | Question: 10



Consider the languages

$$L_1 = \{a^m b^n c^p \mid (m = n \vee n = p) \wedge m + n + p \geq 10\}$$

$$L_2 = \{a^m b^n c^p \mid (m = n \vee n = p) \wedge m + n + p \leq 10\}$$

State which of the following is true?

- A. L_1 and L_2 are both regular.
- B. Neither L_1 nor L_2 is regular.
- C. L_1 is regular and L_2 is not regular.
- D. L_1 is not regular and L_2 is regular.
- E. Both L_1 and L_2 are infinite.

Answer key

21.19.5 Regular Language: TIFR CSE 2015 | Part B | Question: 6



Let B consist of all binary strings beginning with a 1 whose value when converted to decimal is divisible by 7.

- A. B can be recognized by a deterministic finite state automaton.
- B. B can be recognized by a non-deterministic finite state automaton but not by a deterministic finite state automaton.
- C. B can be recognized by a deterministic push-down automaton but not by a non-deterministic finite state automaton.
- D. B can be recognized by a non-deterministic push-down automaton but not by a deterministic push-down automaton.
- E. B cannot be recognized by any push down automaton, deterministic or non-deterministic.

Answer key

21.19.6 Regular Language: TIFR CSE 2018 | Part B | Question: 12



Consider the following statements:

- i. For every positive integer n , let $\#n$ be the product of all primes less than or equal to n . Then, $\#p + 1$ is a prime, for every prime p .
- ii. π is a universal constant with value $\frac{22}{7}$.
- iii. No polynomial time algorithm exists that can find the greatest common divisor of two integers given as input in binary.
- iv. Let $L \equiv \{x \in \{0, 1\}^* \mid x \text{ is the binary encoding of an integer that is divisible by } 31\}$. Then, L is a regular language.

Then which of the following is TRUE ?

- A. Only statement (i) is correct.
- B. Only statement (ii) is correct.
- C. Only statement (iii) is correct.
- D. Only statement (iv) is correct.
- E. None of the statements are correct.

Answer key

21.20

Set Theory (1)

21.20.1 Set Theory: TIFR CSE 2024 | Part A | Question: 12



A subset S of the rational numbers is said to be "nice" if for every infinite sequence of $x_1, x_2, \dots$ of elements from S , there is always two indices $i < j$ such that $x_i \leq x_j$. Consider the following statements.

- i. The set of natural numbers $\mathbb{N}$ is "nice".
- ii. The set of integers $\mathbb{Z}$ is "nice".
- iii. The set of positive rational numbers is "nice".

Which of the above statements is/are true?

- A. Only (i).
C. Only (i) and (iii).
E. None of the three statements is true.
- B. Only (i) and (ii).
D. All three statements are true.

tifr2024 set-theory mathematical-logic analytical-aptitude

Answer key 

Answer Keys

21.1.1	C	21.1.2	E	21.1.3	D	21.2.1	A	21.3.1	C
21.4.1	E	21.4.2	E	21.5.1	B	21.6.1	A	21.7.1	D
21.8.1	B	21.8.2	D	21.8.3	E	21.8.4	A	21.9.1	C
21.9.2	C	21.10.1	C	21.11.1	A	21.11.2	D	21.11.3	B
21.11.4	C	21.11.5	E	21.11.6	D	21.11.7	C	21.11.8	B
21.11.9	C	21.11.10	A	21.11.11	D	21.11.12	C	21.12.1	D
21.13.1	C	21.13.2	A	21.13.3	C	21.13.4	A	21.13.5	E
21.14.1	A	21.15.1	B	21.16.1	B	21.16.2	C	21.17.1	C
21.17.2	C	21.18.1	D	21.18.2	D	21.18.3	E	21.18.4	A
21.18.5	C	21.18.6	E	21.18.7	E	21.19.1	B	21.19.2	D
21.19.3	D	21.19.4	D	21.19.5	A	21.19.6	D	21.20.1	A